Materials application completion review packet

Prepared 5 October 2026 | Local review exhibits for task 2026-10-04

Proposed sole inventor: Robert M. Layne, as instructed; legal conception and filing particulars remain unconfirmed.

This packet presents the additive decision guide, eleven-edition evidence index, material/state/process matrix and navigation to 796 later-claim scenarios. The preserved 47-page 3 October application follows as a separate part with its own preparation label and page numbering. Review companions are not silently content of that application or an earlier disclosure.

No applicant physical experiments are performed or planned. No filing, earliest public availability, complete chronological novelty search, final statutory result, legal inventorship confirmation, license grant or universal blocking is asserted.

The editable Markdown/JSON sidecars retain source locators, exact fractions, assumptions and evidence categories. Use those detailed records for an actual comparison; navigation excerpts supply no independent legal or physical support.

Materials application review and decision packet

Prepared **5 October 2026**, for the completion task initiated 4 October. The filename preserves that task identity; it does not backdate this exhibit. Status: **local additive review exhibits; final counsel decisions remain open**. No filing, external publication, priority entitlement, applicant experiment or universal defensive effect is asserted.

The operative editable application remains the 3 October substantive edition (source file: provisional_application_substantive_review_2026-10-03.md), with its 47-page PDF (source file: pdf/provisional_application_substantive_review_2026-10-03.pdf), **99 numbered paragraphs, 199 unchanged candidate claims, two drawings, abstract and administrative worksheet**. Robert M. Layne remains the instructed **proposed sole inventor**. This packet preserves that application's exact bytes and preparation date. Later review examples in companions are not silently adopted into it or an earlier edition.

Reading and review sequence

Step	Artifact and stable locator	Review purpose
1	Application and edition guide (source file: application_substantive_review_2026-10-03.md); specification [0001]-[0099], claim annex and administrative worksheet	Read the current technical disclosure and its qualifications before comparing a claim
2	Current support map (source file: ../data/claim_support_map_substantive_review_2026-10-03.json); /claims	Locate exact textual support without treating a locator as physical or legal proof
3	Current eight-dimension claim screen (source file: ../data/claim_review_2026-10-03.json); /claims, /families, /drafting_issues	Review exact wording, complete parent chain, attribution and open questions for every candidate
4	All-claim later-scenario exhibit (source file: claim_scenarios_2026-10-04.md) and structured assessments (source file: ../data/claim_scenarios_2026-10-04.json)	Compare identified teaching, changed composition/structure, large-list selection and added method/use against each candidate's actual added limitation
5	Material/state/process exhibit (source file: material_state_process_coverage_2026-10-04.md) and structured matrix (source file: ../data/material_state_process_coverage_2026-10-04.json)	Trace family, state, structure, accounting basis, process, property, evidence and failure boundaries; inspect bounded worked records
6	Edition and source index (source file: evidence_edition_index_2026-10-04.md) and hash-bound record (source file: ../data/evidence_edition_index_2026-10-04.json)	Identify the edition and source actually relied on; preserve preparation, publication and observed access as separate events

ESTABLISHED retains source attribution and documented scope. CALCULATED retains assumptions and exact accounting. PROPOSED retains unperformed conditional operations. UNSUPPORTED retains missing technical support. No applicant physical work is performed or planned. A complete review document can identify unresolved fields; it cannot resolve them by renaming their evidence status.

Technical review checklist

Inquiry	Present review support	Decision or evidence still required
Closed support and specimen boundary	Current claims, [0010]-[0022], [0073] and family comparisons distinguish principal lists from complete retained inventory	Identify actual region/time/basis and trace/residual treatment for a relied-on species; nondetection is not exact zero
Exact numerical domains	Independent original floor/grid and unrestricted positive-rational tools; symbolic real domain retained	Choose which domain the actual asserted feature uses; no universal measurement floor or finite exhaustive real catalog follows
State, intact units and normalization	[0095]-[0099], fifteen checked calculation groups and five joint inventories	Supply availability/lifetime, molecular/site/chain constraints, qualified masses and relevant finite inventory for a physical assertion
Preparation and properties	Attributed finite source examples, present accounting and conditional mechanism records	Source route applicability, retained state, observability and property conditions remain composition-specific; no inherited outcome for a changed candidate
Every candidate claim	199 exact current screens, eight separate dimensions and additive scenario records	Full chronological search and final statutory construction/conclusions remain separate; candidate quantity supplies no legal sufficiency
Drafting alternatives	Current /drafting_issues, including duplicate cardinality descriptions and optional clarity amendments	Counsel and inventor may select meaningful fallback claims or revisions; this packet applies none of those optional changes
Future descriptive adoption	Additive companions are explicitly later review material	A later application edition must expressly include relied-on new teaching and preserve an edition boundary; an external companion is not assumed filed content

Inventor, applicant and counsel checklist

ID	Review value or question	Required disposition
I01	Proposed sole inventor: Robert M. Layne, at his instruction	Confirm exact legal filing name and actual human conception for each retained claimed combination; attribution is preserved
I02	No AI inventor is named; assistance is a tool	Preserve human decisions/conception evidence and assess contributors under applicable standards; document authorship alone is insufficient
I03	Published source findings remain their authors' work	Confirm which contribution Robert claims as his own; do not reassign literature discoveries
A01	Residence, applicant/entitlement, correspondence and representative are UNPROVIDED	Complete privately if submission is later instructed; public review files do not need private addresses
A02	Government interest is UNCONFIRMED; micro-entity status is user-reported	Review actual obligations and any needed eligibility certification; no negative declaration or status certification is inferred
A03	Earlier filing/priority particulars are UNPROVIDED	Obtain actual application/receipt details if any; no preparation date creates priority
C01	Novelty, obviousness, written description, enablement, eligibility and definiteness are unresolved final inquiries	Conduct claim-specific review using qualified references, dates, constructions and exceptions; record contrary/critical-range evidence
C02	Unity/restriction and international effects remain open	Review actual common technical contribution and chosen jurisdictions; shared vocabulary does not establish unity
C03	Universal blocking remains aspirational	No percentage blocked, validity, exclusion, freedom-to-operate or license conclusion is assigned

The official guidance distinguishes written description from enablement, and compares the actual claimed scope with the disclosure. Prophetic examples may contribute but remain unperformed. USPTO written-description guidance, USPTO enablement guidance

Anticipation asks about the complete arrangement of limitations in a qualifying reference. Obviousness requires an articulated rationale and appropriate factual findings; containment in a numerical domain alone decides neither. These exhibits identify possible arguments and counterarguments for review. USPTO anticipation guidance, USPTO obviousness guidance

Robert's review designation remains subject to the ordinary human conception inquiry. The current USPTO guidance rescinds its February 2024 AI guidance; source researchers and tools are not reattributed as Robert's conception. USPTO revised inventorship guidance

Separate submission and publication workflow

Event	Current evidence	Action only after a separate instruction
Application preparation	Preserved dated source/PDF editions and authoring records	Select exact submission contents, including any explicitly adopted later disclosure
Public availability	Historical repository access observation; earliest availability uncertified	Preserve an actual public-version/access receipt; do not backdate later material
U.S. provisional filing	No submission/receipt/identifier asserted	Complete actual cover-sheet particulars, necessary drawings and applicable fee; preserve exact submitted bytes and official receipt
Later benefit claim	No entitlement asserted	Review support for the later claimed subject matter and procedural entitlement; optional broad claims do not reserve every species
PCT route	Historical strategy retained separately	Review a distinct PCT package, contents, unity and route; assess whether an express early-publication request fits the defensive objective
Legal effect	No final outcome asserted	Review reference qualification and effective dates against actual later claims and jurisdiction-specific rules

A provisional is a separate U.S. filing route. Its preparation does not perform filing or publish the document. USPTO provisional guidance, USPTO earlier-benefit guidance

Public accessibility requires evidence distinct from private creation/commit dates. PCT Article 21 provides ordinary publication after 18 months from priority and permits an earlier applicant request; a draft or filing alone does not establish immediate public disclosure. USPTO printed-publication guidance, PCT Article 21, PCT application contents

Verification and acceptance boundaries

This review task follows the actual repository's Python 3.12 standard-library verifier, Ruff 0.16.9 correctness configuration, explicit package inventory and deliberate integrity refresh. New comparison and arithmetic checks must pass before a final acceptance receipt is written. Original supplied PDFs, prior editions, claim wording, provenance and local private records are preserved. Verification is evidence of the checked source; no experiment or legal determination follows.

The separate private milestone/intake record accounts for assigned folders, historical excerpts and unavailable/truncated payloads. Those records contain host paths and are excluded from the research package. A complete source-intake claim requires actual accessible contents, preservation and disposition for every scoped input or Robert's explicit deferral. This exhibit does not waive missing intake or certify complete account-wide history.

Within the inspected assigned local scope, all 79 legacy source files were readable and matched current bytes or exact reachable historical Git blobs. The owner rollout's 21,708 valid saved records and 91 user-role messages were screened for relevant decisions and attachments; both referenced text attachments were present and hash-bound, with no known missing local referenced attachment. The raw dated human decisions close the later inventor/no-experiments instruction gap in the older indexed transcript. Complete cloud account history and an optional raw upstream PDG database are outside this bounded intake claim; selected archive data and its limits remain present.

The historical full-scope audit (source file: full_scope_completion_audit.md) retains its older broader outcome language. The later theoretical-disclosure strategy (source file: theoretical_disclosure_strategy_2026-10-01.md), current application preface and this packet govern the theoretical review endpoint: supported teaching, qualified calculation, conditional proposal, failure boundary or explicit unresolved assessment may address a drafting gap. Full physical realization, universal blocking and final legal outcomes are not represented as completed or as an experimental program.

Edition, citation and chronology index

Prepared **5 October 2026** for task 2026-10-04. This is a later local review index, not an application filing or a certification of earliest public availability.

The operative edition is E11, the 3 October substantive application. Every edition below retains its exact application, claim map, PDF and available authoring captures. Full SHA-256 values and byte sizes are in the structured index (source file: ../data/evidence_edition_index_2026-10-04.json). Stable locators are edition IDs, document paragraph numbers, claim numbers and JSON paths. Read the review decision packet (source file: review_readiness_2026-10-04.md) alongside the technical exhibits.

Preserved application sequence

Order below follows the documented adoption sequence and dated authoring records. Same-day edition order is not inferred from copied filesystem times. All eleven editions contain 199 candidate claims; new supporting descriptions are not backdated.

ID / preparation date	Editable application	Paragraphs	Capture / scope
E01 / 2026-09-29	provisional_application_draft.md (source file: provisional_application_draft.md)	80	Original preserved draft; no later authoring capture assumed
E02 / 2026-09-30	provisional_application_working_2026-09-30.md (source file: provisional_application_working_2026-09-30.md)	80	Authoring record (source file: ../data/working_application_2026-09-30.json)
E03 / 2026-09-30	provisional_application_property_evidence_2026-09-30.md (source file: provisional_application_property_evidence_2026-09-30.md)	80	Authoring record (source file: ../data/application_property_evidence_2026-09-30.json)
E04 / 2026-09-30	provisional_application_preparation_evidence_2026-09-30.md (source file: provisional_application_preparation_evidence_2026-09-30.md)	80	Authoring record (source file: ../data/application_preparation_evidence_2026-09-30.json)
E05 / 2026-09-30	provisional_application_electrical_evidence_2026-09-30.md (source file: provisional_application_electrical_evidence_2026-09-30.md)	80	Authoring record (source file: ../data/application_electrical_evidence_2026-09-30.json)

ID / preparation date	Editable application	Paragraphs	Capture / scope
E06 / 2026-09-30	provisional_application_oxide_defect_evidence_2026-09-30.md (source file: provisional_application_oxide_defect_evidence_2026-09-30.md)	80	Authoring record (source file: ../data/application_oxide_defect_evidence_2026-09-30.json)
E07 / 2026-10-01	provisional_application_phase_specific_evidence_2026-10-01.md (source file: provisional_application_phase_specific_evidence_2026-10-01.md)	80	Authoring record (source file: ../data/application_phase_specific_evidence_2026-10-01.json)
E08 / 2026-10-01	provisional_application_family_evidence_2026-10-01.md (source file: provisional_application_family_evidence_2026-10-01.md)	80	Authoring record (source file: ../data/application_family_evidence_2026-10-01.json)
E09 / 2026-10-01	provisional_application_metal_glass_evidence_2026-10-01.md (source file: provisional_application_metal_glass_evidence_2026-10-01.md)	80	Authoring record (source file: ../data/application_metal_glass_evidence_2026-10-01.json)
E10 / 2026-10-01	provisional_application_theoretical_review_2026-10-01.md (source file: provisional_application_theoretical_review_2026-10-01.md)	94	Authoring record (source file: ../data/application_theoretical_review_2026-10-01.json)
E11 / 2026-10-03	provisional_application_substantive_review_2026-10-03.md (source file: provisional_application_substantive_review_2026-10-03.md)	99	Authoring record (source file: ../data/application_substantive_review_2026-10-03.json)

E10's historical support-map label has a separate hash-bound identity correction (source file: ../data/edition_identity_corrections_2026-10-03.json). The original map and authoring inputs remain preserved; E11's own identity is correct. The current application and all previous PDFs are unchanged by these new exhibits.

Claim preservation is edition-relative. The working edition expressly records its prospective changes to [0047] and claims 169 and 191-193. The current substantive bank preserves its inherited later texts; the index does not imply that every original 29 September claim word remained unchanged throughout all eleven editions.

Finite attributed comparators

These are selected source contexts, not a complete prior-art search or current/future materials universe. A DOI, paper date, source measurement or table membership supplies no applicant experiment, discovery or automatic effective date. Follow each preserved source review for acquisition, original-byte, graphics, uncertainty and rights limits.

ID	Primary source / local locator	Recorded teaching boundary
ALLOY	Primary source; Comparison (source file: alloy_claim_comparison_2026-09-30.md)	CoCrFeNi/CoCrFeMnNi principal equiatomic feeds, prepared specimens and reported local impurities; complete support and exact equality remain conditional.
GLASS	Primary source; Comparison (source file: glass_claim_comparison_2026-10-01.md)	Selected metallic-glass source/route and six-principal target; present fine-box and below-floor models are current calculations, not source specimen data.
OXIDE	Primary source; Comparison (source file: oxide_claim_comparison_2026-10-01.md)	TiO2(B) nominal 1:2 formula, selected nanosheet route; retained surfaces/residuals, source-specific precursor issues and exact binary closure remain qualified.
POLYMER	Primary source; specification [0040] polymer evidence block	Su et al. PLLA/PDLA 1:1 feed blend and ideal repeat/conditional chain-end record. Feed mass and nominal repeats are not complete incorporated composition, sequence or topology; source Mn/PDI assignment remains dimensionally unresolved.
HYDROGEL	Primary source; Comparison (source file: hydrogel_claim_comparison_2026-09-30.md)	Source-local PIC/linker operations and response context; present count/topology/record examples remain calculated/proposed.
ELECTRICAL	Primary source; Comparison (source file: electrical_claim_comparison_2026-09-30.md)	SWCNT/PEDOT:PSS/PS stock origins, source preparation and percolation context; stock-component targets are not elemental fractions or performed source specimen compositions.
MOLECULAR	Primary source; specification [0040] and [0086]	Attributed carbamazepine selected connectivity/hydrate source; bulk isolation/mixture proportions and exact retained inventory not established here.
COMPOSITE	Primary source; specification [0040] and [0089]	Attributed nanotube/epoxy example; loading, constituent identity and whole retained elemental/volume inventory differ.
NUCLEAR	Primary source; specification [0046]-[0050], data/nuclear_archive	Dated nuclear identifiers/evaluation fields, not an arbitrary isotope material or current/future complete nuclear universe.
PARTICLE	Primary source; specification [0051]-[0056], data/particle_archive	Selected identity/property source locators include searches/limits; table inclusion is not discovery or full system-record implementation.

Evidence records and source-edition provenance

The structured index fingerprints each record below. An original source hash appearing inside a review is a recorded acquisition fact; availability of the original payload is a separate question. Reader-only, abstract-only, incomplete graphics, source conflicts and uncertain quantities retain their limitations. NUBASE2020 and PDG 2026.0 remain dated selected sources, distinct from recognized/hypothetical status and future completeness.

- data/electrical_source_review_2026-09-30.json (source file: ../data/electrical_source_review_2026-09-30.json)

- data/family_source_review_2026-10-01.json (source file: ../data/family_source_review_2026-10-01.json)

- data/hydrogel_preparation_source_review_2026-09-30.json (source file: ../data/hydrogel_preparation_source_review_2026-09-30.json)

- data/magneli_source_review_2026-10-01.json (source file: ../data/magneli_source_review_2026-10-01.json)

- data/metal_glass_source_review_2026-10-01.json (source file: ../data/metal_glass_source_review_2026-10-01.json)

- data/oxide_defect_source_review_2026-09-30.json (source file: ../data/oxide_defect_source_review_2026-09-30.json)

- documents/reference_entry_support.md (source file: ../documents/reference_entry_support.md)

- documents/composite_comparator_source_review_2026-09-30.md (source file: ../documents/composite_comparator_source_review_2026-09-30.md)

- data/nuclear_archive/source_metadata.json (source file: ../data/nuclear_archive/source_metadata.json)

- data/nuclear_archive/portable_bundle_manifest.json (source file: ../data/nuclear_archive/portable_bundle_manifest.json)

- data/particle_archive/pdg2026_identity_metadata.json (source file: ../data/particle_archive/pdg2026_identity_metadata.json)

- data/publication_observation.json (source file: ../data/publication_observation.json)

- data/edition_identity_corrections_2026-10-03.json (source file: ../data/edition_identity_corrections_2026-10-03.json)

- data/technical_review_2026-10-03.json (source file: ../data/technical_review_2026-10-03.json)

Chronology for later-claim comparisons

Event type	Preserved evidence	Interpretation
Original draft preparation	E01, 29 September 2026	Creation/preparation only
Repository public access observation	Recorded observation (source file: ../data/publication_observation.json), 30 September 2026 02:31:40 UTC, exact observed commit	Access observed then; earliest availability uncertified
Later adoption	E02-E11 dated sources and source/PDF hashes	Added teaching belongs to its actual edition; no retroactive earlier content
New review exhibits	Task initiated 4 October; exhibits prepared 5 October	Local review material; no new external publication observation
Filing and priority	No submitted application, receipt or official identifier supplied	No filing/effective date or benefit entitlement assigned
Actual later claim	Four representative scenario types only	No actual later application/claim or final statutory result assumed

For each proposed comparison, record the exact earlier bytes/locators, actual later claim and complete arrangement, source accessibility/qualification, effective date, applicable exceptions and enabling teaching. Keep section 102, section 103, written description, enablement, eligibility and inventorship separate.

Fresh official-source checks

The issues below were inspected using official readers during this task. The other legal records in the unchanged 3 October review retain their own source-check date; this table does not assert that all legal developments or all jurisdictions were searched.

Source ID	Issue inspected	Official page
L01	complete-limitation anticipation	Official source
L04	articulated obviousness rationale	Official source
L06	written description separate from enablement	Official source
L07	claimed-scope enablement and prophetic examples	Official source
L13	PCT common special technical feature	Official source
L15	ordinary human conception standard; 2024 guidance rescinded	Official source
L16	earlier-benefit support and procedure	Official source
L17	public accessibility distinct from creation	Official source
L20	September 29 eligibility declaration addendum	Official source
PROVISIONAL	U.S. provisional route	Official source
PCT3	accepted international application contents	Official source
PCT21	ordinary and requested earlier international publication	Official source

Scoped intake and unresolved evidence

The private intake accounts for all 79 assigned legacy files, the evidence/guidance inputs, the exact saved owner rollout and both goal attachments referenced there. Known local scoped inputs are readable, hash-bound and preserved; the later no-experiments and inventor/review-endpoint decisions are verified from dated raw user records. Host paths, chat material and administrative facts remain outside the research package. This bounded closure does not certify complete cloud account history, earliest public accessibility, physical outcomes, inventorship or filing. No source deletion, working-copy retirement or relocation is authorized by this index.

The acceptance receipt must distinguish the local technical document checks actually passed from missing sources, factual inventor/applicant choices, counsel review, publication and filing. An indexed excerpt or historical assistant report does not establish those outcomes.

Material, state and process coverage: later review exhibit

Goal batch **4 October 2026**; prepared **5 October 2026**. Reviewed baseline e0233fe2863b2fd9d43be634b6dc6022b5cc0074.

This source-bound exhibit accounts for selected material families, their composition/structure/state/process/property fields, all **199** candidate claim numbers, and four later worked-record workflows. Its machine-readable matrix (source file: ../data/material_state_process_coverage_2026-10-04.json) preserves exact arithmetic and explicit unknowns. It is a bounded review catalog, not complete physical coverage of every indexed combination.

This exhibit was prepared after the 3 October application. It is separate review content, not retroactively adopted into its 99 paragraphs, 199 claims or 47-page PDF, and not an earlier disclosure, filing or public-availability date.

The operative 3 October specification (source file: provisional_application_substantive_review_2026-10-03.md) retains its **99 paragraphs**, **199 unchanged candidate claims**, and **47-page PDF**. Robert M. Layne remains the user-directed proposed sole inventor for review. Attribution, human conception, filing particulars and legal effect remain separate. No applicant physical experiments are performed or planned.

ESTABLISHED means an attributed source teaching or faithfully retained source identity value. CALCULATED means arithmetic under stated assumptions. PROPOSED means an unperformed conditional record or inquiry. UNSUPPORTED marks physical fields without adequate evidence. A row may contain all four categories; its applicant physical outcome stays UNSUPPORTED.

Coverage matrix

Row and family	Application locators	Existing evidence	Remaining physical support
DOM - Composition-domain and quantity-basis framework	[0007], [0008], [0010], [0011], [0016], [0017], [0018], [0029], [0030], [0093], [0098]	Later ideal/conditional record; no new empirical source	Exact targets do not establish availability, stable phases, making conditions, measured absence, useful outcome or practical resolution.
ALLOY - Metals and crystalline alloys	[0039], [0067], [0082], [0083], [0084], [0095], [0099]	source 1 (source file: ../data/metal_glass_source_review_2026-10-01.json); source 2 (source file: ../data/metal_glass_source_review_2026-10-01.json)	Feed purity, full retained support, impurity quantities, segregation, volatility, phase coexistence and candidate transport remain unresolved.
GLASS - Metallic glass and amorphous structure	[0063], [0079], [0084], [0099]	source 1 (source file: ../data/metal_glass_source_review_2026-10-01.json); source 2 (source file: ../data/glass_candidate_record_review_2026-10-01.json)	Whole inventory, retained flux, candidate geometry, kinetic weighting and actual candidate phase/property measurements remain unresolved.
OXIDE-B - Inorganic nominal TiO2(B) nanosheets	[0044], [0045], [0034], [0042]	source 1 (source file: ../documents/provisional_application_substantive_review_2026-10-03.md)	Formula ratios do not establish exhaustive product purity, every Ti/O ratio, all polymorphs or candidate reproducibility.
OXIDE-RED - Reduced oxide, defects and phase-specific host	[0064], [0069], [0070], [0071], [0072], [0085]	source 1 (source file: ../data/oxide_defect_source_review_2026-09-30.json); source 2 (source file: ../data/oxide_candidate_record_review_2026-10-01.json); source 3 (source file: ../data/magneli_source_review_2026-10-01.json)	Reaction feasibility, precursor conflicts, retained molecular material, exact defect identity and candidate property response remain unresolved.
MOLECULAR - Molecular crystals and hydrate mixtures	[0040], [0086], [0095], [0097]	source 1 (source file: ../data/family_source_review_2026-10-01.json); source 2 (source file: ../data/family_source_review_2026-10-01.json)	Actual isolated stocks, whole deposit purity, prescribed phase proportion, native-source graphics and candidate stability remain unresolved.
POLYMER - PLA, repeat units and finite chains	[0040], [0036], [0087], [0096], [0099]	source 1 (source file: ../data/family_source_review_2026-10-01.json); source 2 (source file: ../data/family_source_review_2026-10-01.json)	Actual ends, incorporated composition, molecular-weight distribution, sequence, full specimen assay and candidate response remain unresolved.
HYDROGEL - PIC network and linker topology	[0065], [0068], [0088], [0099]	source 1 (source file: ../data/hydrogel_preparation_source_review_2026-09-30.json); source 2 (source file: ../data/hydrogel_candidate_record_review_2026-09-30.json)	Candidate accessible sites, topology, retained solvent, complete gel composition and candidate Gprime are unsupported.

Row and family	Application locators	Existing evidence	Remaining physical support
CNT-COMPOSITE - Nanotube/epoxy composite and release boundary	[0040], [0036], [0089], [0096]	source 1 (source file: ../data/family_source_review_2026-10-01.json); source 2 (source file: ../data/family_source_review_2026-10-01.json)	Full specimen composition, actual cured loading, density allocation, quantitative interfaces and candidate response remain unresolved.
ELECTRICAL - SWCNT/PEDOT:PSS/PS and electrical loading	[0066], [0089], [0093], [0096]	source 1 (source file: ../data/electrical_source_review_2026-09-30.json); source 2 (source file: ../data/electrical_candidate_record_review_2026-09-30.json)	Complete retained chemistry, independently selected surfactant control, actual loading, morphology and candidate conductivity remain unresolved.
UNARY - Single-element form: later worked record W-U	[0013], [0016], [0033], [0034], [0035], [0042]	Later ideal/conditional record; no new empirical source	No actual specimen, preparation, structure, sample-independent purity, isotope distribution or property is supported.
SPATIAL - Spatial/layered inventory: later worked record W-L	[0013], [0036], [0037], [0083], [0095]	Later ideal/conditional record; no new empirical source	Stable pure-element layers, real density/thickness, interfacial chemistry, process route and response remain unsupported.
NUCLEAR - Isotope/state and reference-time accounting	[0046], [0047], [0048], [0049], [0050], [0090], [0095], [0097]	source 1 (source file: ../data/nuclear_archive/source_metadata.json); source 2 (source file: ../data/nuclear_archive/archive_counts.json); source 3 (source file: ../data/technical_review_2026-10-03.json)	All-current/future-state completeness, existence assessment for every row, stock availability, isolation and persistence remain unresolved.
PARTICLE - Particle/host construction: later worked record W-P	[0051], [0052], [0053], [0054], [0055], [0056], [0091]	source 1 (source file: ../data/particle_archive/pdg2026_identity_metadata.json); source 2 (source file: ../data/particle_archive/pdg2026_identity_archive.json)	The selected PDG identity tables omit masses/lifetimes/measurements, decay/reference tables and discovery assessments; all quantum/physical combinations remain unestablished.
RECORD - Bounded record workflow: later worked record W-R	[0007], [0030], [0031], [0032], [0098]	Later ideal/conditional record; no new empirical source	Finite software execution does not complete every physical record or all rational/real targets; rejecting one support does not reject a lower-support boundary.

Field assessments

Each statement below retains its local evidence label. ESTABLISHED source structure/process fields describe only the attributed teaching; candidate applicability and outcomes are unresolved.

DOM - Composition-domain and quantity-basis framework

Composition - CALCULATED. Strictly positive rational targets have no imposed floor or domain-wide denominator cap. The original module separately retains a 10^-19 floor; D=10^19 is an optional grid. Real irrational coordinates remain symbolic. Normalization correlates coordinate intervals.

Structure - PROPOSED. An overall vector requires independently identified structure; there is no default phase.

State - PROPOSED. Inventory region, time, species and mass allocation must be supplied for a physical record.

Process - PROPOSED. Numerical indexing, conversion and witness assignment are calculation operations, without a universal preparation route.

Property - PROPOSED. No property follows from an address. Bounds and covariance concern stipulated input knowledge, not a measured specimen.

Remaining physical support - UNSUPPORTED. Exact targets do not establish availability, stable phases, making conditions, measured absence, useful outcome or practical resolution.

ALLOY - Metals and crystalline alloys

Composition - CALCULATED. Ideal Cr/Fe/Co/Ni and Cr/Mn/Fe/Co/Ni principal atomic targets are quarters and fifths. Additional carbon changes support and denominator; principal targets are not exhaustive whole-specimen assays.

Structure - ESTABLISHED. Attributed selected source reports FCC alloys and grain-boundary context; no candidate phase is established.

State - PROPOSED. Source-local anneal and diffusion conditions apply only to the attributed specimens. A candidate needs matched temperature, history, impurity and observation fields.

Process - ESTABLISHED. Selected source-local co-melting/remelting and homogenization are attributed. Supplementary Table S1 is an uninspected dependency in the source review.

Property - PROPOSED. Boundary transport comparison remains a conditional source-informed inquiry with separately specified thermal and microstructural state.

Remaining physical support - UNSUPPORTED. Feed purity, full retained support, impurity quantities, segregation, volatility, phase coexistence and candidate transport remain unresolved.

GLASS - Metallic glass and amorphous structure

Composition - CALCULATED. Principal B/Si/P/Cr/Ni/Nb count target is (32,5,160,87,686,30)/1000. Separate below-grid Al and closed-box records retain exact fractions and their own inventories.

Structure - ESTABLISHED. Attributed source-local glass diagnostics and preparation geometry remain distinct from complete retained inventory and candidate local order.

State - PROPOSED. Thermal trajectory, initial nuclei, observation conditions and temperature-dependent nucleation/growth require support. Residence W alone does not order crystallization outcomes.

Process - ESTABLISHED. Source-local melt/flux/quench operations and their inspection limits are attributed; unsupported route conditions are not supplied.

Property - PROPOSED. Lower crystalline signal is a conditional inquiry, with different paths or nuclei as alternatives; no universal critical cooling rule is asserted.

Remaining physical support - UNSUPPORTED. Whole inventory, retained flux, candidate geometry, kinetic weighting and actual candidate phase/property measurements remain unresolved.

OXIDE-B - Inorganic nominal TiO2(B) nanosheets

Composition - CALCULATED. Nominal TiO2 formula gives canonical (O,Ti) atomic fractions (2/3,1/3); denominator three is absent from D=10^19.

Structure - ESTABLISHED. The attributed Xiang et al. example is phase B and nanosheet morphology. Formula, phase and exposed surface are separate features.

State - PROPOSED. Reported source conditions are local to that teaching; actual candidate retained surfaces, dimensions and complete inventory are unknown.

Process - ESTABLISHED. Application [0044] attributes a solvothermal route and isolation/washing to the cited paper; the complete primary method remains a dependency.

Property - PROPOSED. No new candidate property result is assigned from the nominal formula.

Remaining physical support - UNSUPPORTED. Formula ratios do not establish exhaustive product purity, every Ti/O ratio, all polymorphs or candidate reproducibility.

OXIDE-RED - Reduced oxide, defects and phase-specific host

Composition - CALCULATED. TiO_(2-delta) host count and the restricted Ti(III)/Ti(IV), O(2-) model are calculated under declared assumptions. Water, surface organics and other retained material need independent denominator categories.

Structure - ESTABLISHED. Attributed phase/site diagnostics and a nominal host are kept separate. Oxygen loss alone establishes neither rutile vacancy locations nor a Magneli phase.

State - PROPOSED. The observation boundary must include surface material, hydrogen, hydroxyls and time dependence where retained; source inspection remains bounded.

Process - ESTABLISHED. Attributed precursor/diagnostic context is recorded. [0085] gives a stipulated atom balance, with thermodynamic feasibility and isolation unresolved.

Property - PROPOSED. Source-local optical estimates and visible H2 rates remain attributed in the source record. A candidate optical inquiry and alternative remain unperformed.

Remaining physical support - UNSUPPORTED. Reaction feasibility, precursor conflicts, retained molecular material, exact defect identity and candidate property response remain unresolved.

MOLECULAR - Molecular crystals and hydrate mixtures

Composition - CALCULATED. Unsolvated carbamazepine and dihydrate formula-unit counts have totals 30 and 36; molecular-unit fraction and atom-weight fraction differ. Whole-unit populations impose stronger constraints than elemental denominator alone.

Structure - ESTABLISHED. Connectivity, crystal-form labels and hydration are independently specified; arbitrary bulk ratios do not establish a single hydrate.

State - PROPOSED. Water/solvent exchange, warming, substrate retention and phase conversion can invalidate the selected two-phase boundary.

Process - ESTABLISHED. Selected source solution/hold/quench account applies to sampled crystallites; it is not a demonstrated prescribed bulk phase-separation or proportion route.

Property - PROPOSED. Sampled-crystallite occurrence comparisons require matched sampling and identification. No applicant occurrence or useful-property result is assigned.

Remaining physical support - UNSUPPORTED. Actual isolated stocks, whole deposit purity, prescribed phase proportion, native-source graphics and candidate stability remain unresolved.

POLYMER - PLA, repeat units and finite chains

Composition - CALCULATED. Ideal PLA repeat uses canonical (H,C,O)=(4/9,1/3,2/9). Conditional hydroxyl/carboxyl chains add ends; feed-component mass and retained elemental fractions differ.

Structure - ESTABLISHED. Attributed polymer stereocomplex context is separate from actual sequence, tacticity, chain ends, distributions, crystal fraction and network.

State - PROPOSED. Batch, forming/thermal history, conditioning, residuals and voids must be matched for a candidate comparison.

Process - ESTABLISHED. Source-associated forming/thermal route is attributed with XML/graphics inspection limits; candidate applicability remains conditional.

Property - PROPOSED. Mechanical comparison and a separate chain-shortening/packing alternative are unperformed for a candidate; toy average masses are anonymous arithmetic.

Remaining physical support - UNSUPPORTED. Actual ends, incorporated composition, molecular-weight distribution, sequence, full specimen assay and candidate response remain unresolved.

HYDROGEL - PIC network and linker topology

Composition - CALCULATED. Ideal monomer-only feed counts remain distinct from final solvent-containing gel and incorporated composition. Nonnegative integer linker populations obey declared end-count bounds.

Structure - ESTABLISHED. Source context concerns PIC/linker gels. Bridge, loop, pendant and unreacted counts do not by themselves establish effective load-bearing connectivity.

State - PROPOSED. Solvent fraction, batch, crosslinking/stabilization temperatures, rheometry conditions and network state need explicit matching.

Process - ESTABLISHED. Published preparation operations are attributed; unresolved identities and source conflicts remain in the preparation record.

Property - PROPOSED. The ten source storage-modulus rows remain attributed measurements. Increased effective bridging versus inaccessible sites/pendant attachment are separate candidate hypotheses.

Remaining physical support - UNSUPPORTED. Candidate accessible sites, topology, retained solvent, complete gel composition and candidate Gprime are unsupported.

CNT-COMPOSITE - Nanotube/epoxy composite and release boundary

Composition - CALCULATED. Conditional dry input categories use (9/1250,891/1250,7/25); stock-origin categories do not automatically define realized phases or additive volumes.

Structure - ESTABLISHED. Attributed constituents/film context is separate from candidate dispersion, interfaces, dimensions, orientation and post-cure identity.

State - PROPOSED. Exposure dose, humidity, temperature, specimen geometry, history, recovery and collection boundary require specification.

Process - ESTABLISHED. Selected cure/stock-assembly operations are attributed. Complete supplement and final-publisher comparison remain unperformed.

Property - PROPOSED. Release per initial CNT and per whole specimen are different denominators; recovered material is not automatically total release. Mechanism and alternative remain conditional.

Remaining physical support - UNSUPPORTED. Full specimen composition, actual cured loading, density allocation, quantitative interfaces and candidate response remain unresolved.

ELECTRICAL - SWCNT/PEDOT:PSS/PS and electrical loading

Composition - CALCULATED. Normalized dry-input stock-origin targets retain carrier/residue qualifications. The numerical wt-percent fit coordinate is not an atomic fraction.

Structure - ESTABLISHED. Attributed network/film context does not identify candidate dispersion, retained categories or connected paths.

State - PROPOSED. Processing, observation state and threshold coordinate are source bounded; stock density is not a post-cure phase density by default.

Process - ESTABLISHED. Attributed source operations remain distinct from unsupported independent-surfactant shortcut and candidate formulation/retention.

Property - PROPOSED. Source threshold is 0.443 numerical wt-percent. Fitted exponent/prefactor and coupling-length model are not additional measured specimen properties.

Remaining physical support - UNSUPPORTED. Complete retained chemistry, independently selected surfactant control, actual loading, morphology and candidate conductivity remain unresolved.

UNARY - Single-element form: later worked record W-U

Composition - CALCULATED. Seven stipulated carbon atoms give support [6], vector [1] and primitive denominator one. This is a count target, not a purity assay.

Structure - PROPOSED. Structure tags U_A and U_B are unresolved alternatives; no allotrope, lattice or amorphous identity is assigned.

State - PROPOSED. Region and observation labels are record placeholders; no physical size, temperature, pressure or measured state is supplied.

Process - PROPOSED. A candidate would need an applicable source-supported route; no apparatus or recipe is invented.

Property - PROPOSED. Equal unary fraction does not fix any response. A conditional comparison requires identified structures, controls and observable.

Remaining physical support - UNSUPPORTED. No actual specimen, preparation, structure, sample-independent purity, isotope distribution or property is supported.

SPATIAL - Spatial/layered inventory: later worked record W-L

Composition - CALCULATED. Two stipulated disjoint regions contain six carbon atoms and four oxygen atoms. Global canonical (C,O)=(3/5,2/5); arithmetic averaging of local profiles incorrectly gives (1/2,1/2).

Structure - PROPOSED. Two opposite region orders preserve bulk vector while changing an ordering field. Counts do not establish two realizable material layers or phase identities.

State - PROPOSED. Anonymous coordinate cells have equal model volume and number-density ratio 3:2; units, physical thickness, phase state and observation time remain unsupported.

Process - PROPOSED. Assembly is a conditional record request. No adhesion, interface, deposition, containment or persistence result is supplied.

Property - PROPOSED. Any through-thickness observable needs an applicable state and transport model; no sign or numerical response is supplied.

Remaining physical support - UNSUPPORTED. Stable pure-element layers, real density/thickness, interfacial chemistry, process route and response remain unsupported.

NUCLEAR - Isotope/state and reference-time accounting

Composition - CALCULATED. Classical conditional profiles normalize within each element; every joint count must be integral. Source-qualified mass conversion is unavailable for unknown populated masses.

Structure - ESTABLISHED. The NUBASE2020 snapshot preserves 5,843 source state rows, 3,558 (A,Z) pairs, original state indices/suffixes and property markers; it is not every nuclear level.

State - PROPOSED. Source edition, reference time, lifetime, decay/daughters and ensemble versus exact count meanings remain separate.

Process - PROPOSED. No arbitrary state production, separation or coherent-state preparation route is supplied by an archive identity or count witness.

Property - PROPOSED. No candidate isotope/state property result is assigned; measured/estimated source tokens retain provenance and uncertainty.

Remaining physical support - UNSUPPORTED. All-current/future-state completeness, existence assessment for every row, stock availability, isolation and persistence remain unresolved.

PARTICLE - Particle/host construction: later worked record W-P

Composition - CALCULATED. Host atoms, excitation counts and excitation-energy shares use separate denominators; containing atoms/composites are not counted again as independent constituents.

Structure - ESTABLISHED. Source identity rows and aliases are archived observations about the table. They certify neither every listed entity nor an actual populated host.

State - PROPOSED. Occupation, frequency/energy, driver/host or confinement, interactions, lifetime and observation need independent support. Anonymous energy arithmetic is not a measured spectrum.

Process - PROPOSED. No physical production, excitation, containment or observation is performed; a model-qualified hypothetical row retains experimental-search status unknown.

Property - PROPOSED. The energy total and shares in W-P are stipulated noninteracting bookkeeping, with host response and lifetime unsupported.

Remaining physical support - UNSUPPORTED. The selected PDG identity tables omit masses/lifetimes/measurements, decay/reference tables and discovery assessments; all quantum/physical combinations remain unestablished.

RECORD - Bounded record workflow: later worked record W-R

Composition - CALCULATED. A finite rational prefix, a thirds box and a rejected closure-only binary request have separate outcomes. The full unrestricted domain is not exhausted by a finite run.

Structure - PROPOSED. The record requires explicit unresolved structure rather than inferring a phase from a witness.

State - PROPOSED. No specimen boundary, time, assay, thermal state or population is defaulted from numerical inputs.

Process - PROPOSED. The software procedure assigns fields on successful positive feasibility and records failure without completing a positive material record. No physical preparation follows.

Property - PROPOSED. A property field records a conditional inquiry and alternative without an invented response, model or measurement.

Remaining physical support - UNSUPPORTED. Finite software execution does not complete every physical record or all rational/real targets; rejecting one support does not reject a lower-support boundary.

Claim and property accounting

The JSON alignment contains one row for every claim from 1 through 199, with an exact-text hash, direct/transitive parents, current support locators, own/inherited coverage rows and a qualification. These are relationships to descriptive fields. Every complete_physical_feature_match_verified value remains false. The existing current claim screens (source file: ../data/claim_review_2026-10-03.json) retain the individual assumptions and eight separate assessments.

Claims 2-138 demand exact cardinality k equal to their claim number. The alignment does not infer a closed exact-k specimen from a principal target. Above k=118, at least k−118 hypothetical labels are required; these records remain physically unsupported. The unary workflow concerns independent claim 139, and particle workflows 194-196 keep their separate construction level and dependencies.

Property category	Current bounded evidence or gap
mechanical	Selected attributed mechanical/rheological records; no universal mechanism or candidate response. Application [0058], [0065], [0087], [0088]. Candidate outcome: UNSUPPORTED.
electrical	Selected attributed electrical loading/transport record; fitted/model values stay distinct. Application [0059], [0066]. Candidate outcome: UNSUPPORTED.
thermal	Source-bounded thermal preparation and crystallization context; no selected complete thermal-conductivity/expansion/heat-capacity law. Application [0060], [0063], [0084], [0099]. Candidate outcome: UNSUPPORTED.
optical	Selected attributed optical estimates plus conditional candidate inquiry; no optical law across the domain. Application [0060], [0064], [0072]. Candidate outcome: UNSUPPORTED.
magnetic	Named observable category only; no selected worked magnetic source/process/property record in this exhibit. Application [0060]. Candidate outcome: UNSUPPORTED.
chemical stability	Corrosion/oxidation/stability category remains unspecified for a new candidate. Application [0060]. Candidate outcome: UNSUPPORTED.
catalytic	Source-local visible H2 rates are retained in the oxide source review; no generic selectivity/rate prediction. Application [0060], [0064]. Candidate outcome: UNSUPPORTED.
diffusion sorption permeability wetting	Selected grain-boundary diffusion teaching only; sorption/permeability/wetting need separate records. Application [0060], [0039]. Candidate outcome: UNSUPPORTED.

Property category	Current bounded evidence or gap
radiation	Named response category only; nuclear/particle identities do not supply a material radiation-response model. Application [0060]. Candidate outcome: UNSUPPORTED.
biological	Named interaction category only; no selected biological assay, use or response model. Application [0060]. Candidate outcome: UNSUPPORTED.
release recovery	Selected composite recovery/release inquiry with distinct denominators and collection limitations. Application [0040]. Candidate outcome: UNSUPPORTED.

Later worked records

These four records add explicit workflows where the application supplies definitions or isolated examples. They are later exhibit content. They add no physical outcome, source experiment, empirical law or claim amendment.

W-U - unary count record

The application defines unary targets and structural requirements but has no complete concrete unary count/field workflow.

CALCULATED count record. Support [6] identifies recognized carbon using the element register. Seven stipulated atoms give counts [7], target [1], and primitive denominator 1. Every other elemental coordinate is zero in this ideal target. No actual specimen or impurity assay is supplied.

Structure - PROPOSED. Unresolved structure tags U_A and U_B; neither tag establishes a physical allotrope or topology.

State - UNSUPPORTED. No real region dimensions, temperature, pressure, isotope profile or observation time supplied.

Preparation - PROPOSED. Require a separately supported applicable route and retained-boundary evidence before physical use; no recipe or performed result supplied.

Measurement - UNSUPPORTED. Seven stipulated atoms are not a counted specimen or exhaustive purity assay.

Property - PROPOSED. Use the ideal unary U_A record as the conditional baseline and change only the independently identified structure to U_B. Nominate electrical conductance G across matched contacts at independently supplied temperature/pressure and geometry. A different response needs a justified electronic/defect mechanism; unchanged within validated resolution is a separate alternative if the relevant paths remain equivalent. No applicable constitutive model, sign, magnitude or measured outcome is supplied.

Failure boundaries. Additional positive elemental counts change the unary support. Changing structure can retain [1] while changing the material identity.

W-L - spatial inventory record

The application supplies spatial fields and weighting rules but has no count-resolved example comparing layer order with whole-inventory composition.

CALCULATED spatial record. Region A has anonymous volume 2, atom number density 3, counts (C,O)=(6,0), and local profile (1,0). Region B has anonymous volume 2, density 2, counts (0,4), and profile (0,1). The disjoint union has ten atoms and global fractions (3/5,2/5), with primitive denominator 5. An unweighted local-profile mean gives (1/2,1/2) and fails for these unequal atom totals. Orders A/B and B/A preserve the global vector while

changing the recorded order. These are anonymous coordinate cells and density units, with no physical layer or density measurement asserted.

Structure - PROPOSED. Ordered disjoint count regions A/B and B/A. Neither arrangement certifies phase identity, realizable pure layers or interface.

State - UNSUPPORTED. Physical thickness, coordinate scale, temperature, pressure, time and phase state unresolved.

Preparation - PROPOSED. Conditional assembly request; applicable source route, interface, separation and persistence must be supplied before physical assertions.

Measurement - UNSUPPORTED. No local assay or spatial resolution supplied. A measured spatial average requires an applicable density and boundary model.

Property - PROPOSED. Use order A/B as the conditional baseline and B/A as the controlled ordering change, retaining the declared counts. Nominate through-region thermal conductance K under independently supplied boundaries, contacts and states. An ordering effect requires a justified interfacial/transport model; equivalent effective paths support a separate unchanged-within-resolution alternative. No such model or actual response is supplied.

Failure boundaries. Overlapping regions would double count atoms. Retained interfacial material must enter the full inventory. Unequal atom totals invalidate an unweighted local-profile mean.

W-R - bounded record workflow

Definitions and isolated examples exist; this later exhibit assembles validation, a bounded prefix, distinct domain outcomes and unresolved physical fields into one executable record audit trail.

CALCULATED execution record. On support [1,8], a run capped at denominator 4 exports the five distinct primitive targets (1/2,1/2), (1/3,2/3), (2/3,1/3), (1/4,3/4), and (3/4,1/4). This cap bounds the run. The singleton box ([1/3,1/3],[2/3,2/3]) accepts positive target (1/3,2/3) and count inventory (1,2) at N=3, with explicit unsupported physical fields; it has no D=10^19 grid point. The box ([0,0],[1,1]) has nonnegative witness (0,1) but rejects positive target assignment on [1,8]. Its distinct lower-support record [8] has target [1].

Structure - UNSUPPORTED. The accepted numerical record has no selected physical phase.

State - UNSUPPORTED. No specimen, observation state or physically chosen atom inventory is claimed.

Preparation - UNSUPPORTED. No supported source route assigned. Numerical acceptance completes record fields with visible unknowns, not a material preparation.

Measurement - UNSUPPORTED. No assay, uncertainty, precision or detection performance supplied.

Property - PROPOSED. Require a source/model-defined observable, baseline, controlled intervention and conditional alternative; no numerical prediction supplied.

Failure boundaries. The rejected query never completes positive-target assignment on [1,8]. Its lower-support boundary is a different support record. A finite export budget does not cap the defined rational domain.

W-P - particle host record

The application supplies particle fields and an anonymous energy model but no edition-qualified identity join combined with a complete host/population/energy record.

CALCULATED host/population record. Seven stipulated carbon host atoms retain unary elemental vector [1]. Separate photon modes A/B have counts (3,1) and anonymous energies (2,5) in one common unit. Total

excitation count is 4; count shares are (3/4,1/4). Under the stated noninteracting model, excitation energy is 3*2+1*5=11 anonymous units and energy shares are (6/11,5/11). These are separate denominators; neither set of shares is the host elemental vector. No physical frequency, spectrum or observed population is supplied.

ESTABLISHED attribution only. PDG 2026.0 pdgparticle primary key 1, S000, name gamma, cc_type=S, mcid=22; the archived source (source file: ../data/particle_archive/pdg2026_identity_archive.json) row has no discovery assessment or mass/lifetime measurement in this identity selection. The archived quantum-number strings are quantum_i="0,1", quantum_j="1", quantum_p="-", quantum_c="-", with quantum_g=null; they remain source metadata, without a new measurement interpretation. These values do not certify an actual host population.

Structure - UNSUPPORTED. Actual host phase, spatial mode shape, confinement and interaction model unresolved.

State - UNSUPPORTED. Frequency/momentum, occupation restrictions, nonequilibrium driver, temperature, lifetime/linewidth and observation interval unresolved.

Preparation - PROPOSED. A particle record requests independent support for host preparation, excitation/containment and observation; no performed construction asserted.

Measurement - UNSUPPORTED. No detected photon population, spectrum, linewidth or calibrated host observable supplied.

Property - CALCULATED. The stipulated noninteracting energy total is 11 anonymous units. Count shares differ from energy shares; neither is an elemental fraction or measured host response.

Failure boundaries. Adding excitations to the host elemental support would mix construction levels. Source PART/category/MCID does not certify a discovery. Interactions, unknown occupied energies or different occupation models can invalidate this additive energy accounting.

Validation and remaining decisions

The dependency-free Python checker coverage_checks.py (source file: ../tools/coverage_checks.py) checks source hashes and JSON pointers, application locators, all 199 exact claim/dependency links, evidence boundaries and the four worked calculations. Independent separator enumeration checks the bounded rational prefix; exact count/number-density sums check spatial averaging; the archived source row is joined by its edition and primary key. Its deliberate-defect checks operate in memory and write no package files. Package verification and final manifest review are recorded separately by the integrating change.

This exhibit documents selected coverage and missing fields. Physical preparation, characterization and property experiments remain unperformed and not planned. Magnetic, biological and several other property categories have no selected worked teaching here; their unknowns remain explicit and no study schedule is created. Claim-specific physical/legal support, conception, chronology, applicant particulars and counsel decisions remain unresolved review matters. This work neither files nor publishes the application and gives no universal exclusion or permission conclusion.

All-199 scenario navigation

Each row links the exact candidate to four explicit later-claim comparison records. Feature text below is a navigation excerpt; the appended application contains all complete candidate texts. Full disclosed/missing features, contexts, qualified arguments and counterarguments remain in the separately supplied editable scenario exhibit and JSON. No final statutory outcome or actual later claim is asserted.

Claim / family	Added-feature excerpt	Four scenario IDs	Textual locators
1 / G01	A material comprising a closed elemental inventory of k distinct elemental constituents selected from atomic numbers 1...	C001-S1, C001-S2, C001-S3, C001-S4	[0010], [0011], [0014], [0015], [0016], [0080]
2 / G02	wherein k is exactly 2.	C002-S1, C002-S2, C002-S3, C002-S4	[0010], [0011], [0014], [0015], [0016], [0022], [0073], [0080]
3 / G02	wherein k is exactly 3.	C003-S1, C003-S2, C003-S3, C003-S4	[0010], [0011], [0014], [0015], [0016], [0022], [0073], [0080]
4 / G02	wherein k is exactly 4.	C004-S1, C004-S2, C004-S3, C004-S4	[0010], [0011], [0014], [0015], [0016], [0022], [0073], [0080]
5 / G02	wherein k is exactly 5.	C005-S1, C005-S2, C005-S3, C005-S4	[0010], [0011], [0014], [0015], [0016], [0022], [0073], [0080]
6 / G02	wherein k is exactly 6.	C006-S1, C006-S2, C006-S3, C006-S4	[0010], [0011], [0014], [0015], [0016], [0022], [0073], [0080]
7 / G02	wherein k is exactly 7.	C007-S1, C007-S2, C007-S3, C007-S4	[0010], [0011], [0014], [0015], [0016], [0022], [0073], [0080]
8 / G02	wherein k is exactly 8.	C008-S1, C008-S2, C008-S3, C008-S4	[0010], [0011], [0014], [0015], [0016], [0022], [0073], [0080]
9 / G02	wherein k is exactly 9.	C009-S1, C009-S2, C009-S3, C009-S4	[0010], [0011], [0014], [0015], [0016], [0022], [0073], [0080]
10 / G02	wherein k is exactly 10.	C010-S1, C010-S2, C010-S3, C010-S4	[0010], [0011], [0014], [0015], [0016], [0022], [0073], [0080]
11 / G02	wherein k is exactly 11.	C011-S1, C011-S2, C011-S3, C011-S4	[0010], [0011], [0014], [0015], [0016], [0022], [0073], [0080]
12 / G02	wherein k is exactly 12.	C012-S1, C012-S2, C012-S3, C012-S4	[0010], [0011], [0014], [0015], [0016], [0022], [0073], [0080]
13 / G02	wherein k is exactly 13.	C013-S1, C013-S2, C013-S3, C013-S4	[0010], [0011], [0014], [0015], [0016], [0022], [0073], [0080]
14 / G02	wherein k is exactly 14.	C014-S1, C014-S2, C014-S3, C014-S4	[0010], [0011], [0014], [0015], [0016], [0022], [0073], [0080]
15 / G02	wherein k is exactly 15.	C015-S1, C015-S2, C015-S3, C015-S4	[0010], [0011], [0014], [0015], [0016], [0022], [0073], [0080]
16 / G02	wherein k is exactly 16.	C016-S1, C016-S2, C016-S3, C016-S4	[0010], [0011], [0014], [0015], [0016], [0022], [0073], [0080]
17 / G02	wherein k is exactly 17.	C017-S1, C017-S2, C017-S3, C017-S4	[0010], [0011], [0014], [0015], [0016], [0022], [0073], [0080]
18 / G02	wherein k is exactly 18.	C018-S1, C018-S2, C018-S3, C018-S4	[0010], [0011], [0014], [0015], [0016], [0022], [0073], [0080]

Claim / family	Added-feature excerpt	Four scenario IDs	Textual locators
19 / G02	wherein k is exactly 19.	C019-S1, C019-S2, C019-S3, C019-S4	[0010], [0011], [0014], [0015], [0016], [0022], [0073], [0080]
20 / G02	wherein k is exactly 20.	C020-S1, C020-S2, C020-S3, C020-S4	[0010], [0011], [0014], [0015], [0016], [0022], [0073], [0080]
21 / G02	wherein k is exactly 21.	C021-S1, C021-S2, C021-S3, C021-S4	[0010], [0011], [0014], [0015], [0016], [0022], [0073], [0080]
22 / G02	wherein k is exactly 22.	C022-S1, C022-S2, C022-S3, C022-S4	[0010], [0011], [0014], [0015], [0016], [0022], [0073], [0080]
23 / G02	wherein k is exactly 23.	C023-S1, C023-S2, C023-S3, C023-S4	[0010], [0011], [0014], [0015], [0016], [0022], [0073], [0080]
24 / G02	wherein k is exactly 24.	C024-S1, C024-S2, C024-S3, C024-S4	[0010], [0011], [0014], [0015], [0016], [0022], [0073], [0080]
25 / G02	wherein k is exactly 25.	C025-S1, C025-S2, C025-S3, C025-S4	[0010], [0011], [0014], [0015], [0016], [0022], [0073], [0080]
26 / G02	wherein k is exactly 26.	C026-S1, C026-S2, C026-S3, C026-S4	[0010], [0011], [0014], [0015], [0016], [0022], [0073], [0080]
27 / G02	wherein k is exactly 27.	C027-S1, C027-S2, C027-S3, C027-S4	[0010], [0011], [0014], [0015], [0016], [0022], [0073], [0080]
28 / G02	wherein k is exactly 28.	C028-S1, C028-S2, C028-S3, C028-S4	[0010], [0011], [0014], [0015], [0016], [0022], [0073], [0080]
29 / G02	wherein k is exactly 29.	C029-S1, C029-S2, C029-S3, C029-S4	[0010], [0011], [0014], [0015], [0016], [0022], [0073], [0080]
30 / G02	wherein k is exactly 30.	C030-S1, C030-S2, C030-S3, C030-S4	[0010], [0011], [0014], [0015], [0016], [0022], [0073], [0080]
31 / G02	wherein k is exactly 31.	C031-S1, C031-S2, C031-S3, C031-S4	[0010], [0011], [0014], [0015], [0016], [0022], [0073], [0080]
32 / G02	wherein k is exactly 32.	C032-S1, C032-S2, C032-S3, C032-S4	[0010], [0011], [0014], [0015], [0016], [0022], [0073], [0080]
33 / G02	wherein k is exactly 33.	C033-S1, C033-S2, C033-S3, C033-S4	[0010], [0011], [0014], [0015], [0016], [0022], [0073], [0080]
34 / G02	wherein k is exactly 34.	C034-S1, C034-S2, C034-S3, C034-S4	[0010], [0011], [0014], [0015], [0016], [0022], [0073], [0080]
35 / G02	wherein k is exactly 35.	C035-S1, C035-S2, C035-S3, C035-S4	[0010], [0011], [0014], [0015], [0016], [0022], [0073], [0080]
36 / G02	wherein k is exactly 36.	C036-S1, C036-S2, C036-S3, C036-S4	[0010], [0011], [0014], [0015], [0016], [0022], [0073], [0080]
37 / G02	wherein k is exactly 37.	C037-S1, C037-S2, C037-S3, C037-S4	[0010], [0011], [0014], [0015], [0016], [0022], [0073], [0080]

Claim / family	Added-feature excerpt	Four scenario IDs	Textual locators
38 / G02	wherein k is exactly 38.	C038-S1, C038-S2, C038-S3, C038-S4	[0010], [0011], [0014], [0015], [0016], [0022], [0073], [0080]
39 / G02	wherein k is exactly 39.	C039-S1, C039-S2, C039-S3, C039-S4	[0010], [0011], [0014], [0015], [0016], [0022], [0073], [0080]
40 / G02	wherein k is exactly 40.	C040-S1, C040-S2, C040-S3, C040-S4	[0010], [0011], [0014], [0015], [0016], [0022], [0073], [0080]
41 / G02	wherein k is exactly 41.	C041-S1, C041-S2, C041-S3, C041-S4	[0010], [0011], [0014], [0015], [0016], [0022], [0073], [0080]
42 / G02	wherein k is exactly 42.	C042-S1, C042-S2, C042-S3, C042-S4	[0010], [0011], [0014], [0015], [0016], [0022], [0073], [0080]
43 / G02	wherein k is exactly 43.	C043-S1, C043-S2, C043-S3, C043-S4	[0010], [0011], [0014], [0015], [0016], [0022], [0073], [0080]
44 / G02	wherein k is exactly 44.	C044-S1, C044-S2, C044-S3, C044-S4	[0010], [0011], [0014], [0015], [0016], [0022], [0073], [0080]
45 / G02	wherein k is exactly 45.	C045-S1, C045-S2, C045-S3, C045-S4	[0010], [0011], [0014], [0015], [0016], [0022], [0073], [0080]
46 / G02	wherein k is exactly 46.	C046-S1, C046-S2, C046-S3, C046-S4	[0010], [0011], [0014], [0015], [0016], [0022], [0073], [0080]
47 / G02	wherein k is exactly 47.	C047-S1, C047-S2, C047-S3, C047-S4	[0010], [0011], [0014], [0015], [0016], [0022], [0073], [0080]
48 / G02	wherein k is exactly 48.	C048-S1, C048-S2, C048-S3, C048-S4	[0010], [0011], [0014], [0015], [0016], [0022], [0073], [0080]
49 / G02	wherein k is exactly 49.	C049-S1, C049-S2, C049-S3, C049-S4	[0010], [0011], [0014], [0015], [0016], [0022], [0073], [0080]
50 / G02	wherein k is exactly 50.	C050-S1, C050-S2, C050-S3, C050-S4	[0010], [0011], [0014], [0015], [0016], [0022], [0073], [0080]
51 / G02	wherein k is exactly 51.	C051-S1, C051-S2, C051-S3, C051-S4	[0010], [0011], [0014], [0015], [0016], [0022], [0073], [0080]
52 / G02	wherein k is exactly 52.	C052-S1, C052-S2, C052-S3, C052-S4	[0010], [0011], [0014], [0015], [0016], [0022], [0073], [0080]
53 / G02	wherein k is exactly 53.	C053-S1, C053-S2, C053-S3, C053-S4	[0010], [0011], [0014], [0015], [0016], [0022], [0073], [0080]
54 / G02	wherein k is exactly 54.	C054-S1, C054-S2, C054-S3, C054-S4	[0010], [0011], [0014], [0015], [0016], [0022], [0073], [0080]
55 / G02	wherein k is exactly 55.	C055-S1, C055-S2, C055-S3, C055-S4	[0010], [0011], [0014], [0015], [0016], [0022], [0073], [0080]
56 / G02	wherein k is exactly 56.	C056-S1, C056-S2, C056-S3, C056-S4	[0010], [0011], [0014], [0015], [0016], [0022], [0073], [0080]

Claim / family	Added-feature excerpt	Four scenario IDs	Textual locators
57 / G02	wherein k is exactly 57.	C057-S1, C057-S2, C057-S3, C057-S4	[0010], [0011], [0014], [0015], [0016], [0022], [0073], [0080]
58 / G02	wherein k is exactly 58.	C058-S1, C058-S2, C058-S3, C058-S4	[0010], [0011], [0014], [0015], [0016], [0022], [0073], [0080]
59 / G02	wherein k is exactly 59.	C059-S1, C059-S2, C059-S3, C059-S4	[0010], [0011], [0014], [0015], [0016], [0022], [0073], [0080]
60 / G02	wherein k is exactly 60.	C060-S1, C060-S2, C060-S3, C060-S4	[0010], [0011], [0014], [0015], [0016], [0022], [0073], [0080]
61 / G02	wherein k is exactly 61.	C061-S1, C061-S2, C061-S3, C061-S4	[0010], [0011], [0014], [0015], [0016], [0022], [0073], [0080]
62 / G02	wherein k is exactly 62.	C062-S1, C062-S2, C062-S3, C062-S4	[0010], [0011], [0014], [0015], [0016], [0022], [0073], [0080]
63 / G02	wherein k is exactly 63.	C063-S1, C063-S2, C063-S3, C063-S4	[0010], [0011], [0014], [0015], [0016], [0022], [0073], [0080]
64 / G02	wherein k is exactly 64.	C064-S1, C064-S2, C064-S3, C064-S4	[0010], [0011], [0014], [0015], [0016], [0022], [0073], [0080]
65 / G02	wherein k is exactly 65.	C065-S1, C065-S2, C065-S3, C065-S4	[0010], [0011], [0014], [0015], [0016], [0022], [0073], [0080]
66 / G02	wherein k is exactly 66.	C066-S1, C066-S2, C066-S3, C066-S4	[0010], [0011], [0014], [0015], [0016], [0022], [0073], [0080]
67 / G02	wherein k is exactly 67.	C067-S1, C067-S2, C067-S3, C067-S4	[0010], [0011], [0014], [0015], [0016], [0022], [0073], [0080]
68 / G02	wherein k is exactly 68.	C068-S1, C068-S2, C068-S3, C068-S4	[0010], [0011], [0014], [0015], [0016], [0022], [0073], [0080]
69 / G02	wherein k is exactly 69.	C069-S1, C069-S2, C069-S3, C069-S4	[0010], [0011], [0014], [0015], [0016], [0022], [0073], [0080]
70 / G02	wherein k is exactly 70.	C070-S1, C070-S2, C070-S3, C070-S4	[0010], [0011], [0014], [0015], [0016], [0022], [0073], [0080]
71 / G02	wherein k is exactly 71.	C071-S1, C071-S2, C071-S3, C071-S4	[0010], [0011], [0014], [0015], [0016], [0022], [0073], [0080]
72 / G02	wherein k is exactly 72.	C072-S1, C072-S2, C072-S3, C072-S4	[0010], [0011], [0014], [0015], [0016], [0022], [0073], [0080]
73 / G02	wherein k is exactly 73.	C073-S1, C073-S2, C073-S3, C073-S4	[0010], [0011], [0014], [0015], [0016], [0022], [0073], [0080]
74 / G02	wherein k is exactly 74.	C074-S1, C074-S2, C074-S3, C074-S4	[0010], [0011], [0014], [0015], [0016], [0022], [0073], [0080]
75 / G02	wherein k is exactly 75.	C075-S1, C075-S2, C075-S3, C075-S4	[0010], [0011], [0014], [0015], [0016], [0022], [0073], [0080]

Claim / family	Added-feature excerpt	Four scenario IDs	Textual locators
76 / G02	wherein k is exactly 76.	C076-S1, C076-S2, C076-S3, C076-S4	[0010], [0011], [0014], [0015], [0016], [0022], [0073], [0080]
77 / G02	wherein k is exactly 77.	C077-S1, C077-S2, C077-S3, C077-S4	[0010], [0011], [0014], [0015], [0016], [0022], [0073], [0080]
78 / G02	wherein k is exactly 78.	C078-S1, C078-S2, C078-S3, C078-S4	[0010], [0011], [0014], [0015], [0016], [0022], [0073], [0080]
79 / G02	wherein k is exactly 79.	C079-S1, C079-S2, C079-S3, C079-S4	[0010], [0011], [0014], [0015], [0016], [0022], [0073], [0080]
80 / G02	wherein k is exactly 80.	C080-S1, C080-S2, C080-S3, C080-S4	[0010], [0011], [0014], [0015], [0016], [0022], [0073], [0080]
81 / G02	wherein k is exactly 81.	C081-S1, C081-S2, C081-S3, C081-S4	[0010], [0011], [0014], [0015], [0016], [0022], [0073], [0080]
82 / G02	wherein k is exactly 82.	C082-S1, C082-S2, C082-S3, C082-S4	[0010], [0011], [0014], [0015], [0016], [0022], [0073], [0080]
83 / G02	wherein k is exactly 83.	C083-S1, C083-S2, C083-S3, C083-S4	[0010], [0011], [0014], [0015], [0016], [0022], [0073], [0080]
84 / G02	wherein k is exactly 84.	C084-S1, C084-S2, C084-S3, C084-S4	[0010], [0011], [0014], [0015], [0016], [0022], [0073], [0080]
85 / G02	wherein k is exactly 85.	C085-S1, C085-S2, C085-S3, C085-S4	[0010], [0011], [0014], [0015], [0016], [0022], [0073], [0080]
86 / G02	wherein k is exactly 86.	C086-S1, C086-S2, C086-S3, C086-S4	[0010], [0011], [0014], [0015], [0016], [0022], [0073], [0080]
87 / G02	wherein k is exactly 87.	C087-S1, C087-S2, C087-S3, C087-S4	[0010], [0011], [0014], [0015], [0016], [0022], [0073], [0080]
88 / G02	wherein k is exactly 88.	C088-S1, C088-S2, C088-S3, C088-S4	[0010], [0011], [0014], [0015], [0016], [0022], [0073], [0080]
89 / G02	wherein k is exactly 89.	C089-S1, C089-S2, C089-S3, C089-S4	[0010], [0011], [0014], [0015], [0016], [0022], [0073], [0080]
90 / G02	wherein k is exactly 90.	C090-S1, C090-S2, C090-S3, C090-S4	[0010], [0011], [0014], [0015], [0016], [0022], [0073], [0080]
91 / G02	wherein k is exactly 91.	C091-S1, C091-S2, C091-S3, C091-S4	[0010], [0011], [0014], [0015], [0016], [0022], [0073], [0080]
92 / G02	wherein k is exactly 92.	C092-S1, C092-S2, C092-S3, C092-S4	[0010], [0011], [0014], [0015], [0016], [0022], [0073], [0080]
93 / G02	wherein k is exactly 93.	C093-S1, C093-S2, C093-S3, C093-S4	[0010], [0011], [0014], [0015], [0016], [0022], [0073], [0080]
94 / G02	wherein k is exactly 94.	C094-S1, C094-S2, C094-S3, C094-S4	[0010], [0011], [0014], [0015], [0016], [0022], [0073], [0080]

Claim / family	Added-feature excerpt	Four scenario IDs	Textual locators
95 / G02	wherein k is exactly 95.	C095-S1, C095-S2, C095-S3, C095-S4	[0010], [0011], [0014], [0015], [0016], [0022], [0073], [0080]
96 / G02	wherein k is exactly 96.	C096-S1, C096-S2, C096-S3, C096-S4	[0010], [0011], [0014], [0015], [0016], [0022], [0073], [0080]
97 / G02	wherein k is exactly 97.	C097-S1, C097-S2, C097-S3, C097-S4	[0010], [0011], [0014], [0015], [0016], [0022], [0073], [0080]
98 / G02	wherein k is exactly 98.	C098-S1, C098-S2, C098-S3, C098-S4	[0010], [0011], [0014], [0015], [0016], [0022], [0073], [0080]
99 / G02	wherein k is exactly 99.	C099-S1, C099-S2, C099-S3, C099-S4	[0010], [0011], [0014], [0015], [0016], [0022], [0073], [0080]
100 / G02	wherein k is exactly 100.	C100-S1, C100-S2, C100-S3, C100-S4	[0010], [0011], [0014], [0015], [0016], [0022], [0073], [0080]
101 / G02	wherein k is exactly 101.	C101-S1, C101-S2, C101-S3, C101-S4	[0010], [0011], [0014], [0015], [0016], [0022], [0073], [0080]
102 / G02	wherein k is exactly 102.	C102-S1, C102-S2, C102-S3, C102-S4	[0010], [0011], [0014], [0015], [0016], [0022], [0073], [0080]
103 / G02	wherein k is exactly 103.	C103-S1, C103-S2, C103-S3, C103-S4	[0010], [0011], [0014], [0015], [0016], [0022], [0073], [0080]
104 / G02	wherein k is exactly 104.	C104-S1, C104-S2, C104-S3, C104-S4	[0010], [0011], [0014], [0015], [0016], [0022], [0073], [0080]
105 / G02	wherein k is exactly 105.	C105-S1, C105-S2, C105-S3, C105-S4	[0010], [0011], [0014], [0015], [0016], [0022], [0073], [0080]
106 / G02	wherein k is exactly 106.	C106-S1, C106-S2, C106-S3, C106-S4	[0010], [0011], [0014], [0015], [0016], [0022], [0073], [0080]
107 / G02	wherein k is exactly 107.	C107-S1, C107-S2, C107-S3, C107-S4	[0010], [0011], [0014], [0015], [0016], [0022], [0073], [0080]
108 / G02	wherein k is exactly 108.	C108-S1, C108-S2, C108-S3, C108-S4	[0010], [0011], [0014], [0015], [0016], [0022], [0073], [0080]
109 / G02	wherein k is exactly 109.	C109-S1, C109-S2, C109-S3, C109-S4	[0010], [0011], [0014], [0015], [0016], [0022], [0073], [0080]
110 / G02	wherein k is exactly 110.	C110-S1, C110-S2, C110-S3, C110-S4	[0010], [0011], [0014], [0015], [0016], [0022], [0073], [0080]
111 / G02	wherein k is exactly 111.	C111-S1, C111-S2, C111-S3, C111-S4	[0010], [0011], [0014], [0015], [0016], [0022], [0073], [0080]
112 / G02	wherein k is exactly 112.	C112-S1, C112-S2, C112-S3, C112-S4	[0010], [0011], [0014], [0015], [0016], [0022], [0073], [0080]
113 / G02	wherein k is exactly 113.	C113-S1, C113-S2, C113-S3, C113-S4	[0010], [0011], [0014], [0015], [0016], [0022], [0073], [0080]

Claim / family	Added-feature excerpt	Four scenario IDs	Textual locators
114 / G02	wherein k is exactly 114.	C114-S1, C114-S2, C114-S3, C114-S4	[0010], [0011], [0014], [0015], [0016], [0022], [0073], [0080]
115 / G02	wherein k is exactly 115.	C115-S1, C115-S2, C115-S3, C115-S4	[0010], [0011], [0014], [0015], [0016], [0022], [0073], [0080]
116 / G02	wherein k is exactly 116.	C116-S1, C116-S2, C116-S3, C116-S4	[0010], [0011], [0014], [0015], [0016], [0022], [0073], [0080]
117 / G02	wherein k is exactly 117.	C117-S1, C117-S2, C117-S3, C117-S4	[0010], [0011], [0014], [0015], [0016], [0022], [0073], [0080]
118 / G02	wherein k is exactly 118.	C118-S1, C118-S2, C118-S3, C118-S4	[0010], [0011], [0014], [0015], [0016], [0022], [0073], [0080]
119 / G03	wherein k is exactly 119.	C119-S1, C119-S2, C119-S3, C119-S4	[0010], [0011], [0014], [0015], [0016], [0022], [0073], [0080]
120 / G03	wherein k is exactly 120.	C120-S1, C120-S2, C120-S3, C120-S4	[0010], [0011], [0014], [0015], [0016], [0022], [0073], [0080]
121 / G03	wherein k is exactly 121.	C121-S1, C121-S2, C121-S3, C121-S4	[0010], [0011], [0014], [0015], [0016], [0022], [0073], [0080]
122 / G03	wherein k is exactly 122.	C122-S1, C122-S2, C122-S3, C122-S4	[0010], [0011], [0014], [0015], [0016], [0022], [0073], [0080]
123 / G03	wherein k is exactly 123.	C123-S1, C123-S2, C123-S3, C123-S4	[0010], [0011], [0014], [0015], [0016], [0022], [0073], [0080]
124 / G03	wherein k is exactly 124.	C124-S1, C124-S2, C124-S3, C124-S4	[0010], [0011], [0014], [0015], [0016], [0022], [0073], [0080]
125 / G03	wherein k is exactly 125.	C125-S1, C125-S2, C125-S3, C125-S4	[0010], [0011], [0014], [0015], [0016], [0022], [0073], [0080]
126 / G03	wherein k is exactly 126.	C126-S1, C126-S2, C126-S3, C126-S4	[0010], [0011], [0014], [0015], [0016], [0022], [0073], [0080]
127 / G03	wherein k is exactly 127.	C127-S1, C127-S2, C127-S3, C127-S4	[0010], [0011], [0014], [0015], [0016], [0022], [0073], [0080]
128 / G03	wherein k is exactly 128.	C128-S1, C128-S2, C128-S3, C128-S4	[0010], [0011], [0014], [0015], [0016], [0022], [0073], [0080]
129 / G03	wherein k is exactly 129.	C129-S1, C129-S2, C129-S3, C129-S4	[0010], [0011], [0014], [0015], [0016], [0022], [0073], [0080]
130 / G03	wherein k is exactly 130.	C130-S1, C130-S2, C130-S3, C130-S4	[0010], [0011], [0014], [0015], [0016], [0022], [0073], [0080]
131 / G03	wherein k is exactly 131.	C131-S1, C131-S2, C131-S3, C131-S4	[0010], [0011], [0014], [0015], [0016], [0022], [0073], [0080]
132 / G03	wherein k is exactly 132.	C132-S1, C132-S2, C132-S3, C132-S4	[0010], [0011], [0014], [0015], [0016], [0022], [0073], [0080]

Claim / family	Added-feature excerpt	Four scenario IDs	Textual locators
133 / G03	wherein k is exactly 133.	C133-S1, C133-S2, C133-S3, C133-S4	[0010], [0011], [0014], [0015], [0016], [0022], [0073], [0080]
134 / G03	wherein k is exactly 134.	C134-S1, C134-S2, C134-S3, C134-S4	[0010], [0011], [0014], [0015], [0016], [0022], [0073], [0080]
135 / G03	wherein k is exactly 135.	C135-S1, C135-S2, C135-S3, C135-S4	[0010], [0011], [0014], [0015], [0016], [0022], [0073], [0080]
136 / G03	wherein k is exactly 136.	C136-S1, C136-S2, C136-S3, C136-S4	[0010], [0011], [0014], [0015], [0016], [0022], [0073], [0080]
137 / G03	wherein k is exactly 137.	C137-S1, C137-S2, C137-S3, C137-S4	[0010], [0011], [0014], [0015], [0016], [0022], [0073], [0080]
138 / G03	wherein k is exactly 138.	C138-S1, C138-S2, C138-S3, C138-S4	[0010], [0011], [0014], [0015], [0016], [0022], [0073], [0080]
139 / G04	A single-element material having a closed elemental inventory consisting of one constituent selected from atomic numbers...	C139-S1, C139-S2, C139-S3, C139-S4	[0013], [0015], [0016]
140 / G05	wherein every selected constituent has atomic number from 1 through 118.	C140-S1, C140-S2, C140-S3, C140-S4	[0004], [0010], [0011], [0014], [0015], [0016], [0080]
141 / G06	wherein at least one selected constituent is selected from atomic numbers 119 through 138.	C141-S1, C141-S2, C141-S3, C141-S4	[0004], [0010], [0011], [0014], [0015], [0016], [0041], [0080]
142 / G07	wherein k is 118 and the inventory includes each atomic number from 1 through 118.	C142-S1, C142-S2, C142-S3, C142-S4	[0010], [0011], [0014], [0015], [0016], [0022], [0080]
143 / G08	wherein k is 138 and the inventory includes each atomic number from 1 through 138.	C143-S1, C143-S2, C143-S3, C143-S4	[0010], [0011], [0014], [0015], [0016], [0019], [0022], [0080]
144 / G09	wherein each x_i is the number of atoms of constituent i divided by the total number of atoms in the specified inventory.	C144-S1, C144-S2, C144-S3, C144-S4	[0010], [0011], [0014], [0015], [0016], [0080]
145 / G09	wherein the proportions are specified as mole-of-atoms fractions using the same atom inventory as an atomic-fraction...	C145-S1, C145-S2, C145-S3, C145-S4	[0010], [0011], [0014], [0015], [0016], [0080]

Claim / family	Added-feature excerpt	Four scenario IDs	Textual locators
146 / G10	wherein each x_i is the mass of constituent i divided by the total mass of the specified inventory.	C146-S1, C146-S2, C146-S3, C146-S4	[0010], [0011], [0014], [0015], [0016], [0080]
147 / G11	wherein an isotope-qualified positive finite effective mass M_i is specified for every selected constituent under a...	C147-S1, C147-S2, C147-S3, C147-S4	[0010], [0011], [0014], [0015], [0016], [0047], [0080]
148 / G12	wherein x_i = n_i/q for a positive integer q and positive integers n_i whose sum is q, with q unrestricted except that q...	C148-S1, C148-S2, C148-S3, C148-S4	[0010], [0011], [0014], [0015], [0016], [0017], [0073], [0074], [0080]
149 / G12	wherein gcd(q,n_1,...,n_k) = 1.	C149-S1, C149-S2, C149-S3, C149-S4	[0010], [0011], [0014], [0015], [0016], [0017], [0073], [0074], [0080]
150 / G13	wherein x_i = n_i/D with D = 10^19, each n_i is a positive integer, and sum(n_i) = D.	C150-S1, C150-S2, C150-S3, C150-S4	[0010], [0011], [0014], [0015], [0016], [0018], [0031], [0080]
151 / G13	wherein k-1 constituents each have fraction 1/D and the remaining constituent has fraction (D-k+1)/D.	C151-S1, C151-S2, C151-S3, C151-S4	[0010], [0011], [0014], [0015], [0016], [0018], [0019], [0031], [0072], [0080]
152 / G14	wherein each fraction satisfies a selected closed interval [L_i,U_i], with 0 <= L_i <= U_i <= 1, and the joint...	C152-S1, C152-S2, C152-S3, C152-S4	[0010], [0011], [0014], [0015], [0016], [0029], [0030], [0080]
153 / G14	wherein each interval is selected by integers p,l,u with 0 <= p <= 19 and 0 <= l <= u <= 10^p, its endpoints being l/10^p...	C153-S1, C153-S2, C153-S3, C153-S4	[0010], [0011], [0014], [0015], [0016], [0023], [0029], [0030], [0080]
154 / G14	wherein at least one interval is an elementary cell [j/10^p,(j+1)/10^p] for an integer j with 0 <= j < 10^p.	C154-S1, C154-S2, C154-S3, C154-S4	[0010], [0011], [0014], [0015], [0016], [0023], [0024], [0029], [0030], [0080]
155 / G15	wherein k = 2 and fractions x_A and x_B obey x_A in [L,U] and x_B = 1-x_A in [1-U,1-L], with 0 < x_A < 1.	C155-S1, C155-S2, C155-S3, C155-S4	[0010], [0011], [0014], [0015], [0016], [0024], [0029], [0069], [0080]
156 / G16	wherein 10^-18 <= x_A <= 10^-17, corresponding to 10^-16% through 10^-15%, inclusive.	C156-S1, C156-S2, C156-S3, C156-S4	[0010], [0011], [0014], [0015], [0016], [0024], [0025], [0026], [0029], [0069], [0070], [0080]

Claim / family	Added-feature excerpt	Four scenario IDs	Textual locators
157 / G16	wherein x_A is exactly 10^-18 and x_B is exactly 1-10^-18.	C157-S1, C157-S2, C157-S3, C157-S4	[0010], [0011], [0014], [0015], [0016], [0024], [0025], [0026], [0029], [0069], [0070], [0080]
158 / G16	wherein x_A is exactly 10^-17 and x_B is exactly 1-10^-17.	C158-S1, C158-S2, C158-S3, C158-S4	[0010], [0011], [0014], [0015], [0016], [0024], [0025], [0026], [0029], [0069], [0070], [0080]
159 / G16	wherein x_A = n/10^19 for an integer n from 10 through 100 and x_B = 1-x_A.	C159-S1, C159-S2, C159-S3, C159-S4	[0010], [0011], [0014], [0015], [0016], [0024], [0025], [0026], [0029], [0069], [0070], [0080]
160 / G16	wherein x_A lies in a selected subrange [l/10^19,u/10^19] for integers 10 <= l <= u <= 100.	C160-S1, C160-S2, C160-S3, C160-S4	[0010], [0011], [0014], [0015], [0016], [0024], [0025], [0026], [0029], [0069], [0070], [0080]
161 / G17	wherein k = 2 and each constituent has fraction 1/2, corresponding to 50% each.	C161-S1, C161-S2, C161-S3, C161-S4	[0010], [0011], [0014], [0015], [0016], [0027], [0080]
162 / G17	wherein k = 2 and the constituent fractions are exactly 1/3 and 2/3.	C162-S1, C162-S2, C162-S3, C162-S4	[0010], [0011], [0014], [0015], [0016], [0017], [0071], [0080]
163 / G17	wherein every constituent has fraction 1/k.	C163-S1, C163-S2, C163-S3, C163-S4	[0010], [0011], [0014], [0015], [0016], [0072], [0080]
164 / G18	wherein every constituent has fraction at least 10^-19.	C164-S1, C164-S2, C164-S3, C164-S4	[0010], [0011], [0014], [0015], [0016], [0018], [0020], [0080]
165 / G18	wherein at least one constituent has positive fraction less than 10^-19.	C165-S1, C165-S2, C165-S3, C165-S4	[0010], [0011], [0014], [0015], [0016], [0020], [0080]
166 / G19	wherein at least one interval is [j times 10^(e-r),(j+1) times 10^(e-r)], with integer e from -19 through -1, nonnegative...	C166-S1, C166-S2, C166-S3, C166-S4	[0010], [0011], [0014], [0015], [0016], [0028], [0029], [0030], [0080]
167 / G19	wherein at least one interval has arbitrary rational endpoints r and s satisfying 0 <= r <= s <= 1.	C167-S1, C167-S2, C167-S3, C167-S4	[0010], [0011], [0014], [0015], [0016], [0027], [0029], [0030], [0080]
168 / G19	wherein the endpoints are specified as real target values and the selected fractions satisfy their interval constraints...	C168-S1, C168-S2, C168-S3, C168-S4	[0010], [0011], [0014], [0015], [0016], [0027], [0029], [0030], [0080]
169 / G20	wherein each selected element i has a conditional atom-count isotope-state distribution y_i,a,s with nonnegative entries...	C169-S1, C169-S2, C169-S3, C169-S4	[0010], [0011], [0014], [0015], [0016], [0046], [0047], [0050], [0080]

Claim / family	Added-feature excerpt	Four scenario IDs	Textual locators
170 / G20	wherein at least one constituent is specified by a single mass number and nuclear state with conditional fraction one.	C170-S1, C170-S2, C170-S3, C170-S4	[0010], [0011], [0014], [0015], [0016], [0046], [0047], [0050], [0080]
171 / G20	wherein at least one constituent has two or more specified isotope-state entries of positive conditional fraction.	C171-S1, C171-S2, C171-S3, C171-S4	[0010], [0011], [0014], [0015], [0016], [0046], [0047], [0050], [0080]
172 / G20	wherein each specified nuclear state has a source-qualified identifier that distinguishes a ground state from any...	C172-S1, C172-S2, C172-S3, C172-S4	[0010], [0011], [0014], [0015], [0016], [0046], [0047], [0048], [0049], [0050], [0080]
173 / G21	wherein a radioactive inventory is identified at a specified time and includes daughter constituents that fall within its...	C173-S1, C173-S2, C173-S3, C173-S4	[0010], [0011], [0014], [0015], [0016], [0046], [0047], [0050], [0080]
174 / G22	wherein a crystalline constituent phase has a specified structural model, site occupations, substitution rules and...	C174-S1, C174-S2, C174-S3, C174-S4	[0010], [0011], [0014], [0015], [0016], [0033], [0034], [0080]
175 / G23	wherein an amorphous constituent phase is identified by a stated preparation history and a specified characterization...	C175-S1, C175-S2, C175-S3, C175-S4	[0010], [0011], [0014], [0015], [0016], [0035], [0080]
176 / G24	wherein molecular identity includes specified connectivity, stereochemistry, salt or solvate state, and hydration state...	C176-S1, C176-S2, C176-S3, C176-S4	[0010], [0011], [0014], [0015], [0016], [0035], [0080]
177 / G25	wherein the material is a polymer specified by repeat units, incorporated ratios, sequence distribution, molecular-weight...	C177-S1, C177-S2, C177-S3, C177-S4	[0010], [0011], [0014], [0015], [0016], [0036], [0080]
178 / G26	wherein the material is a composite specified by constituent identities, dimensions, orientations, interfaces and dispersion.	C178-S1, C178-S2, C178-S3, C178-S4	[0010], [0011], [0014], [0015], [0016], [0036], [0080]

Claim / family	Added-feature excerpt	Four scenario IDs	Textual locators
179 / G27	wherein the material has a specified layer order and layer dimensions, with local composition distinguished from...	C179-S1, C179-S2, C179-S3, C179-S4	[0010], [0011], [0013], [0014], [0015], [0016], [0037], [0080]
180 / G27	wherein the material has a specified spatial composition field x_i(r,t) and a density-weighted inventory convention for...	C180-S1, C180-S2, C180-S3, C180-S4	[0010], [0011], [0014], [0015], [0016], [0037], [0080]
181 / G28	wherein the stated elemental inventory describes a specified single phase.	C181-S1, C181-S2, C181-S3, C181-S4	[0010], [0011], [0014], [0015], [0016], [0033], [0034], [0080]
182 / G28	wherein the stated elemental inventory describes a specified multiphase assemblage and each phase is separately identified.	C182-S1, C182-S2, C182-S3, C182-S4	[0010], [0011], [0014], [0015], [0016], [0033], [0036], [0080]
183 / G29	wherein constituent volume fractions and void fraction are specified separately from elemental atomic fractions under...	C183-S1, C183-S2, C183-S3, C183-S4	[0010], [0011], [0012], [0014], [0015], [0016], [0036], [0080]
184 / G30	wherein a nominal target fraction is accompanied by a separately specified measurement uncertainty or tolerance and the...	C184-S1, C184-S2, C184-S3, C184-S4	[0010], [0011], [0014], [0015], [0016], [0042], [0043], [0080]
185 / G30	wherein the product composition is specified separately from its feed proportions and from local measured composition.	C185-S1, C185-S2, C185-S3, C185-S4	[0010], [0011], [0013], [0014], [0015], [0016], [0038], [0080]
186 / G31	A method of constructing a candidate-material record, comprising selecting a constituent support and composition basis;...	C186-S1, C186-S2, C186-S3, C186-S4	[0005], [0007], [0073], [0074], [0075]
187 / G31	comprising enumerating every increasing k-tuple of selected atomic numbers for a specified k from 1 through 138 and...	C187-S1, C187-S2, C187-S3, C187-S4	[0005], [0007], [0015], [0073], [0074], [0075]

Claim / family	Added-feature excerpt	Four scenario IDs	Textual locators
188 / G32	comprising enumerating positive integer tuples with sum q for successive unrestricted positive denominators q, and...	C188-S1, C188-S2, C188-S3, C188-S4	[0005], [0007], [0017], [0073], [0074], [0075]
189 / G32	comprising enumerating positive integer tuples with sum D = 10^19 and storing their normalized fractions as exact integer...	C189-S1, C189-S2, C189-S3, C189-S4	[0005], [0007], [0018], [0070], [0073], [0074], [0075]
190 / G32	comprising selecting every endpoint pair l,u with 0 <= l <= u <= 10^p for each p from 0 through 19, forming correlated...	C190-S1, C190-S2, C190-S3, C190-S4	[0005], [0007], [0023], [0029], [0032], [0073], [0074], [0075]
191 / G33	comprising selecting component bounds L_i and U_i for the selected support; testing 0 <= L_i <= U_i <= 1 for every...	C191-S1, C191-S2, C191-S3, C191-S4	[0005], [0007], [0029], [0030], [0032], [0073], [0074], [0075]
192 / G33	further comprising, for a box satisfying the tests of claim 191, determining strictly positive feasibility for the...	C192-S1, C192-S2, C192-S3, C192-S4	[0005], [0007], [0029], [0030], [0032], [0073], [0074], [0075]
193 / G33	comprising selecting a k-component box with 0 <= L_i <= U_i <= 1; calculating b_i = max(1, ceiling(D L_i)) and c_i =...	C193-S1, C193-S2, C193-S3, C193-S4	[0005], [0007], [0018], [0029], [0031], [0032], [0073], [0074], [0075]
194 / G34	A method of constructing a particle-system record, comprising identifying a species or model-qualified candidate,...	C194-S1, C194-S2, C194-S3, C194-S4	[0051], [0052], [0053], [0056]
195 / G35	wherein the record concerns a host-dependent excitation and specifies the host material, excitation mechanism,...	C195-S1, C195-S2, C195-S3, C195-S4	[0051], [0052], [0053], [0054], [0056]
196 / G36	wherein an unconfirmed entity is marked hypothetical, with a model identifier, parameter assumptions, conservation...	C196-S1, C196-S2, C196-S3, C196-S4	[0051], [0052], [0053], [0055], [0056]

Claim / family	Added-feature excerpt	Four scenario IDs	Textual locators
197 / G37	comprising recording a property scenario with a baseline material and state, controlled change, proposed response,...	C197-S1, C197-S2, C197-S3, C197-S4	[0005], [0007], [0057], [0058], [0059], [0060], [0073], [0074], [0075]
198 / G37	comprising recording a credible alternative response and its mechanism as a separate conditional hypothesis, rather than...	C198-S1, C198-S2, C198-S3, C198-S4	[0005], [0007], [0057], [0058], [0059], [0060], [0061], [0062], [0063], [0064], [0065], [0066], [0067], [0068], [0073], [0074], [0075]
199 / G38	comprising labeling attributed source teachings ESTABLISHED, stated mathematical or model results CALCULATED, unperformed...	C199-S1, C199-S2, C199-S3, C199-S4	[0005], [0007], [0041], [0042], [0073], [0074], [0075]

Provisional application review draft: composition domains and conditional material embodiments

Prepared **3 October 2026** | **Substantively reviewed unfiled theoretical provisional draft**

Proposed sole inventor: Robert M. Layne, as instructed. Legal filing particulars remain unconfirmed. This is a review designation, not an executed declaration or an independent determination of conception for every candidate claim. Published findings remain attributed to their authors; they are not applicant experiments or an assertion that the proposed inventor originated those findings. AI drafting and calculation assistance are tools, not additional named inventors. Current USPTO inventorship guidance

Purpose and completion criterion. The objective is broad defensive prior art, and universal blocking is aspirational. Completion here means a coherent theoretical provisional specification ready for substantive review, with its drawings, numerical definitions, attributed examples, hypothetical embodiments and all 199 candidate claims present. It does not mean every material is enabled, every claim is allowable, any application is filed or all future material development is legally in the public domain. No ownership, license grant or freedom-to-operate determination is created by that aspiration.

Evidence and hypothetical work. No applicant physical experiments are performed or planned. ESTABLISHED means an attributed source teaching; CALCULATED means a stated result under explicit mathematical/model assumptions; PROPOSED means an unperformed conditional embodiment; UNSUPPORTED means a possibility lacking adequate technical support. New examples address the identified gaps by specifying assumptions, routes, structures, observables, alternatives and failure boundaries. An unresolved field remains visible rather than being filled with a fictitious successful result. Known methods and justified theory can support an example without applicant experimentation; hypothetical wording alone does not establish enablement. USPTO prophetic-example guidance

Version boundary. This review edition derives from the preserved 1 October theoretical specification, SHA-256 61ffc4ec1aa89d32dcc4cd44d35852a4b315b13d514c2898c4aabf460e3235f7, at repository baseline 4ee2a5eb9001d6313df705d66358a9e575b0a04d. All 94 preceding numbered paragraphs, every candidate claim and inherited attributed evidence retain their wording. Paragraphs [0095]-[0099] clarify conditional input domains, constituent accounting and finite inventories, record-algorithm failure and enumeration boundaries, and population and kinetic qualifications. Paragraph [0099] expressly corrects an inherited atom-balance term while preserving the historical wording for comparison. The essential additions are printed here; they are not content of an earlier PDF, filing or priority date. The accompanying substantive claim review is a separate preliminary screening record, not a completed statutory novelty search or a legal determination.

Administrative worksheet for later filing review

Field	Review value	Status
Invention title	Composition domains and conditional material embodiments	Draft technical title
Sole inventor named for review	Robert M. Layne	User-directed; exact legal filing name and claim-specific conception review remain open
Inventor residence	UNPROVIDED	Confirm city, state/province and country before submission
Applicant and entitlement	UNPROVIDED	Inventor naming does not supply applicant/assignee particulars
Correspondence address	UNPROVIDED	Complete privately for the official cover sheet
Attorney/agent and docket, if applicable	UNPROVIDED	No representative appointment inferred
Government property interest	UNCONFIRMED	No negative certification inferred
Micro-entity status	USER-REPORTED	Eligibility and required certification are separate from this review draft
Earlier filing/priority particulars	UNPROVIDED	No earlier filing, date or benefit asserted

The specification and optional claims can be reviewed while these administrative fields remain unresolved. A complete submission requires the actual cover-sheet particulars, necessary drawings and applicable fee; no oath or declaration is required for a provisional. A later PCT needs its own accepted application contents and cannot rely on a general repository link to supply omitted essential teaching. USPTO provisional guidance, PCT application contents

1 Technical field

[0001] The subject matter concerns the description, preparation, characterization, and comparison of materials having specified elemental and isotopic compositions, structures, processing histories, and properties. Materials considered include metals and alloys, inorganic compounds and ceramics, molecular substances, polymers, composites, mixtures, and single-element forms. Particle-containing or excited systems are addressed separately from ordinary chemical constituents.

[0002] A composition framework is provided to identify selected constituents, normalize their proportions, express exact values and intervals, and associate each numerical description with a structural and preparation record. Its purpose is to make the requested scope explicit and reproducible as a specification of candidate compositions. Whether an individual candidate exists, can be synthesized, or has a useful property requires additional technical information.

2 Background and evidence status

[0003] An elemental formula does not uniquely define a material. Connectivity, crystal structure, phase assemblage, isotope distribution, defects, interfaces, orientation, morphology, and processing can distinguish systems sharing the same elemental proportions. A molecular formula can describe different isomers; a ceramic formula can describe different polymorphs; a polymer formula can omit sequence and architecture; a composite's overall percentages can omit its load-bearing or conducting network.

[0004] The element vocabulary has two groups. The recognized group consists of atomic numbers 1 through 118 in the IUPAC periodic table. The requested extension consists of unconfirmed placeholders with atomic numbers 119 through 138. The extension is an indexing convention for hypothetical candidates, not a statement that these elements have been discovered or possess known chemistry. IUPAC periodic table

[0005] Four evidence labels are used locally: ESTABLISHED for an attributed source teaching; CALCULATED for a stated mathematical result or model calculation; PROPOSED for an unperformed preparation or test with specified assumptions; and UNSUPPORTED for a possibility without adequate preparation or property support. These labels do not decide patentability. A calculated fraction is not a measured composition, and a proposed recipe is not a performed experiment.

[0006] No new synthesis, measured yield, verified phase, measured property, or applicant-specific successful embodiment is reported in this draft. Published third-party examples retain their attribution. The requested breadth includes unsupported candidates. These candidates are expressly identified so that the scope is visible without representing them as established teachings.

3 Summary of the composition framework

[0007] A candidate material is described by a record R = (B, S, x, I, U, C, P, M, F, E), where B is the composition basis; S is the constituent support set; x is the normalized composition vector; I is the isotope and nuclear-state description; U is the structural and spatial description; C is the preparation and conditioning record; P is the property scenario; M is the measurement record; F identifies failure boundaries; and E identifies evidence and sources. Missing fields are stated explicitly.

[0008] The framework includes every nonempty subset of the 118 recognized element labels and, separately, every nonempty subset of the 138-label vocabulary including the hypothetical extension. It includes every positive normalized continuous composition as a target description and every rational normalized composition as a separate exact accounting description on the expressly selected basis. A rational atomic-fraction vector is compatible with integer atom counts when the chosen atom inventory N satisfies its primitive-denominator condition; a rational mass-fraction vector does not by itself establish that atom-count condition. A finite decimal grid is a useful additional subset, not a replacement for either domain.

[0009] Each interval embodiment is the intersection of its component bounds with the normalization condition. Component endpoints cannot be selected independently if their sum would differ from one. Numerical enumeration provides a reproducible address for a candidate; it supplies neither missing chemistry nor a proof that later claims to that candidate lack novelty.

4 Composition bases and constituent accounting

[0010] On an atomic basis, let $a_i = N_i / \sum(N_j)$, where N_i counts atoms of element i in the specified inventory, and set the generic record fraction $x_i = a_i$. A mole-of-atoms basis has the same normalized proportions when the same atom inventory is used. A mole-of-molecules, formula-unit, repeat-unit, site-occupancy, or feed-component basis is different and must be expressly named. Denominators must identify the counted entity, sample region, and inclusion of impurities.

[0011] On a mass basis, let w_i = m_i / sum(m_j) and set the generic record fraction x_i = w_i. Conversion from elemental atomic fractions uses w_i = a_i M_i / sum_j(a_j M_j), with positive finite isotope-qualified effective masses M_i under the same expressly stated mass-allocation and compatible-unit convention described in [0047]. The inverse uses a_i = (w_i/M_i) / sum_j(w_j/M_j). Standard atomic weights are not substitutes for unknown masses of hypothetical nuclei or other populated states. Conversion is unavailable where the needed mass information is unavailable. The generic x_i is not treated as atomic on a record expressly using mass fractions.

[0012] Volume fractions describe identified phases or constituents under specified temperature, pressure, density, and volume-assignment conventions. They are not automatically elemental atomic fractions. Mixing volumes need not be additive. A composite record therefore identifies constituent densities and whether volume fractions are measured, calculated from a justified additive model, or nominal feed values.

[0013] Feed proportions, retained product composition, and local measured composition are separate fields. A specimen may contain gradients, layers, inclusions, or surface enrichment. The support set identifies which constituents have positive fractions in the stated denominator. Zero entries are omitted from the canonical support set, while a zero concentration endpoint can describe a boundary shared with a smaller support set.

[0014] A closed composition specification includes all constituents in its denominator. An open specification can permit additional constituents, but must state how their inclusion affects normalization. A record describing selected constituents relative only to their subtotal is not a complete bulk-composition assay. Residual impurities are included as identified amounts, bounded unknown residuals, or an expressly unresolved term; they are not silently set to zero.

5 Exact constituent support and composition domains

[0015] Let E = {E_1, ..., E_118}, with E_Z corresponding to the recognized element of atomic number Z. Let H = {E_119, ..., E_138} denote the hypothetical placeholders. For either vocabulary V, choose an ordered support S = (E_z1, ..., E_zk), with strictly increasing atomic numbers. The ordering removes duplicate permutations; it does not restrict spatial ordering within a material.

[0016] For each integer k from 1 through the vocabulary size, the continuous target domain is Delta(S) = {x : x_i > 0 for every member of S, sum(x_i) = 1}. Its closure permits x_i = 0, with the resulting candidate reassigned to its positive support. Every value is dimensionless; percent values equal 100 x_i. For k = 1, the only normalized vector is (1).

[0017] The rational domain consists of all x_i = n_i/q, where q is any positive integer, each n_i is a positive integer, and sum(n_i) = q. For k constituents, q must be at least k. A primitive representation has gcd(q, n_1, ..., n_k) = 1. This removes duplicate representations such as (1/2, 1/2) and (2/4, 2/4). The unrestricted denominator includes (1/3, 2/3), which is absent from a grid whose denominator is 10^19.

[0018] The requested decimal composition grid has accounting denominator D = 10^19. Its vectors satisfy x_i = n_i/D, n_i >= 1, and sum(n_i) = D on the selected composition basis. The integers n_i are grid numerators; D does not automatically equal the specimen's atom count N, particularly on a mass basis. Atomic count compatibility is evaluated using the atomic vector and the chosen N. The smallest positive grid fraction is a = 1/10^19 = 0.0000000000000000001. In percent units it is 0.00000000000000001%. This is a chosen grid increment, not a universal physical concentration limit.

[0019] For k positive constituents on this grid, no single constituent can exceed 1 - (k-1)/D. A two-component extreme is (1/D, (D-1)/D). A 138-component extreme has 137 constituents each at 1/D and the remaining constituent at (D-137)/D. These vectors normalize exactly. They do not establish that a chemically homogeneous or stable 138-element material can be made.

[0020] The broad continuous and unrestricted rational domains remain available below the selected grid floor. If a particular embodiment imposes a positive floor a, then k a <= 1 is necessary and sufficient for an unconstrained normalized vector to exist. For a fixed total atom inventory N, the least positive atomic count fraction is 1/N. Since N depends on the sample, no sample-independent lower fraction follows from atom counting.

[0021] The SI Avogadro constant is exactly 6.02214076 x 10^23 per mole. An inventory of 10^19 atoms is approximately 0.00001660539067 mole of atoms. One specified atom among that inventory gives a count fraction of 10^-19. The arithmetic does not establish controlled placement, detectability, representative sampling, or a reproducible bulk property. BIPM definition of the mole

[0022] The recognized vocabulary has 2^118 - 1 nonempty supports. Its supports of size at least two number 2^118 - 1 - 118. The extended vocabulary has 2^138 - 1 nonempty supports and 2^138 - 1 - 138 supports of size at least two. For a fixed selected support of size k on the D grid, the number of positive vectors is binomial(D-1, k-1). These are counts of mathematical descriptions, not counts of enabled substances.

6 All endpoint ranges and complementary proportions

[0023] For each precision index p = 0, 1, ..., 19, let Q_p = 10^p. Select integers l and u with 0 <= l <= u <= Q_p. The closed interval I(p,l,u) = [l/Q_p, u/Q_p] defines every pair of decimal-grid endpoints at that precision, including exact-point intervals l = u. Percent endpoints are [100 l/Q_p, 100 u/Q_p]. Every elementary cell [j/Q_p, (j+1)/Q_p] is included for 0 <= j < Q_p, as is every union represented by a contiguous pair of endpoints.

[0024] At p = 2 the elementary cells are [0%,1%], [1%,2%], ..., [99%,100%]. For a binary composition, if x_A lies in [L,U], x_B = 1 - x_A lies in [1-U,1-L]. Thus [1%,2%] pairs with [98%,99%], [2%,3%] with [97%,98%], and [49%,50%] with [50%,51%]. Corresponding endpoints describe correlated compositions: (1%,99%) and (2%,98%), not (1%,98%).

[0025] Closed cells share their endpoints. This intentional overlap avoids losing boundary values. Canonical indexing may assign a shared boundary to one cell for retrieval, but the physical composition is unchanged. Exact boundary points, every intermediate target value, each contiguous subrange with specified endpoints, and each complement are contemplated as numerical descriptions. This language does not assert that all such subranges are legally disclosed with sufficient specificity.

[0026] The requested interval 0.0000000000000001% through 0.000000000000001% equals [10^-18,10^-17] as a fraction. On the D grid its numerators are 10 through 100 inclusive: 91 exact grid values and 90 elementary bins. The binary complement is [(D-100)/D,(D-10)/D], or [99.999999999999999%,99.999999999999999%]. Fraction and percent units must not be interchanged.

[0027] The 50/50 binary vector is exactly (1/2,1/2). The entire decimal hierarchy connects ordinary percent cells to the requested very small cells. Additional rational intervals [r,s], for all rational 0 <= r <= s <= 1, and real endpoint target intervals may be specified outside the decimal hierarchy. Real targets, rational finite-count descriptions, decimal grid values, and measured tolerances remain distinct.

[0028] A decade refinement supplies optional addresses for very small positive values. For e from -19 through -1 and any nonnegative integer r, define intervals [j x 10^(e-r), (j+1) x 10^(e-r)] with j = 10^r, ..., 10^(r+1)-1. These partition each selected decade at the specified refinement, with shared closed boundaries. Refinement beyond the D grid is symbolic or rational coverage, not an assertion of greater attainable manufacturing precision.

[0029] For a selected k-component support choose an interval [L_i,U_i] for every component, with 0 <= L_i <= U_i <= 1, after intersecting any supplied bounds with [0,1]. The candidate domain is {x : L_i <= x_i <= U_i, sum(x_i) = 1}, intersected with the applicable positivity or grid conditions. A box with closed component bounds is feasible on the nonnegative continuous simplex exactly when each L_i <= U_i and sum(L_i) <= 1 <= sum(U_i). This test is applied after intersecting with any selected positive floor and corresponding upper bound.

[0030] A preliminary nonnegative normalized witness z can be constructed exactly for a box satisfying the ordinary conditions of [0029]: start with z_i = L_i; let residual r = 1 - sum(L_i); in support order add min(r, U_i-L_i) to each component and reduce r accordingly until sum(z_i) = 1. With rational inputs the calculation uses exact fractions. Assign this preliminary witness as a positive target only when every coordinate is positive. Otherwise a separate strictly positive witness within the same bounds is required before assigning a positive target for that selected support. A greedy witness containing zero does not prove that another positive witness is unavailable. Under the ordinary feasibility conditions, positive feasibility requires every U_i > 0. If S = sum(L_i) = 1, the lower-bound vector is the only feasible vector, and every L_i must be positive. If S < 1, choose epsilon = min(min_i U_i, (1-S)/(2k)), replace each lower bound by L'_i = max(L_i,epsilon), and allocate residual 1-sum(L'_i) within the unchanged upper bounds. Epsilon is positive, each raised lower bound remains within its upper bound, and sum(L'_i) <= S + k epsilon <= (1+S)/2 < 1; sum(U_i) >= 1 supplies the remaining capacity. The result is a strictly positive normalized witness within the original box. This local epsilon is not a global concentration floor or a domain-wide denominator cap. With rational inputs all operations remain exact. Removing zero coordinates describes a separate smaller-support candidate with its own support and bounds, rather than completion of the original positive-support request.

[0031] For the D grid, set b_i = max(1,ceiling(D L_i)) and c_i = min(D-k+1,floor(D U_i)), and test b_i <= c_i for every component and sum(b_i) <= D <= sum(c_i). Every positive integer vector summing to D has n_i <= D-k+1, so the cap excludes no such vector. On acceptance, start with integer n_i = b_i and allocate residual D-sum(b_i) within the upper integer bounds until sum(n_i) = D, then assign x_i = n_i/D as the positive normalized target on the selected basis. The inequalities are necessary and this allocation supplies sufficiency. D is a composition accounting denominator, not automatically an atom inventory N. A box may be continuously feasible and contain no D-grid vector. For example the exact binary vector (1/3,2/3) is rationally feasible but is not on the D = 10^19 grid. Rejection on that grid does not reject the wider rational or real target domain.

[0032] Independent selection of all endpoint ranges is permitted as an indexing operation. Requests failing ordinary feasibility, strictly positive feasibility for the selected support, or a selected grid are recorded as rejected numerical requests with the particular reason. Rejection alone does not complete positive-target assignment or the remaining record operations of claim 186. A box admitting only a zero-containing vector cannot supply that selected positive support. Example: three lower bounds of 40% cannot normalize. A valid ternary example has atomic intervals A = [10%,20%], B = [30%,40%], C = [40%,60%], with positive target (10%,30%,60%). No product phase or preparation follows from that witness.

7 Structural and spatial descriptions

[0033] A material record identifies whether the composition describes a single phase, multiple phases, a molecular species, a solution, a dispersed mixture, a layered article, or an ensemble. Where the distinction is unknown, it is marked unknown. A numerical vector does not imply every constituent occupies a common lattice or remains uniformly distributed.

[0034] For a crystalline candidate, specify the phase identity, space group or structural model where known, lattice parameters with conditions, occupied sites, substitutions, vacancy rules, ordering, and charge-balance constraints. A proposed site assignment is labeled PROPOSED. Unknown structure is not filled by saying all crystal structures are present.

[0035] For an amorphous candidate, identify the preparation history, detectable crystalline fraction, relevant local-order evidence, glass transition or relaxation behavior where measured, and the test's resolution. For molecular candidates identify connectivity, stereochemistry, salt/solvate and hydration state, and purity. Distinct connectivity is separately indexed even if elemental fractions match.

[0036] For polymers specify repeat units, incorporated comonomer ratios, sequence distribution, tacticity, molecular-weight distribution, branching, end groups, crosslink topology, additives, and conditioning. Feed

composition alone does not establish incorporated composition. For composites specify constituent identities, dimensions, orientation, interfaces, dispersion, layer order, void fraction, and the basis of each fraction.

[0037] Spatial variations can be represented by a composition field x_i(r,t) with a defined coordinate region and sampling or averaging convention. A whole-specimen fraction is an integral over the appropriate count or mass density, not a simple average of unequal regions. Gradient, core-shell, layered, patterned, and porous descriptions require their dimensions and fabrication information. None is assumed to arise from every overall vector.

8 Preparation and confirmation records

[0038] Each preparation record states starting substances, grades, quantities, apparatus, order of operations, temperature, pressure, atmosphere, time, cooling or curing history, isolation, purification, post-treatment, scale, and relevant storage. Distinguish actual parameters from suggested values. A route class such as melting or deposition is insufficient as the preparation instruction for an unfamiliar composition.

[0039] Metals and alloys may be investigated through documented melting, powder consolidation, electrodeposition, or vapor deposition, followed by specified thermal and mechanical treatment. The selected route must address volatilization, immiscibility, reaction, contamination, competing phases, and phase stability for the actual constituents. This is a route-selection framework, not an assertion that one process works for every support. The adjoining alloy-evidence block records a selected attributed preparation, principal-inventory accounting and a conditional transport inquiry.

Alloy evidence associated with [0039]: preparation, principal inventory and conditional transport

Attributed alloy teaching (ESTABLISHED report; fitted quantities remain qualified). Vaidya et al. prepared nominally equiatomic CoCrFeNi and CoCrFeMnNi from 99.99 wt.% metal pieces: evacuate the arc-melting chamber to 10^-5 mbar, purge with purified Ar, remelt four or five times and homogenize at 1473 K for 50 h. XRD/EBSD reported FCC phases; atom-probe measurements used 60 K, 250 kHz and 20 pJ. Selected quaternary atom-probe volume contained 0.0087 at.% carbon; the quinary analysis also detected C/N/O. For diffusion, polish 8-mm-diameter, 1-mm-thick cylinders, deposit diluted acidic 63Ni solution, dry, seal under 5N Ar, anneal at 673, 723, 873, 1073 or 1173 K, then reduce diameter by 1 mm. Serial sections (~0.3 micrometres) were scintillation-counted. B-regime P=s*delta*D_gb (m^3/s) and C-regime D_gb (m^2/s) require different analyses; 723-K quaternary data require transition correction. Fitted Ni grain-boundary transport ordering crosses near 800 K. Main text omits charge masses, melting energy, homogenization atmosphere/cooling and tracer solution concentration/volume; annealing times and parameters depend on Supplementary Table S1. Primary archived XML, Methods and Results; Supplementary Table S1 is a further dependency.

Acquisition and candidate boundaries. Main XML text and captions were reviewed; original bytes have a recorded hash. Source figure pixels and the supplement were not inspected in this review. The article permissions state CC BY 4.0 with third-party credit-line exceptions. No source original or graphic is republished. Keep an untraced alloy, dried tracer-coated coupon, annealed coupon and sampled sections as distinct boundaries. A candidate at any of these stages needs independently supported retention/loss and isotope-state accounting. Acid, carrier/counterion residues and applicable decay products remain unresolved; a nickel isotope label does not create another elemental label. No new specimen is represented as measured.

CALCULATED principal targets and whole-inventory transfer. Choose increasing principal supports S_P=(Cr,Fe,Co,Ni) and S_P=(Cr,Mn,Fe,Co,Ni), respectively. Their ideal atom targets r_target=(1/4,1/4,1/4,1/4) and (1/5,1/5,1/5,1/5,1/5) each sum to one. Equal positive integer counts m give totals 4*m and 5*m. These are nominal principal-inventory targets, not exact source atom counts, an FCC-cell decoration or a whole-coupon assay. For an independently defined finite actual atom inventory, let N_P be the principal subtotal, N the whole count, f=N_P/N and r_i=N_i/N_P. For N_P>0, N>0, actual whole fractions are x_i=f*r_i within S_P; additional constituents collectively occupy 1-f. If the additional subtotal is positive, its separately normalized fractions y_j give

x_j=(1-f)*y_j. The target r_target is not substituted for measured r_i, and f is not assumed one. Unknown constituents and unknown f remain unknown.

CALCULATED stipulated additional-carbon inventory. A separate hypothetical model uses m>0 principal counts for each of Cr,Fe,Co,Ni and c>=0 carbon atoms. If c=0 the canonical support has four labels. If c>0 the increasing support is (C,Cr,Fe,Co,Ni) with vector (c,m,m,m,m)/(4*m+c); it has five positive coordinates summing to one. This is not the manganese-containing principal target or a source-carbon assay. For c=m the model gives five equal fractions without establishing any alloy phase. For a mathematical below-grid witness, let L=10^20, c=4 and m=L-1. Then total N=4*L, carbon fraction is 1/L and each principal fraction is (L-1)/(4*L). The primitive denominator is 4*L because gcd(4,L-1)=1. This rational vector lies below the original 10^-19 positive floor for carbon, but remains inside the unrestricted rational domain. No coupon size, successful preparation, trace assay or source-carbon concentration is inferred from the witness.

Mass and isotope conventions. Principal-only ideal mass coordinates require resolved positive finite isotope-qualified effective masses M_i: v_i=r_target_i*M_i/sum_j_in_S_P(r_target_j*M_j). Their denominator differs from a whole-specimen mass allocation including additional constituents. A whole-inventory conversion uses its actual atom fractions and the same justified allocation convention. No numerical masses, isotope proportions, mass equality, tracer amount or residue-free product are assigned here.

Present alloy record operations. The two principal supports, ideal targets, declared basis and additional-carbon model are selected and assigned here. Structural fields retain attributed context, with actual candidate phase, site occupation, boundaries, segregation and spatial inventory UNSUPPORTED. Preparation records the attributed operations, while candidate applicability, missing operating details and successful implementation are PROPOSED and unperformed. Measurement fields retain attributed context and distinguish counting from full uncertainty; actual candidate retained composition, isotope populations and transport outcomes are UNSUPPORTED. Property fields record the following PROPOSED inquiry; actual candidate outcomes UNSUPPORTED. These record operations do not melt, homogenize, coat, anneal or measure a new alloy.

PROPOSED transport inquiry and credible alternative. Hold independently identified feed lots and nominal principal target fixed while comparing two declared conditioning histories. Record the actual thermal path, specimen stage, chemical retention, phase, grain size, boundary character, segregation and isotope-tracer history before interpreting a change. Prespecify one compatible transport observable, temperature, sampling geometry, penetration-profile treatment and fitting regime. A regime-dependent product and a diffusivity cannot be compared as though their units and physical meanings coincide; resolve any conversion or transition correction first. Under supported matched conditions and a valid population-specific analysis, increased mobility in the analyzed boundary population motivates a conditional greater-transport-coefficient hypothesis. A change in fast-path area alone may alter flux or profile amplitude without increasing that fitted coefficient. Lower or unchanged transport is a separate credible alternative if trapping/barriers increase or the relevant pathways remain equivalent. Characterize those alternatives and retain uncertainties, calibration, replicate/sampling choices and a comparison resolution. No new curve, crossover, magnitude or universal effect is inferred, and no comparison has been performed.

UNSUPPORTED alloy extent. Actual exhaustive composition, confirmed phase/occupancy, complete isotope inventory and new property response remain unprovided. Principal-element equality does not certify exact whole-inventory equality or cardinality; unidentified residuals do not establish physical absence. The restricted models do not supply preparation for every non-equiatomic alloy, added element, phase, processing history or later use. Record construction and attributed source support establish no applicant contribution, novelty finding, earlier entitlement or universal defensive coverage.

[0040] Inorganic and ceramic candidates may use documented solid-state, solution, hydrothermal, solvothermal, deposition, or precursor-conversion routes. The actual precursors, stoichiometry, atmosphere, thermal schedule, isolation and phase-confirmation steps are necessary. Molecular candidates require a defined reaction or

assembly and purification. Polymers require a defined polymerization, incorporated composition and forming history. Composites require constituent preparation, mixing or assembly, consolidation, and interface controls. The following selected molecular, polymer and composite records supply attributed operations, present composition accounting and conditional inquiries; candidate realization and broader applicability remain unresolved.

Molecular evidence associated with [0040]: identity, crystallization and accounting

Attributed molecular teaching (ESTABLISHED report). Broadhurst et al. report freshly preparing saturated carbamazepine solution (0.1125 g in 25.6257 g ethanol; reported water 0.03%), then gravity-filtering undissolved material. Quoted grades were CBZ >=98.5% and ethanol >=99.8%. A 3 microlitre aliquot went onto a Quantifoil R2/2 grid plasma-treated for 45 s. After 20-180 s at 298 K and 21% humidity, pressure-assisted blotting through filter paper over a water-aspirator-connected Buchner flask preceded immediate liquid-ethane vitrification. CryoTEM/3DED used 200 kV and a 100 K holder. Selected 30 s crystallites included dihydrate and forms II, III and IV; only dihydrate was identified at 180 s. These are sampled-crystal observations, not measured bulk phase fractions. Supplemental powder-diffraction analysis warns that the 30 s assignment is nonunique. Primary article, Experimental/Results; supporting information, Powder Diffraction, sup-4-sup-5.

Identity and acquisition. Carbamazepine, C15H12N2O, CAS 298-46-4, has standard connectivity identifier InChI=1S/C15H12N2O/c16-15(18)17-13-7-3-1-5-11(13)9-10-12-6-2-4-8-14(12)17/h1-10H,(H2,16,18) in the NIST identity record. This identifies connectivity rather than a crystal form or specimen purity. Official article/SI text and the identity block were inspected; original molecular source bytes and graphics were not acquired or verified. The main article states CC BY 4.0; a separate SI grant is unverified. The SI header says IUCrJ (2022), 9, whereas the main record is 2021, 8; the header does not establish a later publication date. The powder preparations are separate from the grid specimens.

CALCULATED formula-unit accounting. On ordered ideal atom support (H,C,N,O), the unsolvated unit has counts (12,15,2,1), total 30, and target (2/5,1/2,1/15,1/30). An ideal dihydrate unit has counts (16,15,2,3), total 36, and target (4/9,5/12,1/18,1/12). Each vector is positive and sums to one. Distinct unsolvated crystal forms retain the same vector; packing is a separate field. For m counted unsolvated molecular units and q counted water units, m>0 and q>=0 integers, the count vector is (12*m+2*q,15*m,2*m,m+q), total 30*m+3*q. This defines a count-conserving molecular/water inventory, not a stable hydrate for every q/m or an asserted position for water. q=2*m gives the ideal dihydrate accounting. Solvent, support grid, other residues and impurities are excluded from this stipulated subinventory, not declared absent from a real deposit. Numerical mass conversion requires qualified masses and a consistent state/allocation convention.

Present molecular record operations. The support, ideal formula-unit basis, unsolvated target and separate hydrated target are assigned here. The structural field records the connectivity identifier and distinguished hydration/form labels (ESTABLISHED attribution), with any new candidate structure, purity and phase proportions UNSUPPORTED. Preparation records the attributed local operations, while applicability to a new batch or isolated bulk minor phase is PROPOSED and unperformed. Measurement fields retain source observations as attributed; actual candidate inventory, representative phase fractions, classification uncertainty and warming/storage behavior are UNSUPPORTED. The property field records the following PROPOSED comparison; actual candidate outcomes UNSUPPORTED. Record construction does not prepare a molecular specimen.

PROPOSED occurrence comparison and alternative. Define an independently supported short-hold and longer-hold comparison using matched solution/lot, environmental history and quench procedure. Prespecify region selection, number of grids, independent preparations, crystallite-selection rule, diffraction classification, unknown class and uncertainty. Report identified non-CBZ crystals separately and exclude them from the prespecified occurrence denominator. Let n be the remaining selected population of classified CBZ and unresolved crystallites, a the classified unsolvated CBZ count and u the unclassified count, with 0<=a<=a+u<=n. For n>0 report classified occurrence a/n and the unresolved-count interval [a/n,(a+u)/n] under that sampling convention. For n=0 report no denominator rather than a zero fraction. These are selected-crystallite count

quantities, not bulk mass or volume phase fractions. A decrease in occurrence is a conditional hypothesis; persistence or increase is a credible separate alternative if nucleation, transformation or sampling differs. Define the comparison resolution before evaluating a difference. No new comparison, mechanical, electrical or solubility result is performed or inferred.

UNSUPPORTED molecular breadth. Whole-deposit composition and impurity budget, a representative bulk phase fraction, pure bulk isolation of every minor form, retained hydration after warming and complete physical scope of claims 1/174/176 remain unresolved. Molecular identity, local crystallization and field curation do not establish every molecule, hydrate, ratio, processing method or later use, nor an applicant contribution, novelty finding or earlier entitlement.

Polymer evidence associated with [0040]: stereochemistry, processing and accounting

Attributed polymer teaching (ESTABLISHED report; derived crystallinity CALCULATED). Su et al. compared PLLA 4032D with a 1:1 PDLA/PLLA mass blend. Overnight vacuum drying at 40 degrees C preceded melt-blending at 190-220 degrees C/15 rpm, cooling to 20 degrees C, granulation, pressing at 230 degrees C/15 min and cooling to 20 degrees C. DSC used 10 mg under nitrogen, 10 degrees C/min over 30-250 degrees C, with 3-minute holds at 250 and 30 degrees C; second-heating results were reported. PLLA/blend melting peaks were 171.2/227.7 degrees C; blend SC crystallinity was estimated at 43.7%. Tensile testing at 20 degrees C, 20 mm/min, ISO-527-2, at least five tests gave strength/modulus 52.98 MPa/1.13 GPa versus 83.70 MPa/2.04 GPa. PDLA Mn/PDI are assigned "1.53 and 1.7 x 10^5 g/mol, respectively", an unresolved dimensional inconsistency. Batch masses, vacuum/press pressures, residence time/cooling rate, optical purity/end groups and complete retained inventory remain unspecified. Primary archived XML, sections 2.1-2.3.1/2.3.3 and 3.1/3.5.

Acquisition and specimen state. Ordinary public XML acquisition retained the primary article, whose license states CC BY 4.0. Native figure pixels, a complete PDF and separate supplements were not inspected. No source-assignment correction is invented. The DSC-derived crystallinity is source calculation rather than a measured exact phase fraction. Second-heating state does not automatically describe an as-pressed mechanical-test specimen, and a crystal-type assignment is not an exhaustive inventory. Mn/PDI even if resolved do not specify a full molecular-weight distribution. This is attributed third-party work; no new specimen is represented as measured.

CALCULATED ideal repeat and conditional finite-chain accounting. The constitutional repeat -O-CH(CH3)-C(=O)- contains C3H4O2. Its ideal repeat-inventory atom fractions on (C,H,O) are (1/3,4/9,2/9). The two stereochemical repeat types share this bookkeeping while chain stereochemistry and packing remain separate identity fields. For an expressly hypothetical linear chain with only hydroxyl/carboxyl termini, n>=1 repeats give C3n H(4n+2) O(2n+1), total 9n+3 atoms. For R repeats distributed over K such chains, R>=K>=1 integers, totals are (3*R,4*R+2*K,2*R+K) and 9*R+3*K. Divide each by that total for a normalized ideal atom vector. Termini are conditional assumptions, not source-polymer endpoint identities. Other termini, initiators, residuals, additional elemental labels or degradation need their own inventory. The infinite-repeat vector is a limiting model, not a finite-chain assay or complete product composition.

CALCULATED feed and chain-number distinction. A nominal two-polymer feed mass target assigns PLLA/PDLA component fractions (1/2,1/2), distinct from elemental fractions, repeat populations and final retained composition. With independently justified pure-polymer masses and resolved number-average molar masses M_L and M_D, equal feed masses imply PLLA chain-number fraction M_D/(M_L+M_D) and PDLA fraction M_L/(M_L+M_D). No numerical chain-count conversion uses the unresolved source assignment. Two stereochemical polymer component labels do not make elemental k=2; the stipulated ideal repeat support has three elements. Neither the mass ratio nor the repeat formula supplies chain-length/sequence distributions, covalent branching or a specimen-wide impurity budget.

Present polymer record operations. The ideal repeat basis and vector, separate equal component-mass target and conditional chain model are assigned here. The structural field records chain types and repeat connectivity as attributed context, while actual candidate incorporated ratios, sequence/molecular-weight distributions, endpoints, crystallinity and architecture are UNSUPPORTED. Preparation retains attributed forming operations, with completed candidate implementation PROPOSED and unperformed. Measurement fields distinguish source reports/calculations from actual candidate inventory, test-state phase fractions and mechanical outcomes UNSUPPORTED. The property field records the following PROPOSED comparison; actual candidate outcomes UNSUPPORTED. These record operations do not polymerize chains, form a film or establish all limitations of claim 177.

PROPOSED mechanical inquiry and credible alternative. Hold independently identified lots and the nominal component-mass ratio fixed while comparing two declared thermal histories. Define actual histories, retention, molecular-weight/end-group evidence, phase fractions in the mechanical-test state, porosity, conditioning, geometry, local strain and a modulus-fitting convention before interpretation. Under matched justified covariates, greater effective crystalline junctions motivate a conditional higher-modulus hypothesis. A separate credible alternative is unchanged or lower modulus if chain shortening, voids or ineffective crystallization offsets that change. Source control/blend results are not this fixed-feed comparison. Record calibration, replicates, sampling, uncertainty and an a priori comparison resolution; modulus does not substitute for strength, ductility or toughness. No new comparison, magnitude, significance, universal response or covalent-branching result is asserted.

UNSUPPORTED polymer breadth. Actual full retained inventory, incorporated ratios, sequence and molecular-weight distributions, network/end-group identity and new candidate properties remain unprovided. This selected teaching, conditional accounting and record construction do not supply every polymer, arbitrary elemental support/ratio, isotope/particle state, forming route or later use. Complete physical claim scope, applicant contribution, novelty, inventorship and earlier entitlement remain unresolved.

Composite evidence associated with [0040]: constituents, preparation and accounting

Attributed composite teaching (ESTABLISHED report). Nguyen et al. describe 0.72 mass% MWCNT in stoichiometric DGEBA/polyoxypropylenetriamine epoxy, using a commercial 1 mass% resin dispersion without UV stabilizers/additives. Mix with curing agent by magnetic stirring for 1 h, degas 1 h at room temperature, and bar-draw on PET to approximately 150 micrometres. Toluene was used; its quantity is unspecified. Cure three days at 24 degrees C/45% RH, then 110 degrees C/4 h in circulating air. UV exposure used 295-400 nm, approximately 140 W/m2, 50 degrees C/75% RH. Surface analysis also identified contaminants. An entangled surface network was observed, and collector SEM found neither fragments nor individual MWCNT after 4865 MJ/m2. This is method-bounded nondetection, not zero release or a quantitative detection limit. NIST author manuscript, sections 2.1-2.3 and 3.4, printed lines 87-181/344-374.

Source access and applicability. Ordinary public acquisition retained the 31-page author manuscript; the selected preparation/exposure and release text was inspected, including native pages 5-7/19-20. The NIST bibliographic record identifies the 2017 study. No complete supplement, graphical verification, final-publisher comparison or broad reuse grant is asserted. Source originals and graphics are not republished. Batch formulation, solvent/lot details, quantitative collection limits and specimen-wide retained inventory remain incomplete; justified existing evidence may support them without a categorical new-experiment requirement.

CALCULATED loading representation and conditional balance. The nominal 0.72 mass% label converts to 9/1250; its arithmetic representation does not convert a rounded formulation statement into an exact assay. Loading is a constituent mass coordinate, not an elemental atomic fraction or a volume fraction. A distinct ideal dry-input model assumes the dispersion contains exactly 1/100 MWCNT by mass, other stock mass is resin, curing agent is the only additional dry input, no loss occurs, and the target MWCNT mass coordinate is exactly 9/1250. For total model dry mass M>0, the dispersion mass is (18/25)*M and curing-agent mass is (7/25)*M. Thus modeled dry categories (MWCNT,resin,curing agent) have fractions (9/1250,891/1250,7/25), which sum to one.

This is a conditional conservation calculation, not the source's actual batch masses, a stoichiometric cure prescription or retained cured-product assay. Additional components, solvent retention or losses invalidate that three-category closure unless separately accounted. Category coordinates are not outputs of the elemental composition generators, and no elemental fractions are calculated from unspecified component chemistry.

Present composite record operations. The constituent-loading basis and its nominal representation, plus the separate conditional input model, are recorded here. The structural field identifies the attributed constituents and film context, with actual candidate dimensions, orientation, interfaces and dispersion UNSUPPORTED. Preparation retains the reported operations as ESTABLISHED attribution, while candidate applicability, completed formulation and retention are PROPOSED and unperformed. Measurement fields retain the method-bounded source observations; actual candidate complete inventory, morphology, interfacial properties and release measurements are UNSUPPORTED. The property field records the following PROPOSED inquiry; actual candidate outcomes UNSUPPORTED. A nominal loading and conservation model do not certify claim 1's entire closed elemental inventory or successful cure.

PROPOSED stress comparison and alternative. Prepare or rely on independently supported, separately characterized composite and neat-matrix controls. Define UV-only, stress-only, combined and unexposed histories using calibrated dose, temperature, humidity, specimen dimensions, water/mechanical conditions, order and conditioning. Prespecify collection boundaries, blanks, recovery, sampling schedule, replicates, uncertainty and detection/quantification limits. Let m_CNT,0 be independently supported initial embedded nanotube mass, m_0 initial whole-specimen mass, and m_rec recovered CNT mass, including CNT within fragments under a stated, supported allocation convention. Gross CNT-containing-fragment mass is a separate quantity. For positive denominators report r_CNT=m_rec/m_CNT,0 and r_whole=m_rec/m_0 separately; they obey r_whole=r_CNT*(m_CNT,0/m_0) for consistent boundaries. m_rec is recovered material rather than automatically all material released. Apply an independently justified correction only when supported, retaining its uncertainty and missed collection. Distinguish nanotube-containing fragments from individual tubes.

PROPOSED mechanism and credible alternative. Additional stress may increase collected release if it disrupts an exposed network or remaining attachments. Unchanged or lower collected release is a separate credible alternative if the network/attachments persist, the treatment changes fragment form or collection differs. Compare microscopy, matrix chemical changes and interfacial evidence before assigning causation; a release difference alone cannot isolate a mechanism. Prespecify resolution and order controls. No quantitative response, numerical zero, universal sign, new applicant result or successful comparison is asserted.

UNSUPPORTED composite breadth. Complete specimen elemental inventory, exact retained loading, quantitative dispersion/interface states and candidate response remain unresolved. Surface measurements cannot supply an exhaustive bulk assay. This selected teaching and record do not enable every matrix, filler, ratio, architecture, isotope, particle system, manufacturing method or use. Candidate contribution, novelty, earlier entitlement and universal defensive coverage are not established.

[0041] An unperformed recipe is a PROPOSED example stated prospectively. It reports no fabricated yield, measurement, or success. If a step depends on discovering an unavailable precursor, synthesizing a hypothetical element, stabilizing an unobserved nucleus, or establishing an unknown reaction, that dependency is an unresolved research problem rather than routine implementation.

[0042] Confirmation is selected to distinguish the target from realistic alternatives. Composition analysis alone may not identify a polymorph; diffraction alone may not detect a trace constituent; nominal instrument resolution is not a validated limit of detection in the actual matrix. Record calibration, reference materials, uncertainty, sampling, specimen history, and the method's detection and quantification limits. A nondetection is not proof of absence. IUPAC limit of detection

[0043] A tiny exact fraction is a mathematical target until preparation and measurement support a physical assertion. Experimental tolerance is specified separately, for example an interval or uncertainty with confidence

convention. No instrument is claimed here to distinguish compositions separated by 10^-19 fraction. At trace levels, isotope signatures, contamination control, adsorption, sampling statistics, and atom-count discreteness can dominate the interpretation.

[0044] An established teaching for comparison is TiO2(B) nanosheets reported by Xiang and colleagues in 2013. Their nominal formula has Ti:O = 1:2; the B phase, nanosheet morphology, and exposed surface are additional features. Their paper reports a solvothermal preparation from titanium tetrachloride, water, and ethylene glycol, at 150 degrees C for six hours, with isolation and washing and diffraction/microscopy characterization. This is an attributed source summary; the complete methods remain necessary, and no applicant repetition is claimed. Xiang et al., Scientific Reports 3, 1411

[0045] The TiO2 formula corresponds to rational atomic fractions (1/3,2/3). Its inclusion demonstrates why the unrestricted rational domain is needed and why a decimal grid alone is incomplete. It does not establish exact product purity, all titanium oxide polymorphs, every Ti/O ratio, or every process and use. Reproducing old literature is not represented as creating a newly available teaching.

9 Isotopes and nuclear states

[0046] An isotope-resolved constituent is identified by atomic number Z, mass number A, neutron count N = A-Z, and nuclear-state designation. For the elemental vocabulary, Z is 1 through 118 or an explicitly hypothetical 119 through 138. A is an integer at least Z. This parameterization includes candidate identifiers beyond observed nuclides; it does not assert that every (Z,A) nucleus is bound, observable, stable, or preparable.

[0047] For each element i, let y_i,a,s be the conditional atom-count fraction of its atoms assigned to isotope a and nuclear state s in the specified inventory, with y_i,a,s >= 0 and sum_a,s(y_i,a,s) = 1. This conditional distribution remains atom-count based regardless of the composition basis selected for the material. Let a_i denote the elemental atomic fraction and w_i the elemental mass fraction; x_i of the selected composition record equals a_i on an atomic basis or w_i on a mass basis. The whole-inventory atomic fraction of a nuclide-state entry is a_i y_i,a,s. Where mass conversion is supported, specify qualified masses mu_i,a,s per counted atom under a consistent, expressly stated mass-allocation convention, including the relevant isotope, nuclear/electronic state and any precision-dependent binding or allocation qualification. Define M_i = sum_{a,s : y_i,a,s > 0}(y_i,a,s mu_i,a,s), with positive finite M_i, summing only over populated entries. For each such entry define the derived conditional mass fraction r_i,a,s = y_i,a,s mu_i,a,s/M_i. The whole-inventory nuclide-state mass fraction is w_i r_i,a,s. Entries with y_i,a,s = 0 have r_i,a,s = 0 without requiring a mass value for an unpopulated entry. Each element's nuclide-state atomic fractions sum to a_i and its nuclide-state mass fractions sum to w_i. Elemental conversions use w_i = a_i M_i/sum_j(a_j M_j) and a_i = (w_i/M_i)/sum_j(w_j/M_j) under the same convention. Unknown masses for positively populated entries prevent numerical mass conversion; they are not supplied by standard atomic weights, placeholders, blanks or assumed zeros. An atomic target and its conditional atom-count distribution can still be described without asserting an unavailable mass conversion. Natural, enriched, depleted, pure-isotope and mixed-isotope targets can be recorded. These equations are compositional accounting, not proof of state existence, availability, preparation, persistence or measured fractions; lifetime, decay products, observation time and preparation history remain separate requirements.

[0048] Ground states, isomers, resonances, and evaluated excited levels are not collapsed into one identifier. NUBASE2020 and AME2020 provide dated evaluation resources for nuclear states and atomic masses. Their measured, estimated, and unobserved entries must retain those distinctions. The 2020 edition's experimental cutoff is not a statement that it contains every nuclide known in September 2026. NUBASE2020, AME2020 Part I, AME2020 Part II

[0049] An archival isotope record retains the source edition, nuclide identifier, state, existence-status evidence, property-estimation markers, units, uncertainties, half-life, decay modes, and retrieval date. A systematics marker on a mass is not by itself an existence-status classification. LiveChart API records likewise require their evaluation

provenance; its guide identifies the underlying snapshots. No complete current nuclide export is represented as included in this draft. IAEA LiveChart API guide

[0050] A changing radioactive inventory is a function of time and preparation history. Its composition vector is identified at a specified time or over a stated averaging period. Daughter products enter the support when included in the denominator. A short-lived nuclear species is not assumed to persist through an ordinary bulk preparation or measurement.

10 Subatomic and emergent particle systems

[0051] Established elementary-particle categories are distinguished from composite particles, nuclei, material excitations, and hypothetical particles. The Standard Model register includes six quark flavors, six lepton flavors, gauge bosons, and the Higgs boson, with antiparticle and charge-state conventions where applicable. A category listing does not teach arbitrary stable macroscopic mixtures of those particles. CERN Standard Model

[0052] A particle-system record identifies species and experimental status, quantum numbers, charge, mass where established, energy/momentum distribution, population or occupation convention, confinement or host, temperature or non-equilibrium state, lifetime, interaction model, preparation, and observation. Quarks within a nucleus, that nucleus, and the atom containing it are not independently added to an elemental count unless a physical construction and separate accounting basis are defined.

[0053] Composite particles such as protons, neutrons, mesons, and nuclei require their own identities and preparation or host descriptions. Photons and other field excitations require frequency or energy and occupation conditions rather than an unqualified chemical mass fraction. Antimatter-containing candidates require containment and annihilation considerations; no ordinary stable alloy is inferred from a particle list.

[0054] Phonons, magnons, excitons, polarons, and related quasiparticles are host-dependent excitations. Their record specifies the underlying material, excitation mechanism, field or carrier conditions, lifetime, and observable. They do not extend the periodic table and are not interchangeable elemental feedstocks. IUPAC quasiparticle definition

[0055] Hypothetical classes include gravitons, axion-like particles, sterile-neutrino candidates, dark photons, magnetic-monopole candidates, supersymmetric partners, and other model-defined entities. Each is an UNSUPPORTED candidate unless evidence and a preparation/observation route are supplied. A model identifier, parameters, conservation constraints, and search status must accompany it. No finite list can establish all future theoretical particle species or enable their material combinations.

[0056] PDG2026 tables and the versioned PDG API are source locators for particle identities and properties. They include experimental searches and limits as well as established particles. Inclusion in a table is not proof of discovery. The cited 2026 listings have a stated cutoff, and updates need their own date. PDG 2026 tables, PDG API

11 Conditional properties and alternative responses

[0057] A property scenario identifies a baseline material and state, a controlled change, the expected response with mechanism and conditions, a credible alternative response with its mechanism, a discriminating test, and evidence status. The baseline and alternative are separate conditional embodiments or hypotheses. They are not asserted simultaneously for the same specimen under identical conditions.

[0058] Mechanical scenarios include stiffness, shear modulus, yield and ultimate strength, torsional response, fracture toughness, ductility, hardness, fatigue, creep, wear, and stress relaxation. These quantities have distinct definitions. A proposed strength increase cannot be inferred from stiffness alone, and specimen geometry can

affect torque resistance. Tests specify direction, temperature, strain rate, environment, conditioning, and failure criterion.

[0059] Electrical scenarios include DC and AC conductivity, resistivity, carrier type and density, mobility, dielectric response, capacitance, breakdown, piezoresponse, thermoelectric response, and superconducting behavior where supported. Frequency, field, temperature, contacts, and geometry are specified. Negative differential resistance under driven conditions is not treated as negative passive DC conductivity. No assertion of superconductivity or a transition temperature is made for an unidentified composition.

[0060] Additional scenario fields include thermal conductivity and expansion; heat capacity and transitions; optical absorption, emission, refractive and nonlinear response; magnetic ordering, susceptibility and coercivity; corrosion, oxidation and chemical stability; catalytic selectivity and rate; diffusion, sorption, permeability and wetting; radiation response; and biological interaction where relevant. Each requires a defined observable, conditions, model or measurement, and local evidence label. A category name is not a predicted performance value.

[0061] A scenario may propose increased, decreased, approximately unchanged, threshold, nonmonotonic, hysteretic, anisotropic, transient, or sign-changing response, only with an explicit baseline and physically meaningful definition. An unchanged result means within stated resolution or tolerance, not mathematically identical under every test. These response categories enumerate research questions; they do not establish all future unexpected results.

[0062] ESTABLISHED mechanism example with PROPOSED extrapolation: in suitable precipitation-strengthened alloys, fine precipitates can impede dislocations; altered aging can coarsen precipitates and reduce hardness. For a new alloy record, specify its phases and aging path, then test hardness and microscopy rather than asserting every added element strengthens it. NBS study of 2024 aluminum alloy processing

[0063] ESTABLISHED mechanism example with PROPOSED extrapolation: a cooling path in a glass-forming system may suppress crystallization, while a different path or geometry permits nucleation and growth. A candidate record specifies the melt and measured thermal history and distinguishes glass from crystalline product using diffraction and calorimetry. This does not identify glass-forming behavior for every support. Johnson et al., metallic-glass formation The adjoining glass-evidence block records a selected attributed preparation, principal-inventory accounting and a conditional thermal-path inquiry.

Glass evidence associated with [0063]: preparation, phase identity and conditional response

Attributed glass teaching (ESTABLISHED report). Na et al. report nominal Ni68.6Cr8.7Nb3P16B3.2Si0.5 (atomic percentages), using Ni/Cr/Nb/P/B/Si feedstock grades 99.995/99.996/99.95/99.9999/99.5/99.9999%. Inductively alloy elemental charges in silica under stated high-purity argon. For the demonstrated 17-mm rod, remelt the ingot with dehydrated boron-oxide flux in a 1-mm-wall silica tube at 1350 degrees C under that atmosphere; hold approximately two hours, then rapidly quench into stirred, room-temperature water. Cross-sectional CuK-alpha diffraction and calorimetry at 20 K/min support the reported amorphous assignment. Nominal feed ratios and diffraction do not certify exhaustive product inventory or zero crystallinity. Batch/flux quantities, dehydration conditions, gas pressure/flow, ramp, transfer delay, cooling trajectory and analytical detection limits remain unspecified. Author-repository article, Materials and Methods and Figure 3; its supplement retains additional context.

Access and diagnostic scope. Original primary and supplemental PDF bytes were acquired with recorded hashes from the official author repository. Selected native pages were inspected; complete graphics verification and source replication are not asserted. Source originals and graphics are not republished, and no broad reuse grant is assumed. Keep a reported diagnostic classification, its measurement history and a new candidate's unconfirmed identity as different fields. Nondetection at a method's resolution does not establish an exact

crystalline fraction zero or a complete whole-specimen elemental assay. No new specimen is represented as measured.

CALCULATED nominal principal inventory. On increasing elemental support (B,Si,P,Cr,Ni,Nb), the ideal source-nominal principal atom target is r_target=(4/125,1/200,4/25,87/1000,343/500,3/100). Equivalently the integer vector (32,5,160,87,686,30) has total 1000 and gcd one; every coordinate is positive and the fractions sum to one. Compatible ideal inventories multiply all six counts by any positive integer m, with total 1000*m. The denominator 1000 divides 10^19, so this nominal vector is compatible with that decimal grid as well as the unrestricted rational domain. A count-compatible target does not assign atoms to a glass network, a rod size, exact source charge masses or an assay.

Whole-product and mass boundaries. Let N_P>0 count atoms in the six-principal-element subset and N>0 count all atoms in the independently declared retained specimen; let f=N_P/N. Its actual principal ratios r_i and whole fractions x_i obey x_i=f*r_i, while other constituents occupy 1-f. The source-nominal r_target does not supply measured r_i or f. Flux, tube/contact material, atmosphere, conditioning, surface material and unknown residues require their own retention/allocation evidence instead of a default zero. Principal-only ideal mass conversion v_i=r_target_i*M_i/sum_j_in_S_P(r_target_j*M_j) requires qualified positive finite masses and a consistent allocation; it is not a complete specimen mass assay. No numerical isotope populations or mass fractions are supplied.

Present glass record operations. The six-label principal support, ideal basis and normalized target are assigned here. Structural fields retain attributed diagnostic context; actual candidate local order, phase assemblage, crystalline fraction and spatial inventory are UNSUPPORTED. Preparation retains attributed operations; their applicability to a new candidate, complete retained composition and successful implementation remain PROPOSED and unperformed. Measurement fields distinguish reported diagnostics from candidate assays and resolutions, with actual candidate retained inventory, isotope state and phase/property outcomes UNSUPPORTED. The property fields record the following PROPOSED inquiry; actual candidate outcomes UNSUPPORTED. This record construction does not melt, flux, quench or characterize a new rod.

PROPOSED thermal-path inquiry and credible alternative. Compare independently supported specimens with the same nominal principal target, defined geometry and material lots under two declared cooling histories. Measure or otherwise justify the actual temperature histories at relevant positions; a bath description or modeled critical cooling rate is not an observed centerline trajectory. Record melt/flux contact, initial state, losses, nucleation/contamination evidence, conditioning and the specimen region examined. Prespecify diffraction/calorimetry protocols, reference controls, calibration, detectable-crystalline-signal criteria and uncertainty. Under independently supported matched conditions, a path spending less time in the relevant crystallization window motivates a conditional lower-crystalline-signal hypothesis. An unchanged signal is credible if both paths meet the same diagnostic boundary; a greater signal is credible if the actual local path or heterogeneous-nucleation population instead differs in a crystallization-favoring way. Investigate those differences before assigning cause. Retain spatial sampling, replicates and a comparison resolution. No complete phase fraction, new critical diameter/rate, universal property sign or performed comparison is asserted.

UNSUPPORTED glass extent. Complete actual elemental/isotope retention, quantitative crystalline fraction, network identity and new response remain unprovided. An elemental vector alone does not select amorphous versus crystalline arrangement. The selected operation does not demonstrate glass formation for arbitrary elemental supports, proportions, dimensions, cooling histories or uses. Source-reported diagnostic context and present accounting supply no applicant experiment, contribution, novelty finding, earlier entitlement or universal physical/legal coverage.

[0064] Source observations, derived defect models and unperformed response proposals are separate evidence categories. The adjacent rutile block supplies particular preparation, diagnostic and assay contexts plus CALCULATED ideal-host accounting. A source-associated mechanism does not establish every carrier pathway,

composition, structure or response. Proposed comparisons require a stated technical basis and appropriately separate carrier-dynamics and reaction evidence; no categorical requirement for a new experiment or universal opposite outcome is imposed. The adjoining phase-specific block records a particular attributed teaching, present ideal-host targets and evidence fields, and a conditional optical inquiry; its candidate realization remains unresolved.

Phase-specific evidence and worked record associated with [0064]

Source account (ESTABLISHED attribution; estimates and causal interpretations remain qualified).

Zhang et al. describe 2 g Ti (>=99.5 wt%; stated average particle size 0.25 mm) stirred with 40 mL 28 wt% aqueous ammonia and 80 mL 30 wt% peroxide at 0-5 °C for 3 h. Add 100 mL 2 wt% F68 solution; stir 2 h, rest 12 h, heat 70 °C/0.5 h, add 100 mL ethanol, centrifuge, and calcine 200 °C/2 h. The subsequent reduction feed is still called H4TiO5. Hydrogen reduction reportedly gives Ti4O7 at 850 °C/5 h or 860 °C/4 h, supported by diffraction. Diffuse-reflectance observations concern UV and visible/near-infrared response, with a proposed vacancy interpretation. Hydrogen pressure/purity/flow, ramps, cooling/passivation and exhaustive retained inventory remain unspecified. This account does not resolve the post-calcination precursor or establish elemental exclusivity, measured conductivity or a candidate optical assay. Primary DOI.

Access and attribution boundary. Source access is reader-only: official publisher HTML and three-page PDF text were inspected, but no original article bytes or PDF hash were acquired and no usable page-pixel/graphic inspection was completed. No permissive reuse grant is verified.

CALCULATED ideal target and denominators. For this record choose recognized Ti/O support on a stipulated occupied-atom basis, not a feed, site or whole-product assay basis. The target is Ti4O7 = TiO_(2-delta) with delta = 1/4, x_Ti = 4/11 and x_O = 7/11. Both fractions are positive and sum exactly to one. Compatible ideal counts are N_Ti = 4*m, N_O = 7*m and total atoms = 11*m for any positive integer m; this assigns no specimen size or physical atom count. Against the assumed 2*N_Ti oxygen-site reference, V = m gives V/(2*N_Ti) = 1/8 missing reference sites and 7/8 occupied reference sites. These differ from the oxygen occupied-atom fraction 7/11 and do not establish a rutile vacancy map or observed site order. Vacancies are not atoms.

Under the expressly restricted assumptions of a neutral ideal host, formal O(2-) and Ti(III)/Ti(IV) only, the Ti populations are half each among Ti atoms and 2/11 each among all ideal atoms, with 3*(2*m)+4*(2*m)-2*(7*m)=0. These model populations are not measured oxidation states, carriers or additional elemental labels. A mass-basis conversion instead gives w_Ti = 4*M_Ti/(4*M_Ti+7*M_O) and w_O = 7*M_O/(4*M_Ti+7*M_O), subject to positive finite isotope-qualified effective masses under one stated allocation and compatible-unit convention. No numerical masses or mass fractions are assigned. The primitive target denominator is 11, which does not divide 10^19: this exact vector belongs to the unrestricted rational domain, not the finite decimal grid; no rounding is used to change membership.

Present record construction; physical implementation unperformed. The support, basis and normalized target are selected and assigned here. The structural field records a nominal Ti4O7 phase target (PROPOSED), while actual phase assemblage, site order, morphology, surfaces and spatial homogeneity remain UNSUPPORTED for a new specimen. The preparation field records the above sequence (ESTABLISHED attribution), with candidate applicability and conditioning prerequisites PROPOSED; no candidate preparation or successful repetition is asserted. The measurement field records candidate inventory, phase, isotope-state and optical results as UNSUPPORTED rather than treating a fraction as an assay. The property field records the conditional comparison below, with actual candidate outcomes UNSUPPORTED. These present record operations provide the claim-186 context without performing physical synthesis or demonstrating the complete physical scope of any material claim. Each field retains its stated evidence category; no unresolved preparation is recorded as successful.

Boundary-specific verification (PROPOSED). Keep the stipulated ideal host, complete conditioned product and optical specimen as separate declared boundaries. Before implementation resolve the precursor stage, route applicability and critical operating gaps rather than invent defaults; record identified inputs, equipment, actual delivery, isolation, losses, thermal/atmosphere history and conditioning. Use appropriate attributable support or supplied characterization to evaluate phase, Ti/O ratio, other retained species, surface chemistry and spatial representativeness. Record calibration, sampling, limits and uncertainty when evidence is obtained. Diffraction phase identity does not certify elemental exclusivity, and nondetection does not certify exact zero outside Ti/O. Additional retained labels require a different complete inventory; they cannot be silently deleted from the ideal two-label model. Appropriate existing evidence can support a field: no categorical new-experiment requirement is imposed for every example, but unprovided evidence remains unprovided.

Conditional optical scenario and alternative (PROPOSED). The baseline is a future conditioned TiO2 reference associated with ideal atomic fractions (Ti,O) = (1/3,2/3), not an already-measured source specimen. Compare a future candidate targeting (4/11,7/11): the complementary change is (Ti,O) = (+1/33,-1/33). This comparison changes composition and phase/structure and may change morphology and surfaces; it is not a demonstrated defect-only perturbation.

One hypothesis is increased absorptance in a preselected visible/near-infrared band if the realized electronic state supplies additional optically active transitions. Before implementation specify wavelength limits and weighting, equilibrated temperature, illumination/collection geometry, specimen thickness or areal loading, packing, calibration and uncertainty. Match or quantify surface, morphology and scattering covariates. If using spectral A(lambda)=1-R(lambda)-T(lambda), R and T must be compatible total reflected/transmitted power fractions relative to incident power for the declared optical specimen. Instrument absorbance or diffuse reflectance is not silently substituted for absolute absorptance; band and normalization choices remain unresolved until specified.

The discriminating inquiry compares independently supported phase/inventory evidence with calibrated optical measurements, retaining raw evidence and uncertainty and checking whether a response survives geometry/scattering controls. A separate credible alternative is unchanged response within a defined resolution or lower band absorptance if the realized transitions contribute weakly in that band or the changed morphology/light interaction offsets them. Its mechanism and conditions are conditional, not simultaneous opposite outcomes for one identical specimen and state. Neither a numerical magnitude nor a universal sign, source curve or newly observed unexpected effect is asserted. No candidate comparison has been performed.

UNSUPPORTED extent. No complete retained Ti/O-only specimen, measured delta, candidate preparation, phase stability, isotope distribution or property is assigned by this nominal target. It does not establish preparation of every rational composition, hypothetical element or particle system, or every property outcome. The existing broader domains remain the requested scope; this particular attributed example and record do not prove their universal physical accessibility or a universal patent bar. No applicant contribution, inventorship, priority, novelty, filing or historical inclusion is inferred from this present addition.

Oxide and defect evidence associated with [0064]: preparation, diagnostics and conditional response

Published preparation (ESTABLISHED attribution). Li et al., rutile TiO2, Methods: add TiCl4 (10 mL) dropwise under stirring to ice-water (30 mL); stir 30 min; heat 373 K within 5 min; wash resulting solid; air-dry 353 K/24 h; calcine 473/673 K/2 h (T-1/T-2). Missing evaporation/washing endpoints, ramp and atmosphere remain gaps, not operating defaults.

Rutile quantity	T-1	T-2	Source and basis
TEM size	8 nm	27 nm	Main Table 1; TEM
BET area	101 m2/g	86 m2/g	Main Table 1; BET

Rutile quantity	T-1	T-2	Source and basis
Visible H2 rate	932	372	Main Results p5/Figure 4; micromol h^-1 g^-1
UV-Vis gap	2.74 eV	2.84 eV	Supplement Table 2; optical estimate

Diagnostic basis. Published measurements retain ESTABLISHED attribution; source-derived estimates and interpretations retain CALCULATED status. OH is oxygen-species-based. Figure 1f contains T-1/T-3 positrons only. Neither measures whole-specimen elemental fractions or charge-carrier lifetimes. Size and area covary with treatment; neither a diagnostic intensity nor BET area is a calibrated count of electrically active defects.

Hydrogen assay (ESTABLISHED report). Methods: catalyst 100 mg/~100 mL 10% methanol-water; 30-min evacuation; stirred 313 K; Xe 200 W, 400-780 nm/~80 mW/cm2; nominal Pt 1 wt%, in-situ photodeposited from H2PtCl6 1 M; pH 6.5; GC/TCD. This is methanol-assisted hydrogen evolution. The loading is not an assayed retained Pt fraction, and the reported rate does not establish donor-free overall water splitting. Actual deposited Pt, averaging interval, replicate count and rate uncertainty remain unresolved; the separate optical chamber's conditions cannot replace this assay.

Unresolved source differences. The linked source review (source file: ../data/oxide_defect_source_review_2026-09-30.json) records conflicting facet, exposure-time and sample-route labels. Preserve the differences; do not silently relabel samples or select a corrected setting. Model assumptions and source interpretations are not observed charge-carrier dynamics or a universal defect-response rule.

CALCULATED ideal-host accounting. On an expressly ideal Ti/O atom-count basis, write TiO_(2-delta), with rational 0 <= delta < 2. Then x_Ti = 1/(3-delta) and x_O = (2-delta)/(3-delta); both are positive and sum to one. For N Ti atoms and integer V missing oxygen sites, delta = V/N, 0 <= V < 2N, and the ideal counts are (N,2N-V). With N = 10^20 and V = 1, the exact target is (10^20,2*10^20-1)/(3*10^20-1). With V = 2N-1, x_O = 1/(10^20+1), below the original 10^-19 floor. These are finite ideal atomic inventories on recognized Ti/O support, distinct from the source's catalyst, OH convention, feed and measurement mixtures. The near-2 limit does not establish a rutile lattice or any stable phase. Irrational delta remains symbolic and has no exact finite integer inventory.

Conditional charge model (CALCULATED, not an assay). If the ideal host is neutral, every O is formally O^2-, and only Ti^3+/Ti^4+ populations provide compensation, average Ti charge is 4-2*delta and the Ti^3+ fraction among Ti atoms would be 2*delta. This restricted model permits 0 <= delta <= 1/2; above 1/2 it would demand a fraction greater than one. Other charge states, delocalized carriers, hydrogen, adsorbates and additional species require different accounting. Source OH percentages and positron intensities cannot be substituted for delta or that Ti population. Availability, bonding, sites and structural stability are separate from normalization.

PROPOSED present verification and alternative. A future source-type comparison would retain separately characterized preparation batches and declared specimen boundaries. It would record morphology, crystallinity, area, surface chemistry, retained Pt and irradiation history before associating any defect descriptor with an optical or reaction observable. To isolate a mechanism, additional controls must address those covariates instead of treating the source T-1/T-2 contrast as defect-only. One conditional hypothesis is increased hydrogen evolution when an independently supported change reduces recombination while leaving accessible reaction pathways; a credible alternative is unchanged or lower evolution if trapping or blocked surface transfer dominates, even with greater absorption. Existing justified models or separate carrier-dynamics and reaction evidence may support the inquiry; new tests remain unperformed and their operating, calibration and uncertainty details must be specified before execution.

UNSUPPORTED extensions. No preparation, retained inventory, measured delta, carrier lifetime, assay precision, isotope-state distribution or property is assigned to the ideal trace targets. No earlier source prediction or applicant contribution is inferred from this present calculation. The selected source example and conditional

comparison do not enable all Ti/O phases, all other elements or every unexpected property. No applicant experiment, novelty determination, inventorship ruling, filing or universal patent bar is asserted.

[0065] ESTABLISHED attributed preparation and property example, with the source-specific identity and operation record below: Schoenmakers, Rowan and Kouwer report azide-functionalized polyisocyanide (PIC) 1 with bifunctional crosslinker 2a. Cold polymer/linker solutions are mixed and heated into the gel phase for a one-hour crosslinking period. Their loading comparison uses PIC at 1 mg/mL, crosslinking at 37 °C and subsequent stabilization for 11 hours at 5 °C. Each linker supplies two DBCO groups; the varied DBCO/azide ratio is a functional-group ratio. Storage modulus G′ is measured using 40 mm sandblasted parallel plates, a 500 µm gap, strain amplitude 0.04 and frequency 1 Hz; temperature-dependent measurements use a 1 °C/min ramp. Supplementary Table 1 reports the absolute moduli below, including the zero-linker control. These values show increases and decreases across different loading comparisons, not simultaneous opposite responses for one specimen under identical conditions. This is a published third-party example; no applicant repetition or universal polymer response is asserted. Crosslinking of fibrous hydrogels, Nature Communications 9, 2172 (2018), DOI 10.1038/s41467-018-04508-x.

Table associated with [0065]: reported absolute storage moduli

DBCO/azide ratio	Linker 2a concentration (µM)	G′ at 37 °C (Pa)	G′ at 5 °C (Pa)
0	0	136	<1
0.1	5.2	157	7
0.2	10.4	140	20
0.5	26.1	211	78
0.9	46.8	148	72
1	52.1	201	93
1.25	65.1	184	95
1.5	78.2	125	48
1.75	91.0	115	22
2.0	104.1	125	20

Source and quantity basis. Transcribed from Supplementary Table 1, page 4, of 41467_2018_4508_MOESM1_ESM.pdf, obtained from the primary supplementary archive. The article is licensed CC BY 4.0; this account paraphrases the authors' teaching and attributes their data. The ratio concerns DBCO and azide functional groups; it is neither an elemental atomic/mass fraction nor a measured active-bridge fraction. Source methods, molecular structures and precursor references must be consulted together; this summary is not a complete precursor synthesis instruction or whole-product elemental assay.

Observable and uncertainty. Article Figure 4 reports G′(5 °C)/G′(37 °C), whose denominator varies with loading. Its error bars are standard deviations with n = 2 or 3; the absolute-value table gives no uncertainty per row. Listed 5 °C values are not monotonic at every lower-loading step, and the largest listed value occurs at ratio 1.25. No continuous response curve, exact optimum or significance for every difference is inferred. A ratio of the listed values is not asserted to reproduce a mean of individual normalized measurements.

Model-derived interpretation. The authors estimate active-crosslink concentration from measured conversion and an assumption requiring two reactions of the same linker to form an active crosslink. They explain poorer

stabilization at excess linker through singly attached linker. This remains a source-reported interpretation with its assumptions, not a direct count of every bridge or an applicant measurement. The source describes the near-stoichiometric optimum as expected; no applicant-specific statutory unexpected-results showing is made here.

PROPOSED extrapolation. A study of another polymer, linker or processing history remains an unperformed conditional proposal. It separately identifies its material, controlled change, proposed response, mechanism and distinguishing measurements. Conversion, swelling or modulus measurements for that proposed study are not supplied by this table. G′ does not establish ultimate strength, fracture toughness, torque resistance, swelling or every mechanical property. Unsupported extensions retain UNSUPPORTED status; this example supplies no universal property rule for arbitrary indexed compositions.

Preparation associated with [0065]: identities, operations and composition basis

Identity and evidence basis. The following is an attributed account of the 2018 source's Figure 1b, main Methods and Supplementary Methods, page 1; it records reported operations, not an applicant experiment. The two neutral monomer connectivities can be written [C-]#[N+]-CH(CH3)-C(=O)-NH-CH(CH3)-C(=O)-O-T, where # denotes a triple bond, the brackets identify formal isocyanide charges, and T is the tail below. The full monomer remains neutral; this is condensed connectivity notation rather than a stereochemical SMILES specification. This connectivity notation leaves stereochemistry unresolved as explained below. The primary Figure 1 graphic and primary supplementary archive identify the specific source. This account paraphrases the CC BY 4.0 article; its definitions and quantities are printed here, rather than presumed incorporated from links.

Source monomer	Tail T, from the ester oxygen	CALCULATED neutral formula	Atoms per neutral monomer
2018 methoxy	CH2CH2-O-CH2CH2-O-CH2CH2-O-CH3	C14H24N2O6	46
2018 azide	CH2CH2-O-CH2CH2-O-CH2CH2-O-CH2CH2-N3	C15H25N5O6	51

Structural limits. These formulas are CALCULATED from the source graphs, not elemental assays. Literal CIP interpretation of the drawn solid wedges gives R at the isocyanide-bearing alpha carbon and S at the ester-bearing alpha carbon. The supplement's azide-product and formamido-precursor names instead assign S and R, respectively. This discrepancy is unresolved; no corrected stereoisomer or identity-specific reproduction is asserted. Figure 1b defines linker 2a with two N-acyl dibenzocyclooctyne groups, amide connections and a spacer bearing four bracketed ethylene-oxide repeats plus its depicted flanking groups. The article estimates a roughly 3 nm spacer and calls 2a commercially available. Those facts do not identify a verified vendor product, CAS number, grade or exact supplier specification. Longer linker 2b, its 14-26 repeat range and PEG1000 preparation are distinct and cannot replace 2a automatically.

Precursor dependency. The cited Mandal et al. 2013 supporting information, sections 2.1.1-2.1.6, describes glycol tosylation, Boc-L-alanine esterification, deprotection/Boc-D-alanine coupling, deprotection/formylation, azide substitution and isocyanide formation. Its methoxy route uses four ethylene segments, unlike the three-segment 2018 methoxy graph; the longer counterpart would have CALCULATED formula C16H28N2O7. The cached text contains inconsistent reported mass/amount/yield numbers; none is corrected here. Cached primary text was inspected, but live download and page-image requests failed; no raw PDF hash or graphical verification is available. This citation documents a dependency, not an automatically incorporated complete route or proof of applicability to the shorter methoxy monomer.

Reported modified first step. The 2018 supplement changes the cited route's first and last operations. It reports tetraethylene glycol (52.68 g, 271 mmol) in THF (10 mL), cooled to 0 °C; aqueous NaOH (1.81 g, 45.25 mmol in 10 mL water), five minutes' vigorous stirring; then dropwise tosyl chloride (8.08 g, 42.4 mmol in 70 mL THF),

followed by 2.5 hours at room temperature. Workup uses 200 mL ice water, 50 mL DCM for layer separation and four 100 mL DCM extractions; flash silica/EtOAc purification reportedly yields 11.9 g, 80%. The drying-agent formula is printed as NaSO4; its identity is unresolved and is not silently rewritten. Reported source yields are third-party observations, not guaranteed outcomes.

Reported modified final step. The named azido formamido precursor (640 mg, 1.64 mmol) is reported in freshly distilled DCM (60 mL), with Burgess reagent, methyl N-(triethylammoniumsulfonyl) carbamate (594 mg, 2.49 mmol). The mixture is stirred six hours at room temperature with consumption followed by TLC; solvent removal and silica chromatography, DCM:MeCN 3:1, reportedly yield 0.436 g, 72%. The reported name retains the unresolved stereochemical assignment. This modified dehydration is not replaced by the cited older diphosgene operation.

Reported PIC preparation. The commercially available methoxy monomer is first purified using silica, MeCN:DCM 1:3. Purified methoxy monomer (250 mg, 0.79 mmol) and azide monomer (10 mg, 27 µmol) are dissolved in freshly distilled toluene (4 mL). The source prepares a 1 mM Ni(ClO4)2·6H2O stock in toluene:ethanol 9:1; 81.7 µL of stock is diluted to 1 mL with toluene and added. After 72 hours' stirring at room temperature, DCM (5 mL) is added and polymer precipitated in diisopropyl ether (100 mL), filtered, redissolved in DCM (5 mL) and reprecipitated in diisopropyl ether (100 mL). The source states that this procedure is repeated two more times. It reports 244 mg, 94%, and Mv = 599 kDa. Mv is an empirical viscometric estimate using Mark-Houwink parameters borrowed from other polyisocyanides; it is not a measured Mn, Mw or complete molecular-weight distribution. A supplier specification for the actual shorter methoxy monomer remains unverified.

CALCULATED monomer-only feed basis. Treating the nominal 790 µmol methoxy and 27 µmol azide charges as exact arithmetic inputs gives the following ideal atom-amount balance. The common amount scale is µmol of atoms; 37,717 is a proportional inventory total, not the atom count of an actual specimen. Reported charge precision is not increased by exact arithmetic. The azide monomer mole/number feed fraction is 27/817, a different quantity from the elemental atomic fractions below.

Element	Ideal feed amount (µmol of atoms)	Exact ideal atomic fraction
C	11465	11465/37717
H	19635	19635/37717
N	1715	1715/37717
O	4902	4902/37717

Feed calculation boundary. The four ideal fractions sum to one. They exclude solvent, catalyst, linker, water and other process inputs and do not measure conversion, relative incorporation, sequence, end groups, residual species or the final polymer/gel's complete elemental inventory. Both competing stereochemical assignments have the same formulas, so this accounting does not resolve them. The graphical x = 0.03 and n = 1900 labels are schematic source values, not a whole-product assay or every-chain count. A feed vector cannot automatically satisfy claim 1's expressly identified complete-inventory basis or claim 177's additional polymer limitations.

Reported gel assembly. Main Methods describes 2 mg/mL PIC stock in Milli-Q water, dissolved overnight at 4 °C and stored at -20 °C. Linker 2a stock is 2 mg/mL in DMSO and diluted with water; cold diluted linker and polymer solutions are mixed in equal volumes in a precooled glass vial, briefly homogenized and used immediately, then heated quickly to the chosen crosslinking temperature for one hour. Linker dilution varies loading rather than fixing one DMSO fraction. For the preceding property table, use the separately reported final PIC concentration, crosslinking/stabilization schedule and rheometry conditions. This stock procedure does not erase the source-specific conditions of each measurement.

Support boundary. Established precursor knowledge may assist a skilled reader; a complete synthesis of every upstream reagent and applicant repetition are not categorical prerequisites. Actual identity, applicability of the referenced operations and preparation without undue experimentation remain substantive inquiries. The unresolved stereo assignment, shorter-methoxy availability and source inconsistencies are retained rather than filled with invented facts. This specific attributed preparation does not establish every composition, trace fraction, polymer architecture or property scenario, or the applicant's technical contribution. Citations are not automatic incorporation of unprinted essential disclosure. Claims 1-199 and their assessment statuses are unchanged.

[0066] Attributed electrical-composite evidence and PROPOSED verification distinguish source preparation, measured profiles, fitted parameters and local transport models. The following record fixes loading coordinates and preserves conditional alternatives, missing conditions and whole-product inventory limits. The same named constituents alone do not fix a transport response.

Electrical evidence associated with [0066]: preparation, coordinates and conditional response

The attributed source is Kyrylyuk et al., Controlling electrical percolation in multicomponent carbon nanotube dispersions, published 10 April 2011. Its coauthor-uploaded original text was inspected through cached primary-document text; native acquisition failed. No main-PDF bytes or full graphical verification are claimed.

Reported preparation uses aqueous SWCNT/SDS and PS/PEDOT:PSS latexes, freeze-dried overnight at 0.25 mbar. After degassing, compression moulding uses 180 °C and 100 bar for 2 minutes; separate degassing conditions remain unspecified. Four-point DC measurements use graphite contacts and Keithley 6512/220 instruments. No applicant repetition is asserted.

Reported inputs include Merck SDS (90%), H.C. Starck Clevios P stock (0.4 wt% PEDOT; 0.8 wt% PSS) and HiPCO nanotubes (10-15 wt% organic impurities; 5 wt% metal catalyst). Stock percentages are not dry-composite loadings.

Electrical quantity	Reported value	Basis and source	Evidence
Fixed SWCNT loading	0.35 wt%; 0.26 vol%	Figure 2 composite loading	ESTABLISHED report
PEDOT:PSS threshold	0.443 wt%	Figure 2 given threshold; numerical wt%	ESTABLISHED source coordinate
Exponent t	1.92	Figure 2 fitted exponent	CALCULATED source fit
Prefactor sigma_0	0.952	Figure 2; S/m in numerical wt% convention	CALCULATED source fit
Moulding temperature	453 K	Supplement page 3	ESTABLISHED report
xi_rr; xi_ss; xi_sr	About 1; 10; 5 nm	Supplement page 3 effective coupling lengths	CALCULATED model

Figure 2 uses log10 conductivity in S/m and a two-parameter fit, sigma = sigma_0 (p - p_c)^t, at given p_c, with numerical wt% coordinate p. Negative plotted ordinates represent positive conductivity. The CALCULATED fraction representation is p_c/100 = 443/100000; changing to f = p/100 requires sigma_0,f = sigma_0,p 100^t. No curve is generated or extrapolated below the threshold. A reported threshold is not an exact product assay.

The publisher supplement, pages 1-3, discusses surfactant redistribution, thermal network equilibration and cooling before probing. Its proposed local relation sigma_mu_nu proportional to exp(-r/xi_mu_nu) uses effective model lengths. Quantitative macroscopic-conductivity prediction is outside the source's scope. Failed

final-composite domain imaging and uncertainty about processing remain limitations; observations of PEDOT:PSS-only films do not establish the composite's morphology.

The source's experimental mixtures were cooperative; antagonistic percolation was predicted by theory/Monte Carlo, with experimental confirmation outside its scope. PROPOSED present verification: after confirming input identities, retained constituents and the dry-loading convention, compare specified PEDOT:PSS targets on opposite sides of the reported threshold at fixed nominal SWCNT loading. Balance the matrix and hold retained surfactant, processing, temperature, geometry and current conditions constant. The hypothesis is increased positive DC conductivity when additional effective mixed contacts connect transport paths. A separate credible alternative is unchanged or reduced conductivity if added particles disrupt effective tube paths while mixed contacts transfer charge poorly. These are conditional, unperformed hypotheses informed by existing literature, not simultaneous opposite outcomes or earlier applicant predictions. Matched four-point measurements, contact controls, network characterization, replicate uncertainty and retained-material analysis would discriminate them; threshold and conductivity are separate observables.

UNSUPPORTED extensions include exact new conductivity, a complete elemental inventory inferred from component loading, and every future unexpected electrical response. Lot identifiers, impurity chemical identities and retained-product amounts remain unresolved despite reported grades/levels. Blend denominator, contact spacing, thickness, applied current, raw data and uncertainty also require resolution. [0059] remains applicable. The source's known result is not new applicant work, and this addition is not presumed present in an earlier PDF or filing.

[0067] ESTABLISHED mechanism example with PROPOSED extrapolation: aggregation can increase emission for suitable luminogens by changing nonradiative pathways, while other packing and excited-state interactions can quench it. A new candidate needs molecular identity, packing evidence, quantum yield, lifetime, and conditions. The alternatives do not apply indiscriminately to every compound. Tetrathienylethene emission study

[0068] For an otherwise identical material, an unrecognized property may already be inherent, but necessity must be established for the relevant conditions. Stating a property and its opposite as possible does not establish inherency. A later process or use with additional steps is separately compared. The property-scenario framework is not a promise to prevent patents on every later finding. USPTO inherency guidance

12 Worked numerical embodiments

Binary percent cells

[0069] CALCULATED example NUM-001 selects an arbitrary ordered pair of distinct constituent labels A and B, on an atomic basis. For every j = 0, ..., 99, x_A is in [j/100,(j+1)/100] and x_B = 1-x_A. Every composition satisfying 0 < x_A < 1 has two-element support, including shared internal cell endpoints. Only x_A = 0 or x_A = 1 is unary. No particular chemical preparation is inferred. The pair can be replaced by any ordered distinct pair from the specified vocabulary, with hypothetical labels retaining their status.

Fine trace range

[0070] CALCULATED example NUM-002 selects x_A = 10/D, 11/D, ..., 100/D and x_B = 1-x_A, with D = 10^19. For instance x_A = 10/D = 10^-18 fraction = 10^-16%, and x_B = (D-10)/D. Exact fractions are stored as integer numerator and denominator strings, avoiding floating-point rounding of the complement. Every contiguous subrange [l/D,u/D] with 10 <= l <= u <= 100 is separately addressable. Preparation and detectability remain UNSUPPORTED.

Rational composition outside the decimal grid

[0071] CALCULATED example NUM-003 uses (1/3,2/3). No integer n satisfies n/10^19 = 1/3 because 3 does not divide 10^19. The example is explicitly included through unrestricted rational denominators. A molecular stoichiometry, phase, and product purity require additional descriptions; the vector alone is not the TiO2(B) literature product.

Multiple components and rejected intervals

[0072] CALCULATED example NUM-004 uses the feasible ternary box in paragraph [0032]. NUM-005 rejects three lower bounds of 40% because their sum is 120%. NUM-006 selects any k from 2 through 138, assigns 1/D to k-1 constituents and (D-k+1)/D to the last. For k = 138 the small constituents total 137/D. An equal-fraction candidate uses x_i = 1/k in the rational domain; it lies on the D grid only when k divides D.

13 Reproducible enumeration instructions

[0073] To enumerate supports, fix the vocabulary edition, choose k, and list every increasing k-tuple of atomic numbers. Attach each label's recognized or hypothetical status. To enumerate a fixed rational denominator q, list every positive integer k-tuple summing to q. Use the last numerator as q minus the sum of the preceding numerators and reject values below one. This procedure terminates for each finite q and k.

[0074] To enumerate the unrestricted rational domain, repeat for q = k,k+1,... and retain only primitive representations when deduplication is required. This is a countably infinite sequence, not a claim that a finite application prints every tuple. To enumerate decimal intervals, iterate p, l, and u over the finite limits in paragraph [0023], form component boxes, apply normalization feasibility, and construct a witness. The real continuous domain is defined symbolically; it cannot be exhaustively listed by a finite digital enumeration.

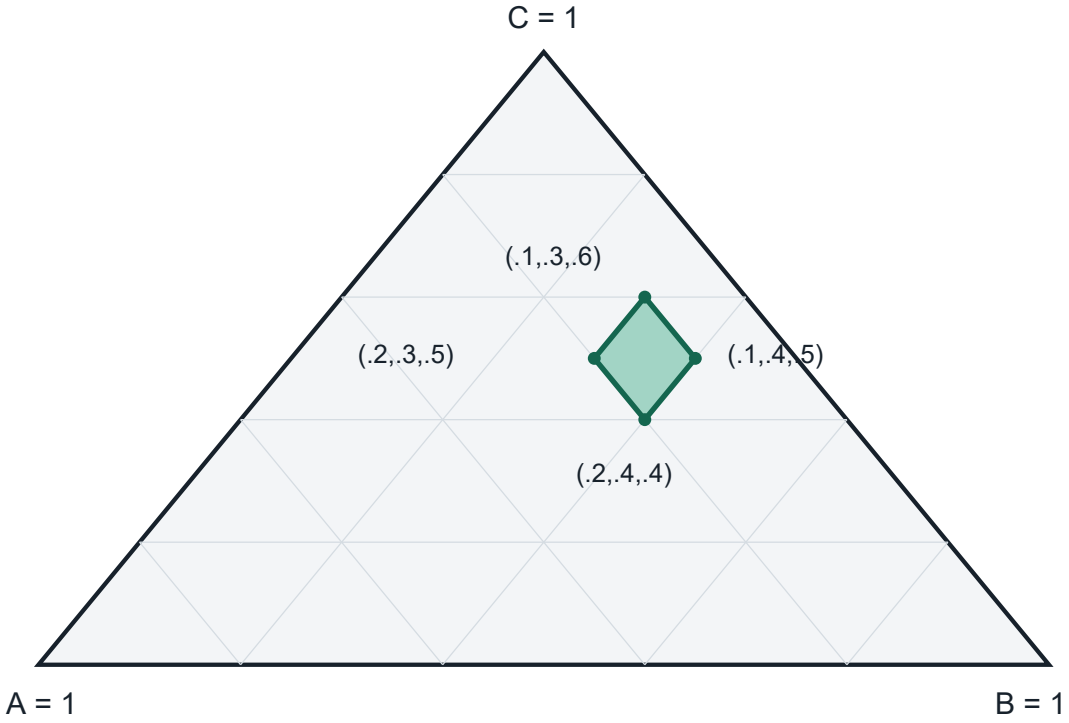
[0075] The complete record architecture specifies vocabulary edition, support tuple, composition basis, exact endpoints, normalization rule, witness or infeasibility reason, and evidence status, together with independently supplied material identity, structure, preparation, measurement, property and source fields. A record identifier encodes its stated numerical inputs, not an assertion of a synthesized material. The included numerical generators implement numerical targets and selected classification/status fields; their actual output is a partial numerical record, not an implementation supplying all fields of the complete architecture or every operation of claim 186. Missing fields remain unresolved and must not default to successful or established. Numerical rejection records remain separate from completed positive-target records.

[0076] A reproducibility package accompanies this draft for exact arithmetic and verification. Its external files are aids for review, not automatically part of a filed or published application. Essential definitions and worked examples are therefore stated in this description. A final filing must carry all relied-on technical content in accepted application formats, with the filed and published versions checked.

14 Drawings

[0077] Figure 1 illustrates normalization for three constituents. The triangle represents the nonnegative simplex; its vertices are unary boundaries. The highlighted feasible interval intersection represents A in [10%,20%], B in [30%,40%], and C in [40%,60%]. It is a numerical diagram, not a measured phase diagram.

Figure 1. Normalized three-component composition

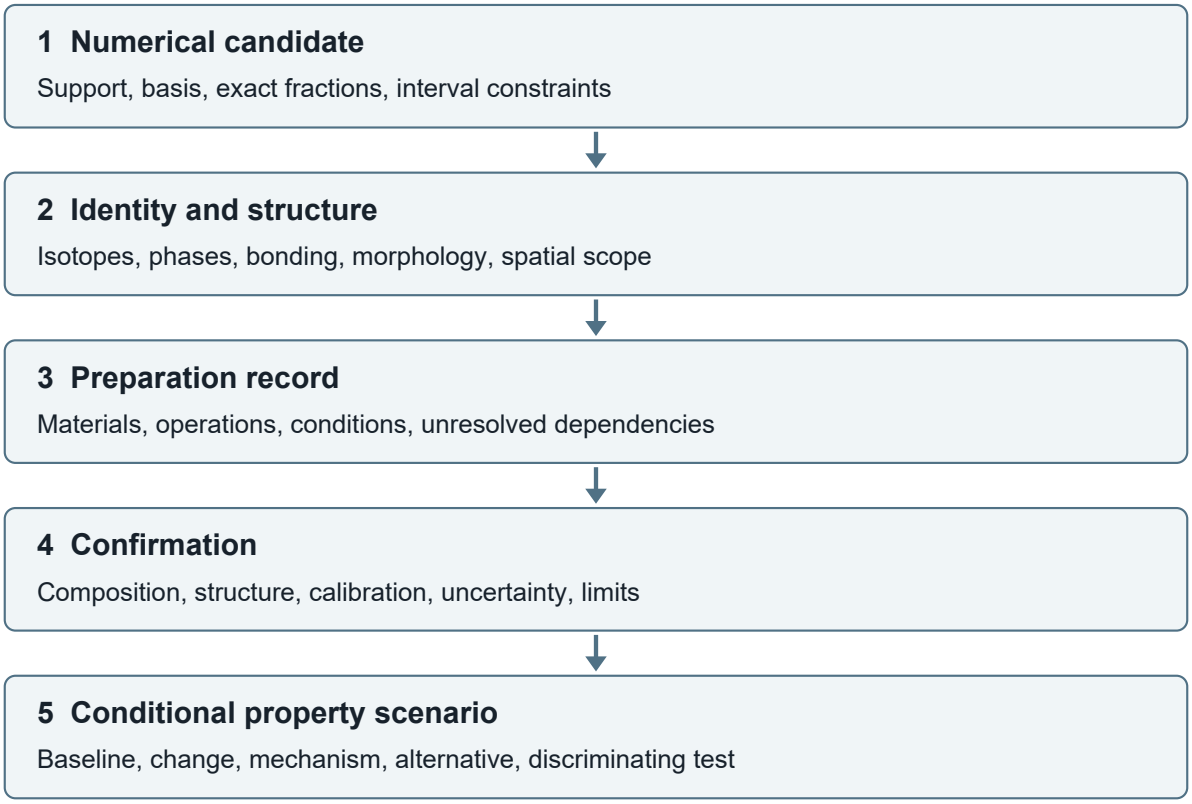


Highlighted: A in [0.1,0.2], B in [0.3,0.4], C in [0.4,0.6].

Every point obeys A + B + C = 1. Numerical region only; not a phase diagram.

[0078] Figure 2 shows the material-record sequence from a numerical candidate to structure, preparation, confirmation, and a conditional property scenario. An unresolved field remains unresolved. The drawing does not assert that this workflow supplies a universal preparation route.

Figure 2. Candidate-material record sequence



At every step: preserve evidence status and leave missing information unresolved.

A completed numerical record is not proof of an available physical material.

15 Industrial applicability and unresolved technical scope

[0079] Supported individual materials may be applicable to structural components, electronic and optical devices, thermal management, chemical processing, coatings, storage, or other uses established for the individual record. This draft does not establish industrial applicability of every hypothetical element, nuclide, particle system, or composition. An actual material-use record must state how the identified product is made and used.

[0080] Unresolved technical scope includes preparation of most indexed compositions; crystal and molecular identity; phase stability and coexistence; isotope availability and lifetime; precision and impurity control; particle confinement; property magnitudes and conditions; and the applicant's actual technical contribution. Numeric refinement cannot resolve these gaps. The candidate claims below express the requested scope for review and are not certified as novel, enabled, unified, or allowable.

Hypothetical embodiments for review

[0081] The following finite set of conditional embodiments supplements the existing preparation, identity and property records. It is theoretical disclosure, not a scheduled physical program. ESTABLISHED labels an attributed existing source teaching; CALCULATED labels the stated bookkeeping consequences under their assumptions; PROPOSED labels a candidate route, structure or response branch; actual new preparation, identity and outcome remain UNSUPPORTED where evidence is absent. Existing published operations and observations remain attributed to their original sources. These embodiments do not claim that every numerical target can be made or that a hypothetical element, state or particle exists.

[0082] CALCULATED feed-to-retained feasibility model: let q_r be a nonnegative integer number of identified input packets, and A_e,r the known integer atoms of element e in each packet. For a declared finite product boundary, let b_e and l_e be nonnegative integer ingress and loss counts. Ordinary chemical processing, excluding nuclear transformation, then requires z_e=sum_r(A_e,r*q_r)+b_e-l_e>=0. A prescribed atomic target x_e additionally requires z_e=N*x_e for a positive integer N=sum_e(z_e), with zero counts outside its complete support. In an ideal closed-input branch b_e=l_e=0 is an express model assumption, not an impurity assay. An unwanted element in every input packet selected with q_r>0 cannot disappear by choosing nonnegative packet quantities; an independently supported removal term or different input is necessary. Unknown stock chemistry, ingress or losses leave the inverse input-quantity calculation unresolved. Continuous mole-amount balances are a separate proportional model and do not assign integer counts to a real specimen.

[0083] CALCULATED phase-mixture identity constraint: for specified phases a, let c_e,a be their complete elemental atomic profiles and N_a their counted atom totals. On this expressly atom-weighted boundary, x_e=sum_a(N_a*c_e,a)/sum_a(N_a). Thus a target assembled solely from those profiles lies in their convex hull, subject to finite-count compatibility. This does not establish that the phases coexist, retain their identities or occupy the proposed geometry. A structural proposal additionally identifies each phase model, site or molecular occupations, ordering, local domain, interface and observation state. Unknown surface material, solvent or other phases are not deleted to obtain closure. Mass and volume weighting require different quantities and compatible conversion. A target outside the permitted phase-profile mixture cannot be realized by that particular assemblage merely by changing its proportions.

[0084] PROPOSED alloy and glass branches retain the source-associated route assumptions in [0039] and [0063]. An alloy branch may nominate the documented co-melting/remelting and homogenization class only where the selected feeds can enter that process and a retention/phase model is supported. Charge balance in [0082] does not eliminate volatilization, segregation, chemical incompatibility or contact contamination; FCC identity and boundary transport are additional fields, not consequences of equiatomic input. A glass branch instead nominates the documented melt/flux/quench class with its specified contact boundary and geometry. For a proposed local trajectory T(r,t), define the CALCULATED residence functional W(r)=integral I_window(T(r,t)) dt over a stated interval, where I_window is one inside [T_a,T_b] and zero outside, if that crystallization window is independently supported. Neither W nor a bath description supplies a critical cooling condition by itself. Different nucleation populations, local trajectories or retained flux can defeat the proposed lower-crystalline-signal branch. No new temperature, atmosphere, timescale or threshold is assigned.

[0085] PROPOSED inorganic branch separates precursor conversion, isolated host and retained surface material. For a stipulated ideal TiO2 starting host and 0<=delta<2, TiO2+delta H2 -> TiO_(2-delta)+delta H2O is a CALCULATED atom balance, not a thermodynamic prediction or a replacement for the phase-specific source's unresolved precursor. In a finite count model with positive integer N Ti atoms require V=N*delta to be an integer, so oxygen changes from 2N to 2N-V and V water units carry the transferred oxygen. Product water enters or leaves the denominator according to the declared boundary. The nominal reduced host requires a separate phase/site model; an oxygen count does not prove a rutile vacancy map or Magneli phase. The restricted Ti(III)/Ti(IV), O(2-) compensation model in [0064] remains restricted to its stated delta range. Unknown hydrogen,

hydroxyls, ligands, chloride or other retained material prevent treating the real product as only Ti/O. Reaction feasibility, phase selection, isolation and persistence remain PROPOSED or UNSUPPORTED rather than inferred from the equation.

[0086] PROPOSED molecular branch distinguishes a mixture of fixed molecular phases from arbitrary occupancy of one hydrate. Let m_U and m_D be nonnegative integers, not both zero, counting ideal unsolvated carbamazepine and dihydrate units with their specified connectivity and separate crystal-form labels. The CALCULATED (H,C,N,O) counts are (12*m_U+16*m_D,15*(m_U+m_D),2*(m_U+m_D),m_U+3*m_D), total 30*m_U+36*m_D. The dihydrate formula-unit fraction m_D/(m_U+m_D) differs from its atom-weight fraction lambda=36*m_D/(30*m_U+36*m_D). The aggregate atom vector is (1-lambda)*x_U+lambda*x_D using the formula vectors in [0040]. A source-local solution/hold/quench route supports its stated sampled crystallites; it is not a demonstrated bulk separation or prescribed phase-proportion route. Assembly from separately available phases is a conditional alternative only if their isolation and retention are supported. Water/solvent exchange, polymorphic conversion, warming and substrate retention can change this two-phase boundary or invalidate its fixed-unit assumption.

[0087] PROPOSED polymer branch retains incorporated-repeat, chain-end, sequence and network variables separately from feed. The conditional hydroxyl/carboxyl chain model in [0040] gives (C,H,O)=(3R,4R+2K,2R+K) for R repeats in K chains; it does not identify the source's actual ends. On a proposed chain population with nonnegative integer counts K_n whose total is positive and positive finite chain masses or molar masses M_n under one consistent convention, number-average mass is sum_n(K_n*M_n)/sum_n(K_n), while weight-average mass is sum_n(K_n*M_n^2)/sum_n(K_n*M_n). These CALCULATED averages do not specify the entire distribution or sequence. A source-associated forming/thermal route remains conditional on input identity and applicable processing; equal feed mass does not fix chain number, retained molecular weight, packing or crystallinity. Chain shortening, altered ends, residuals, voids or a different thermal state can invalidate the proposed mechanical branch. Physical junctions and covalent network connections remain distinct structural descriptions.

[0088] CALCULATED bifunctional-linker topology branch: in an expressly defined population let L linkers be partitioned into B links between nominated distinct network nodes, C doubly attached nonconnecting/loop links, P singly attached pendant links and U unreacted links, with L=B+C+P+U. If A reactive host groups are available and no other reactions are admitted, the consumed-end count is R_end=2B+2C+P<=A and <=2L, and B<=min(L,floor(A/2)). Identifying two reacted ends does not establish a load-bearing bridge. For L=10 and A>=8, the hypothetical populations (B,C,P,U)=(4,0,0,6) and (0,4,0,6) both consume eight ends while having different B. These are count examples, not source observations. The source-associated PIC/linker route supplies context; candidate site accessibility, incorporation, reaction side-products, topology and mechanical effectiveness remain unresolved. A higher-modulus branch is conditional on increased effective load-bearing connectivity with relevant state variables matched. Excess pendant attachment, loops or altered bundling support a distinct unchanged/lower-response hypothesis, not a universal modulus law.

[0089] PROPOSED composite branch starts from disjoint constituent or explicitly allocated stock-origin categories. In a no-ingress retention model m_j,ret=h_j*m_j,input, 0<=h_j<=1, and w_j,ret=m_j,ret/sum_l(m_l,ret) when the total is positive. The h_j are unsupported parameters unless evidence supplies them; a lost category leaves its positive support. Reactive transformations require a justified origin allocation or additional product categories, rather than pretending all constituents retain their input identities. For compatible additive constituent volumes, V_j=m_j,ret/rho_j with rho_j>0, and phi_j=V_j/(sum_l(V_l)+V_void); void fraction uses that same denominator. Nonadditivity requires a different model. A proposed reinforcement orientation tensor A_orient=average(u*u^T), for unit directions u in a nonempty selected population with declared nonnegative weights summing to one, is symmetric positive semidefinite with trace one. It does not specify dispersion, interfaces or a connected transport network. The documented cure or stock-assembly routes remain attributed; no retained loading, alignment or conductivity follows solely from nominal input fractions.

[0090] PROPOSED isotope-stock branch assumes identified, independently supported stocks of one element at a specified reference time, with conditional state profiles y_j,s and integer atom totals N_j. CALCULATED mixture counts are n_s=sum_j(N_j*y_j,s), requiring each contributed finite sub-count to be integral; the conditional profile is n_s/sum_j(N_j) when that denominator is positive. It is a convex combination of the available stock profiles. A positively requested state absent from every stock requires a separately supported production/separation route; elemental charge mixing cannot create it. This classical count model does not describe arbitrary coherent-state preparation. Processing losses, decay and daughters require time-dependent boundaries, not assumed persistence of the initial profile. Unknown positive-population masses prevent the mass conversion of [0047]; an unpopulated state's unknown mass does not. Nuclear identifiers and count compatibility remain separate from existence, lifetime and route feasibility.

[0091] PROPOSED particle/host branch first fixes its construction level. For a nominated host with an atomic inventory, added excitations are separately recorded rather than added as new elemental labels or counted again as constituents of their containing composites. In an expressly noninteracting excitation model with nonnegative integer counts n_s and supported positive energies epsilon_s for populated entries, excitation energy above the selected reference is E_exc=sum_(s:n_s>0)(n_s*epsilon_s). Zero populations contribute zero without requiring an energy value. If E_exc>0, energy-share coordinates are n_s*epsilon_s/E_exc; they are not atomic or unqualified chemical mass fractions. A zero-excitation branch has no positive excitation-energy denominator. The selected quantum model must supply occupation restrictions, charge/conservation conditions, driver/host or confinement, lifetime and observation domain. Unknown species, mode energies, production or containment keep the physical branch UNSUPPORTED. A label or normalized classical population does not satisfy those gates, and the example assigns no stable arbitrary particle mixture.

[0092] PROPOSED property branch uses a nominated scalar observable P and preparation/intervention coordinate u, with independently parameterized linked retained-composition, structural and state variables z_l. On an applicable differentiable model branch, P=P(u,z(u)) gives the CALCULATED relation dP/du=partial P/partial u+sum_l((partial P/partial z_l)*(dz_l/du)). A positive direct term does not fix the total sign if a linked structural or retention term is negative and larger. This supplies a conditional alternative mechanism without assigning incompatible outcomes to an identical state. A separate observable Q has its own relation; its sign is not inferred from P. Phase transitions, discontinuities or failure of the chosen model require a separate branch. Actual functions, parameter values, domains, uncertainties and response magnitudes remain UNSUPPORTED unless justified sources or theory provide them; no invented universal constitutive law is supplied.

[0093] CALCULATED precision branch distinguishes exact targets from bounded input knowledge. For complete mass categories with justified bounds 0<a_i<=m_i<=b_i and positive denominators, normalized w_i=m_i/sum_j(m_j) obeys a_i/(a_i+sum_(j!=i)(b_j))<=w_i<=b_i/(b_i+sum_(j!=i)(a_j)). These are deterministic model bounds, not measured confidence intervals; different coordinate extremes need not occur together. Unknown residual mass requires an additional category or unresolved term, not zero. For a justified local uncertainty model the normalization Jacobian is J_i,j=(delta_i,j-w_i)/T, T=sum_j(m_j), where delta_i,j is one for i=j and zero otherwise, and the first-order covariance is J*C_m*J^T. This linearization requires applicable input covariance and small-enough departures; it does not invent them or replace exact bounds. Percent covariance is 10000 times fraction covariance. Exact interval refinement, finite atom-count compatibility, sampling, detection and this uncertainty model remain separate. No globally finest measurable interval or resolved adjacent-grid distinction is certified.

[0094] These finite hypothetical embodiments provide reviewable assumptions, linked variables and failure boundaries. Existing preparation/source sections remain operative, and no critical operating gap is filled by a default. A particular field can rely on adequate literature, justified theory and skilled knowledge without categorical applicant experimentation; absent applicable support remains absent. All new physical implementations and outcomes are unperformed and no applicant experiment program is planned. The resulting draft is suitable for

technical and legal review with unresolved truth made visible, while universal defensive preemption remains aspirational rather than an achieved result.

[0095] CALCULATED mixture-domain clarification of [0083] and [0090]: each complete phase atomic profile c_e,a is nonnegative and sums to one over the declared full elemental inventory; phase atom totals N_a are nonnegative integers with positive sum, with each N_a*c_e,a integral for an exact finite-count phase profile. Whole formula units, molecules and required site multiplicities impose their own stronger count constraints. Each isotope-stock profile y_j,s is nonnegative and sums to one over its declared states at the reference time; N_j are nonnegative integer atom totals with positive sum, and every N_j*y_j,s must be an integer for an exact finite-count stock description. An expected or ensemble-average profile is a distinct model, not an exact stock population. These qualifications make the stated convex-combination calculations well-defined; they do not establish phase coexistence, stock availability, state persistence or a preparation route.

[0096] CALCULATED accounting clarification of [0040], [0087] and [0089]: the polymer tuples displayed as (C,H,O) use an explicit local display order. The canonical ascending-atomic-number support is (H,C,O), with ideal-repeat fractions (4/9,1/3,2/9) and the correspondingly permuted finite-chain counts (4R+2K,3R,2R+K). Reordering coordinates changes neither inventory nor composition. In the composite retention model an identified category is removed from positive support only when its retained mass is zero. Stock-origin allocation after reaction is bookkeeping; it does not automatically define realized physical phases, component volumes or post-cure densities. The additive volume branch applies only to disjoint realized retained phases or constituents with justified finite positive densities under the stated observation conditions, finite nonnegative V_void, and positive total volume. Feedstock densities cannot be carried into an origin-allocation model without a separate justification. A zero-total or nonadditive case is unresolved under this particular volume formula, not silently normalized.

[0097] CALCULATED joint finite-inventory clarification: elemental count compatibility is necessary but can be insufficient for a more fully specified identity. For a declared finite atomic inventory N and complete within-element isotope/nuclear-state profiles y_i,a,s, every joint count N*a_i*y_i,a,s must be a nonnegative integer. For positive rational joint atomic coordinates, N must be a positive multiple of their primitive denominator, which may exceed the elemental denominator. For example a=(1/2,1/2) and the first element's profile (1/2,1/2), with the second element in one state, give joint coordinates (1/4,1/4,1/2). N=2 permits the elemental counts (1,1) but gives nonintegral joint counts (1/2,1/2,1); N=4 gives (1,1,2). Likewise an intact hypothetical benzene-only molecular inventory uses C6H6 units: its (H,C) fractions (1/2,1/2) allow elemental count arithmetic at N=2, while whole intact molecules require N=12*m for a positive integer molecule count m. Qualified conversion must first supply the atomic vector if the stated composition basis is mass. This accounting does not impose specimen-independent minima, permit arbitrary nuclei, prove molecular integrity or establish a physical realization.

[0098] PROPOSED record-operation clarification of [0030]-[0038] and the candidate method claims: unrestricted rational enumeration is an incremental generator over successive denominators, with an expressly declared finite stopping budget or requested prefix for a particular run. No finite run completes an infinite domain or enumerates all irrational real targets. For a bounded composition query, a preliminary nonnegative witness does not establish feasibility on the chosen strictly positive support. A record procedure applies the applicable positive-feasibility conditions, assigns a qualified positive witness only on success, and records infeasibility without assigning such a target on failure. A lower-support boundary may instead be recorded under its distinct support. For example bounds ([0,0],[1,1]) have the nonnegative vector (0,1) but no strictly positive binary witness. Exact integer-grid membership and unrestricted positive rational feasibility remain separate tests. These recording rules do not manufacture the indexed material or certify a technical improvement, novelty, eligibility or defensive legal result.

[0099] CALCULATED domain and terminology qualification of [0084], [0087] and [0088]: the phrase "Charge balance in [0082]" in [0084] refers to the elemental atom balance in [0082], not to electrical charge conservation; that equation includes no ionic-charge, electron or current term. In the conditional finite-chain population, n is a positive integer repeat count, K_n are nonnegative integers with finite nonempty support, K=sum_n(K_n)>0 and

R=sum_n(n*K_n)>=K. The linker categories L, B, C, P, U and available host-group count A in [0088] are nonnegative integers under its declared finite-population convention. These count domains do not identify actual chain ends, accessible sites or load-bearing connections. The unweighted thermal-window residence W in [0084] is a descriptive path functional, not a universal crystallization predictor: a comparative response needs support for the applicable temperature-dependent nucleation and growth behavior, initial nuclei, local thermal paths and matched observation conditions. Equal W can coexist with different trajectories and outcomes. No unsupported kinetic weighting, cooling threshold or measured response is assigned.

16 Draft abstract

A framework describes candidate materials through elemental support sets, normalized composition vectors, interval constraints, isotope distributions, structures, preparation records, and conditional property scenarios. The vocabulary separates 118 recognized elements from hypothetical atomic-number placeholders through 138. Composition descriptions include positive continuous targets, unrestricted rational proportions, a decimal composition grid with accounting denominator 10^19 on the selected basis, and correlated component ranges intersected with normalization. Exact endpoints, binary complements, and feasibility witnesses support reproducible indexing. Nuclear states and particle systems use separate identity and accounting conventions. Material records associate numerical candidates with preparation, characterization, evidence status, and unresolved conditions. The framework distinguishes calculated descriptions and unperformed proposals from established material teachings.

17 Candidate claims

The following complete claim annex expresses requested coverage for review. Hypothetical constituents and broad material classes lack demonstrated full-scope preparation support. Adding separate count or interval claims does not cure that deficiency.

Claims 1-199 receive the separate preliminary feature screening identified in the companion review; no complete chronological novelty search or final statutory novelty determination is asserted. The inherited historical novelty-assessment flags remain unchanged. Claims 1-185 are physical-material review candidates; claims 186-193 and 197-199 concern record operations, and claims 194-196 concern separate particle-system records. No claim represents an assertion that its full physical scope is enabled.

Claim 1. A material comprising a closed elemental inventory of k distinct elemental constituents selected from atomic numbers 1 through 138, wherein k is an integer from 2 through 138, each selected constituent has a positive normalized fraction x_i on an expressly identified atomic-fraction or mass-fraction basis, the fractions of the selected constituents sum to one, and constituents outside the selected inventory have zero fraction on that basis.

Claim 2. The material of claim 1, wherein k is exactly 2.

Claim 3. The material of claim 1, wherein k is exactly 3.

Claim 4. The material of claim 1, wherein k is exactly 4.

Claim 5. The material of claim 1, wherein k is exactly 5.

Claim 6. The material of claim 1, wherein k is exactly 6.

Claim 7. The material of claim 1, wherein k is exactly 7.

Claim 8. The material of claim 1, wherein k is exactly 8.

Claim 9. The material of claim 1, wherein k is exactly 9.

Claim 10. The material of claim 1, wherein k is exactly 10.

Claim 11. The material of claim 1, wherein k is exactly 11.

Claim 12. The material of claim 1, wherein k is exactly 12.

Claim 13. The material of claim 1, wherein k is exactly 13.

Claim 14. The material of claim 1, wherein k is exactly 14.

Claim 15. The material of claim 1, wherein k is exactly 15.

Claim 16. The material of claim 1, wherein k is exactly 16.

Claim 17. The material of claim 1, wherein k is exactly 17.

Claim 18. The material of claim 1, wherein k is exactly 18.

Claim 19. The material of claim 1, wherein k is exactly 19.

Claim 20. The material of claim 1, wherein k is exactly 20.

Claim 21. The material of claim 1, wherein k is exactly 21.

Claim 22. The material of claim 1, wherein k is exactly 22.

Claim 23. The material of claim 1, wherein k is exactly 23.

Claim 24. The material of claim 1, wherein k is exactly 24.

Claim 25. The material of claim 1, wherein k is exactly 25.

Claim 26. The material of claim 1, wherein k is exactly 26.

Claim 27. The material of claim 1, wherein k is exactly 27.

Claim 28. The material of claim 1, wherein k is exactly 28.

Claim 29. The material of claim 1, wherein k is exactly 29.

Claim 30. The material of claim 1, wherein k is exactly 30.

Claim 31. The material of claim 1, wherein k is exactly 31.

Claim 32. The material of claim 1, wherein k is exactly 32.

Claim 33. The material of claim 1, wherein k is exactly 33.

Claim 34. The material of claim 1, wherein k is exactly 34.

Claim 35. The material of claim 1, wherein k is exactly 35.

Claim 36. The material of claim 1, wherein k is exactly 36.

Claim 37. The material of claim 1, wherein k is exactly 37.

Claim 38. The material of claim 1, wherein k is exactly 38.

Claim 39. The material of claim 1, wherein k is exactly 39.

Claim 40. The material of claim 1, wherein k is exactly 40.

Claim 41. The material of claim 1, wherein k is exactly 41.

Claim 42. The material of claim 1, wherein k is exactly 42.

Claim 43. The material of claim 1, wherein k is exactly 43.

Claim 44. The material of claim 1, wherein k is exactly 44.

Claim 45. The material of claim 1, wherein k is exactly 45.

Claim 46. The material of claim 1, wherein k is exactly 46.

Claim 47. The material of claim 1, wherein k is exactly 47.

Claim 48. The material of claim 1, wherein k is exactly 48.

Claim 49. The material of claim 1, wherein k is exactly 49.

Claim 50. The material of claim 1, wherein k is exactly 50.

Claim 51. The material of claim 1, wherein k is exactly 51.

Claim 52. The material of claim 1, wherein k is exactly 52.

Claim 53. The material of claim 1, wherein k is exactly 53.

Claim 54. The material of claim 1, wherein k is exactly 54.

Claim 55. The material of claim 1, wherein k is exactly 55.

Claim 56. The material of claim 1, wherein k is exactly 56.

Claim 57. The material of claim 1, wherein k is exactly 57.

Claim 58. The material of claim 1, wherein k is exactly 58.

Claim 59. The material of claim 1, wherein k is exactly 59.

Claim 60. The material of claim 1, wherein k is exactly 60.

Claim 61. The material of claim 1, wherein k is exactly 61.

Claim 62. The material of claim 1, wherein k is exactly 62.

Claim 63. The material of claim 1, wherein k is exactly 63.

Claim 64. The material of claim 1, wherein k is exactly 64.

Claim 65. The material of claim 1, wherein k is exactly 65.

Claim 66. The material of claim 1, wherein k is exactly 66.

Claim 67. The material of claim 1, wherein k is exactly 67.

Claim 68. The material of claim 1, wherein k is exactly 68.

Claim 69. The material of claim 1, wherein k is exactly 69.

Claim 70. The material of claim 1, wherein k is exactly 70.

Claim 71. The material of claim 1, wherein k is exactly 71.

Claim 72. The material of claim 1, wherein k is exactly 72.

Claim 73. The material of claim 1, wherein k is exactly 73.

Claim 74. The material of claim 1, wherein k is exactly 74.

Claim 75. The material of claim 1, wherein k is exactly 75.

Claim 76. The material of claim 1, wherein k is exactly 76.

Claim 77. The material of claim 1, wherein k is exactly 77.

Claim 78. The material of claim 1, wherein k is exactly 78.

Claim 79. The material of claim 1, wherein k is exactly 79.

Claim 80. The material of claim 1, wherein k is exactly 80.

Claim 81. The material of claim 1, wherein k is exactly 81.

Claim 82. The material of claim 1, wherein k is exactly 82.

Claim 83. The material of claim 1, wherein k is exactly 83.

Claim 84. The material of claim 1, wherein k is exactly 84.

Claim 85. The material of claim 1, wherein k is exactly 85.

Claim 86. The material of claim 1, wherein k is exactly 86.

Claim 87. The material of claim 1, wherein k is exactly 87.

Claim 88. The material of claim 1, wherein k is exactly 88.

Claim 89. The material of claim 1, wherein k is exactly 89.

Claim 90. The material of claim 1, wherein k is exactly 90.

Claim 91. The material of claim 1, wherein k is exactly 91.

Claim 92. The material of claim 1, wherein k is exactly 92.

Claim 93. The material of claim 1, wherein k is exactly 93.

Claim 94. The material of claim 1, wherein k is exactly 94.

Claim 95. The material of claim 1, wherein k is exactly 95.

Claim 96. The material of claim 1, wherein k is exactly 96.

Claim 97. The material of claim 1, wherein k is exactly 97.

Claim 98. The material of claim 1, wherein k is exactly 98.

Claim 99. The material of claim 1, wherein k is exactly 99.

Claim 100. The material of claim 1, wherein k is exactly 100.

Claim 101. The material of claim 1, wherein k is exactly 101.

Claim 102. The material of claim 1, wherein k is exactly 102.

Claim 103. The material of claim 1, wherein k is exactly 103.

Claim 104. The material of claim 1, wherein k is exactly 104.

Claim 105. The material of claim 1, wherein k is exactly 105.

Claim 106. The material of claim 1, wherein k is exactly 106.

Claim 107. The material of claim 1, wherein k is exactly 107.

Claim 108. The material of claim 1, wherein k is exactly 108.

Claim 109. The material of claim 1, wherein k is exactly 109.

Claim 110. The material of claim 1, wherein k is exactly 110.

Claim 111. The material of claim 1, wherein k is exactly 111.

Claim 112. The material of claim 1, wherein k is exactly 112.

Claim 113. The material of claim 1, wherein k is exactly 113.

Claim 114. The material of claim 1, wherein k is exactly 114.

Claim 115. The material of claim 1, wherein k is exactly 115.

Claim 116. The material of claim 1, wherein k is exactly 116.

Claim 117. The material of claim 1, wherein k is exactly 117.

Claim 118. The material of claim 1, wherein k is exactly 118.

Claim 119. The material of claim 1, wherein k is exactly 119.

Claim 120. The material of claim 1, wherein k is exactly 120.

Claim 121. The material of claim 1, wherein k is exactly 121.

Claim 122. The material of claim 1, wherein k is exactly 122.

Claim 123. The material of claim 1, wherein k is exactly 123.

Claim 124. The material of claim 1, wherein k is exactly 124.

Claim 125. The material of claim 1, wherein k is exactly 125.

Claim 126. The material of claim 1, wherein k is exactly 126.

Claim 127. The material of claim 1, wherein k is exactly 127.

Claim 128. The material of claim 1, wherein k is exactly 128.

Claim 129. The material of claim 1, wherein k is exactly 129.

Claim 130. The material of claim 1, wherein k is exactly 130.

Claim 131. The material of claim 1, wherein k is exactly 131.

Claim 132. The material of claim 1, wherein k is exactly 132.

Claim 133. The material of claim 1, wherein k is exactly 133.

Claim 134. The material of claim 1, wherein k is exactly 134.

Claim 135. The material of claim 1, wherein k is exactly 135.

Claim 136. The material of claim 1, wherein k is exactly 136.

Claim 137. The material of claim 1, wherein k is exactly 137.

Claim 138. The material of claim 1, wherein k is exactly 138.

Claim 139. A single-element material having a closed elemental inventory consisting of one constituent selected from atomic numbers 1 through 138, wherein the constituent has normalized atomic fraction one and every other elemental constituent has fraction zero.

Claim 140. The material of claim 1, wherein every selected constituent has atomic number from 1 through 118.

Claim 141. The material of claim 1, wherein at least one selected constituent is selected from atomic numbers 119 through 138.

Claim 142. The material of claim 1, wherein k is 118 and the inventory includes each atomic number from 1 through 118.

Claim 143. The material of claim 1, wherein k is 138 and the inventory includes each atomic number from 1 through 138.

Claim 144. The material of claim 1, wherein each x_i is the number of atoms of constituent i divided by the total number of atoms in the specified inventory.

Claim 145. The material of claim 1, wherein the proportions are specified as mole-of-atoms fractions using the same atom inventory as an atomic-fraction description.

Claim 146. The material of claim 1, wherein each x_i is the mass of constituent i divided by the total mass of the specified inventory.

Claim 147. The material of claim 1, wherein an isotope-qualified positive finite effective mass M_i is specified for every selected constituent under a stated mass-allocation and compatible-unit convention, elemental atomic fractions a_i and elemental mass fractions w_i satisfy w_i = a_i M_i/sum_j(a_j M_j) and a_i = (w_i/M_i)/sum_j(w_j/M_j), and the fractions x_i of claim 1 equal a_i when the expressly identified basis is atomic fraction or w_i when it is mass fraction.

Claim 148. The material of claim 1, wherein x_i = n_i/q for a positive integer q and positive integers n_i whose sum is q, with q unrestricted except that q is at least k.

Claim 149. The material of claim 148, wherein gcd(q,n_1,...,n_k) = 1.

Claim 150. The material of claim 1, wherein x_i = n_i/D with D = 10^19, each n_i is a positive integer, and sum(n_i) = D.

Claim 151. The material of claim 150, wherein k-1 constituents each have fraction 1/D and the remaining constituent has fraction (D-k+1)/D.

Claim 152. The material of claim 1, wherein each fraction satisfies a selected closed interval [L_i,U_i], with 0 <= L_i <= U_i <= 1, and the joint composition satisfies sum(x_i) = 1 and x_i > 0.

Claim 153. The material of claim 152, wherein each interval is selected by integers p,l,u with 0 <= p <= 19 and 0 <= l <= u <= 10^p, its endpoints being l/10^p and u/10^p.

Claim 154. The material of claim 153, wherein at least one interval is an elementary cell [j/10^p,(j+1)/10^p] for an integer j with 0 <= j < 10^p.

Claim 155. The material of claim 1, wherein k = 2 and fractions x_A and x_B obey x_A in [L,U] and x_B = 1-x_A in [1-U,1-L], with 0 < x_A < 1.

Claim 156. The material of claim 155, wherein 10^-18 <= x_A <= 10^-17, corresponding to 10^-16% through 10^-15%, inclusive.

Claim 157. The material of claim 156, wherein x_A is exactly 10^-18 and x_B is exactly 1-10^-18.

Claim 158. The material of claim 156, wherein x_A is exactly 10^-17 and x_B is exactly 1-10^-17.

Claim 159. The material of claim 156, wherein x_A = n/10^19 for an integer n from 10 through 100 and x_B = 1-x_A.

Claim 160. The material of claim 159, wherein x_A lies in a selected subrange [l/10^19,u/10^19] for integers 10 <= l <= u <= 100.

Claim 161. The material of claim 1, wherein k = 2 and each constituent has fraction 1/2, corresponding to 50% each.

Claim 162. The material of claim 1, wherein k = 2 and the constituent fractions are exactly 1/3 and 2/3.

Claim 163. The material of claim 1, wherein every constituent has fraction 1/k.

Claim 164. The material of claim 1, wherein every constituent has fraction at least 10^-19.

Claim 165. The material of claim 1, wherein at least one constituent has positive fraction less than 10^-19.

Claim 166. The material of claim 152, wherein at least one interval is [j times 10^(e-r),(j+1) times 10^(e-r)], with integer e from -19 through -1, nonnegative integer r, and integer j from 10^r through 10^(r+1)-1.

Claim 167. The material of claim 152, wherein at least one interval has arbitrary rational endpoints r and s satisfying 0 <= r <= s <= 1.

Claim 168. The material of claim 152, wherein the endpoints are specified as real target values and the selected fractions satisfy their interval constraints and exact normalization.

Claim 169. The material of claim 1, wherein each selected element i has a conditional atom-count isotope-state distribution y_i,a,s with nonnegative entries summing to one over its specified isotope-state inventory, and wherein the whole-inventory nuclide-state fractions are specified on the same basis as the fractions x_i of claim 1: on an atomic-fraction basis, the nuclide-state fraction is x_i y_i,a,s; and on a mass-fraction basis, for each positively populated isotope-state entry the nuclide-state fraction is x_i y_i,a,s mu_i,a,s/M_i, where mu_i,a,s is a qualified positive mass per counted atom under an expressly identified mass-allocation convention and M_i is the positive finite sum of y_i,a,s mu_i,a,s over the positively populated entries of element i; a zero-population entry has whole-inventory fraction zero on either basis.

Claim 170. The material of claim 169, wherein at least one constituent is specified by a single mass number and nuclear state with conditional fraction one.

Claim 171. The material of claim 169, wherein at least one constituent has two or more specified isotope-state entries of positive conditional fraction.

Claim 172. The material of claim 169, wherein each specified nuclear state has a source-qualified identifier that distinguishes a ground state from any specified isomer or excited state.

Claim 173. The material of claim 169, wherein a radioactive inventory is identified at a specified time and includes daughter constituents that fall within its stated denominator.

Claim 174. The material of claim 1, wherein a crystalline constituent phase has a specified structural model, site occupations, substitution rules and charge-balance constraints.

Claim 175. The material of claim 1, wherein an amorphous constituent phase is identified by a stated preparation history and a specified characterization convention for detectable crystalline fraction.

Claim 176. The material of claim 1, wherein molecular identity includes specified connectivity, stereochemistry, salt or solvate state, and hydration state where present.

Claim 177. The material of claim 1, wherein the material is a polymer specified by repeat units, incorporated ratios, sequence distribution, molecular-weight distribution and network architecture.

Claim 178. The material of claim 1, wherein the material is a composite specified by constituent identities, dimensions, orientations, interfaces and dispersion.

Claim 179. The material of claim 1, wherein the material has a specified layer order and layer dimensions, with local composition distinguished from whole-specimen composition.

Claim 180. The material of claim 1, wherein the material has a specified spatial composition field x_i(r,t) and a density-weighted inventory convention for deriving the whole-specimen fractions.

Claim 181. The material of claim 1, wherein the stated elemental inventory describes a specified single phase.

Claim 182. The material of claim 1, wherein the stated elemental inventory describes a specified multiphase assemblage and each phase is separately identified.

Claim 183. The material of claim 178, wherein constituent volume fractions and void fraction are specified separately from elemental atomic fractions under stated temperature, pressure, density and volume-assignment conventions.

Claim 184. The material of claim 1, wherein a nominal target fraction is accompanied by a separately specified measurement uncertainty or tolerance and the associated sampling convention.

Claim 185. The material of claim 1, wherein the product composition is specified separately from its feed proportions and from local measured composition.

Claim 186. A method of constructing a candidate-material record, comprising selecting a constituent support and composition basis; assigning positive normalized target fractions whose sum is one; recording structural, preparation, measurement and property fields; and recording an evidence status for each field without treating an unresolved field as a successful preparation.

Claim 187. The method of claim 186, comprising enumerating every increasing k-tuple of selected atomic numbers for a specified k from 1 through 138 and attaching a recognized or hypothetical status to each label.

Claim 188. The method of claim 186, comprising enumerating positive integer tuples with sum q for successive unrestricted positive denominators q, and retaining primitive common-denominator representations.

Claim 189. The method of claim 186, comprising enumerating positive integer tuples with sum D = 10^19 and storing their normalized fractions as exact integer numerator and denominator strings.

Claim 190. The method of claim 186, comprising selecting every endpoint pair l,u with 0 <= l <= u <= 10^p for each p from 0 through 19, forming correlated component boxes, and rejecting boxes that fail the applicable normalization and positivity tests.

Claim 191. The method of claim 186, comprising selecting component bounds L_i and U_i for the selected support; testing 0 <= L_i <= U_i <= 1 for every component and sum_i(L_i) <= 1 <= sum_i(U_i); when those tests are satisfied, constructing a preliminary nonnegative normalized witness z_i within the bounds by starting at z_i = L_i and allocating the residual 1-sum_i(L_i) in support order without exceeding any U_i until sum_i(z_i) = 1; and assigning as the positive target fractions of claim 186 a strictly positive normalized witness within those bounds, using the preliminary witness as that target only when every coordinate is positive and otherwise constructing a separate strictly positive witness before target assignment.

Claim 192. The method of claim 191, further comprising, for a box satisfying the tests of claim 191, determining strictly positive feasibility for the selected support by requiring U_i > 0 for every component; when sum_i(L_i) = 1, additionally requiring L_i > 0 for every component and using the lower-bound vector as the positive witness; and when sum_i(L_i) < 1, setting epsilon = min(min_i(U_i), (1-sum_i(L_i))/(2k)), replacing each lower bound by L'_i = max(L_i, epsilon), and constructing a strictly positive normalized witness by allocating the residual 1-sum_i(L'_i) within the unchanged upper bounds; and assigning the resulting positive witness as the target fractions of claim 186.

Claim 193. The method of claim 186, comprising selecting a k-component box with 0 <= L_i <= U_i <= 1; calculating b_i = max(1, ceiling(D L_i)) and c_i = min(D-k+1, floor(D U_i)) for D = 10^19; accepting the box as numerically feasible on the positive D-grid only when b_i <= c_i for every component and sum_i(b_i) <= D <= sum_i(c_i); and, on acceptance, constructing integers n_i within those bounds with sum_i(n_i) = D by residual allocation and assigning x_i = n_i/D as the positive normalized target fractions of claim 186.

Claim 194. A method of constructing a particle-system record, comprising identifying a species or model-qualified candidate, experimental status, quantum numbers, population or occupation convention, energy conditions, host or confinement, lifetime information, preparation and observation fields, while specifying a construction level that avoids counting a particle and its containing composite as independent copies of the same inventory.

Claim 195. The method of claim 194, wherein the record concerns a host-dependent excitation and specifies the host material, excitation mechanism, conditions, lifetime or linewidth, and observable.

Claim 196. The method of claim 194, wherein an unconfirmed entity is marked hypothetical, with a model identifier, parameter assumptions, conservation constraints and experimental-search status, without assigning unsupported measured properties.

Claim 197. The method of claim 186, comprising recording a property scenario with a baseline material and state, controlled change, proposed response, mechanism, conditions, discriminating test and evidence status.

Claim 198. The method of claim 197, comprising recording a credible alternative response and its mechanism as a separate conditional hypothesis, rather than asserting opposite responses for the same specimen under identical conditions.

Claim 199. The method of claim 186, comprising labeling attributed source teachings ESTABLISHED, stated mathematical or model results CALCULATED, unperformed tests or preparations PROPOSED, and possibilities without adequate support UNSUPPORTED, and retaining source and condition information with the labels.

18 Element and particle register

The following 138 rows are self-contained identity entries. Recognized rows follow the IUPAC table dated 4 May 2022, checked 29 September 2026. E119-E138 are local hypothetical placeholders, not accepted symbols. No row supplies a making method or atomic mass.

Z	Symbol / label	Name / local description	Status
1	H	hydrogen	Recognized
2	He	helium	Recognized
3	Li	lithium	Recognized
4	Be	beryllium	Recognized
5	B	boron	Recognized
6	C	carbon	Recognized
7	N	nitrogen	Recognized
8	O	oxygen	Recognized
9	F	fluorine	Recognized
10	Ne	neon	Recognized

Z	Symbol / label	Name / local description	Status
11	Na	sodium	Recognized
12	Mg	magnesium	Recognized
13	Al	aluminium	Recognized
14	Si	silicon	Recognized
15	P	phosphorus	Recognized
16	S	sulfur	Recognized
17	Cl	chlorine	Recognized
18	Ar	argon	Recognized
19	K	potassium	Recognized
20	Ca	calcium	Recognized
21	Sc	scandium	Recognized
22	Ti	titanium	Recognized
23	V	vanadium	Recognized
24	Cr	chromium	Recognized
25	Mn	manganese	Recognized
26	Fe	iron	Recognized
27	Co	cobalt	Recognized
28	Ni	nickel	Recognized
29	Cu	copper	Recognized
30	Zn	zinc	Recognized
31	Ga	gallium	Recognized
32	Ge	germanium	Recognized
33	As	arsenic	Recognized
34	Se	selenium	Recognized
35	Br	bromine	Recognized
36	Kr	krypton	Recognized
37	Rb	rubidium	Recognized
38	Sr	strontium	Recognized
39	Y	yttrium	Recognized
40	Zr	zirconium	Recognized

Z	Symbol / label	Name / local description	Status
41	Nb	niobium	Recognized
42	Mo	molybdenum	Recognized
43	Tc	technetium	Recognized
44	Ru	ruthenium	Recognized
45	Rh	rhodium	Recognized
46	Pd	palladium	Recognized
47	Ag	silver	Recognized
48	Cd	cadmium	Recognized
49	In	indium	Recognized
50	Sn	tin	Recognized
51	Sb	antimony	Recognized
52	Te	tellurium	Recognized
53	I	iodine	Recognized
54	Xe	xenon	Recognized
55	Cs	caesium	Recognized
56	Ba	barium	Recognized
57	La	lanthanum	Recognized
58	Ce	cerium	Recognized
59	Pr	praseodymium	Recognized
60	Nd	neodymium	Recognized
61	Pm	promethium	Recognized
62	Sm	samarium	Recognized
63	Eu	europium	Recognized
64	Gd	gadolinium	Recognized
65	Tb	terbium	Recognized
66	Dy	dysprosium	Recognized
67	Ho	holmium	Recognized
68	Er	erbium	Recognized
69	Tm	thulium	Recognized
70	Yb	ytterbium	Recognized

Z	Symbol / label	Name / local description	Status
71	Lu	lutetium	Recognized
72	Hf	hafnium	Recognized
73	Ta	tantalum	Recognized
74	W	tungsten	Recognized
75	Re	rhenium	Recognized
76	Os	osmium	Recognized
77	Ir	iridium	Recognized
78	Pt	platinum	Recognized
79	Au	gold	Recognized
80	Hg	mercury	Recognized
81	Tl	thallium	Recognized
82	Pb	lead	Recognized
83	Bi	bismuth	Recognized
84	Po	polonium	Recognized
85	At	astatine	Recognized
86	Rn	radon	Recognized
87	Fr	francium	Recognized
88	Ra	radium	Recognized
89	Ac	actinium	Recognized
90	Th	thorium	Recognized
91	Pa	protactinium	Recognized
92	U	uranium	Recognized
93	Np	neptunium	Recognized
94	Pu	plutonium	Recognized
95	Am	americium	Recognized
96	Cm	curium	Recognized
97	Bk	berkelium	Recognized
98	Cf	californium	Recognized
99	Es	einsteinium	Recognized
100	Fm	fermium	Recognized

Z	Symbol / label	Name / local description	Status
101	Md	mendelevium	Recognized
102	No	nobelium	Recognized
103	Lr	lawrencium	Recognized
104	Rf	rutherfordium	Recognized
105	Db	dubnium	Recognized
106	Sg	seaborgium	Recognized
107	Bh	bohrium	Recognized
108	Hs	hassium	Recognized
109	Mt	meitnerium	Recognized
110	Ds	darmstadtium	Recognized
111	Rg	roentgenium	Recognized
112	Cn	copernicium	Recognized
113	Nh	nihonium	Recognized
114	Fl	flerovium	Recognized
115	Mc	moscovium	Recognized
116	Lv	livermorium	Recognized
117	Ts	tennessine	Recognized
118	Og	oganesson	Recognized
119	E119	hypothetical element 119	Hypothetical placeholder
120	E120	hypothetical element 120	Hypothetical placeholder
121	E121	hypothetical element 121	Hypothetical placeholder
122	E122	hypothetical element 122	Hypothetical placeholder
123	E123	hypothetical element 123	Hypothetical placeholder
124	E124	hypothetical element 124	Hypothetical placeholder
125	E125	hypothetical element 125	Hypothetical placeholder
126	E126	hypothetical element 126	Hypothetical placeholder
127	E127	hypothetical element 127	Hypothetical placeholder
128	E128	hypothetical element 128	Hypothetical placeholder
129	E129	hypothetical element 129	Hypothetical placeholder
130	E130	hypothetical element 130	Hypothetical placeholder

Z	Symbol / label	Name / local description	Status
131	E131	hypothetical element 131	Hypothetical placeholder
132	E132	hypothetical element 132	Hypothetical placeholder
133	E133	hypothetical element 133	Hypothetical placeholder
134	E134	hypothetical element 134	Hypothetical placeholder
135	E135	hypothetical element 135	Hypothetical placeholder
136	E136	hypothetical element 136	Hypothetical placeholder
137	E137	hypothetical element 137	Hypothetical placeholder
138	E138	hypothetical element 138	Hypothetical placeholder

Particle category register

Six quark flavor categories, three charged-lepton particle/antiparticle categories, three neutrino flavor categories, and five boson categories. W+ and W- are grouped; gluon color components are grouped. This is a table convention, not an absolute count of physical particles or quantum states.

Flavor labels differ from neutrino mass eigenstates. The register does not assert that the minimal massless-neutrino Standard Model explains observed neutrino masses, or resolve Dirac versus Majorana nature.

Sources: CERN Standard Model and PDG 2026 tables. The 17 rows are categories under this convention, not all particle states or material ingredients.

Category	Labels	Accounting and status qualification
up quark flavor	u, anti-u	One flavor category; antiquark and three color states are not additional rows. No isolated free-quark ingredient is asserted.
down quark flavor	d, anti-d	One flavor category; antiquark and three color states are not additional rows. No isolated free-quark ingredient is asserted.
charm quark flavor	c, anti-c	One flavor category; antiquark and three color states are not additional rows. No isolated free-quark ingredient is asserted.
strange quark flavor	s, anti-s	One flavor category; antiquark and three color states are not additional rows. No isolated free-quark ingredient is asserted.
top quark flavor	t, anti-t	One flavor category; antiquark and three color states are not additional rows. No isolated free-quark ingredient is asserted.
bottom quark flavor	b, anti-b	One flavor category; antiquark and three color states are not additional rows. No isolated free-quark ingredient is asserted.
electron and positron	e-, e+	Particle and antiparticle grouped in one category.
muon and antimuon	mu-, mu+	Particle and antiparticle grouped in one category.

Category	Labels	Accounting and status qualification
tau and antitau	tau-, tau+	Particle and antiparticle grouped in one category.
electron-flavor neutrino	nu_electron, anti-nu_electron	Weak-interaction flavor category, not a mass eigenstate. Labels for neutrino/antineutrino interactions do not settle Dirac versus Majorana nature.
muon-flavor neutrino	nu_muon, anti-nu_muon	Weak-interaction flavor category, not a mass eigenstate. Labels for neutrino/antineutrino interactions do not settle Dirac versus Majorana nature.
tau-flavor neutrino	nu_tau, anti-nu_tau	Weak-interaction flavor category, not a mass eigenstate. Labels for neutrino/antineutrino interactions do not settle Dirac versus Majorana nature.
photon	gamma	One species category; momentum and polarization states are not separately enumerated.
gluon	g	Eight color components grouped in one category; no free isolated-gluon ingredient is asserted.
charged W bosons	W+, W-	Two charge-conjugate states grouped in one category.
Z boson	Z0	One species category; polarization states are not separate entries.
Higgs boson	H	Observed Higgs-boson category; additional hypothetical Higgs states are not asserted.

Composite particles, nuclear states, quasiparticles and hypothetical particle classes use the separate parameters in paragraphs [0046]-[0056]. Their possible identifiers are not additional recognized elemental rows. No complete current nuclide table or exhaustive theoretical-particle universe is claimed.

19 Claim support and review map

Textual support and physical enablement are different inquiries. Every claim has a separate preliminary substantive screening; a complete chronological novelty search and final statutory determinations remain unresolved. Historical individual assessment flags are preserved. The complete per-claim map is supplied in the companion claim_support_map_substantive_review_2026-10-03.json; the operative definitions and claims are printed here and do not depend on that external file.

Claims	Textual locations	Subject matter	Enablement and review status
1-138	[0010]-[0022], [0073]	Broad normalized material inventory and every exact component count.	Mathematical definitions present; composition-specific full-scope preparation absent. Counts above 118 necessarily include hypothetical labels.
139-147	[0010]-[0016], [0041], [0047]	Unary material, vocabulary restrictions and composition bases.	Identity or basis definitions do not establish novelty, availability or making methods.
148-168	[0017]-[0032], [0069]-[0072]	Rational and grid fractions, all endpoint ranges, complements and explicit boundary targets.	Exact arithmetic present; physical realization, tolerances and phase identity unresolved. Below-grid targets are broader than the floor-limited verifier.

Claims	Textual locations	Subject matter	Enablement and review status
169-173	[0046]-[0050]	Isotope, nuclear-state and time-indexed inventory descriptions.	Parameter identifiers do not establish nuclear existence, isolation, lifetime or synthesis.
174-185	[0012]-[0013], [0033]-[0043]	Structures, phases, spatial forms and measurement conventions.	Structural fields are specified as requirements, not completed material teachings for the universal composition domain.
186-193	[0005]-[0007], [0023]-[0032], [0073]-[0075]	Candidate-record construction and mathematical enumeration/feasibility methods.	Operations are mathematically described and preliminarily screened; a complete chronological novelty search, final eligibility, human inventive contribution and full legal support remain unresolved.
194-196	[0051]-[0056]	Separate particle-system record methods.	Recording method does not prepare or stabilize every particle or hypothetical system.
197-199	[0005], [0041]-[0042], [0057]-[0068], [0075]	Conditional property scenarios and local evidence status.	No measured applicant result, inherent property or universal predicted performance is supplied.

20 Source and version notes

Source links identify attributed teachings, definitions, and evaluation editions. Mathematical models, record designs, numerical examples, and proposed scenarios are this draft's calculations or proposals. Source retrieval and local document creation dates are not patent filing or public-availability dates. Robert M. Layne is named as proposed sole inventor for this review draft at his instruction; no inventor signature, applicant entitlement, priority claim, micro-entity certification, complete current isotope archive, or worldwide patent-status finding is assumed. The later filing memorandum addresses procedure separately from this technical description.

This prospective edition incorporates paragraph [0047] and claims 169 and 191-193 from the reviewed clarification proposal, with accompanying basis and target-assignment consistency changes. The isotope index a in $y_{i,a,s}$ is distinct from the elemental atomic fraction a_i. Its source-specific textual map and rendered review PDF describe this edition only. Adoption into an unfiled working source does not establish support in an earlier filed application; any claimed earlier benefit requires comparison with the actual relied-upon filed version. Applicant evidence and filing facts remain unresolved.